

Inference on Dynamic Spatial Autoregressive Models with Change Point Detection

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Abstract

We analyze a varying-coefficient dynamic spatial autoregressive model with spatial fixed effects. One salient feature of the model is the incorporation of multiple spatial weight matrices through their linear combinations with varying coefficients, which help solve the problem of choosing the most “correct” one for applied econometricians who often face the availability of multiple expert spatial weight matrices. We estimate and make inferences on the model coefficients and coefficients in basis expansions of the varying coefficients through penalized estimations, establishing the oracle properties of the estimators and the consistency of the overall estimated spatial weight matrix, which can be time-dependent. We further consider two applications of our model in change point detections in dynamic spatial autoregressive models, providing theoretical justifications in consistent change point locations estimation and practical implementations. Simulation experiments demonstrate the performance of our proposed methodology, and a real data analysis is also carried out.

Key words and phrases: Time-varying spatial weight matrix; spillover effects; instruments; adaptive LASSO; structural changes.

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1 Introduction

The study of spatial dependence in regional science gives rise to the techniques in spatial econometrics that we commonly use nowadays. Restricting to cross-sectional data only, a very general form of a model describing spatial dependence can be $\mathbf{y} = f(\mathbf{y}) + \boldsymbol{\epsilon}$ (Anselin, 1988), where $\mathbf{y}$ denotes a vector of d observed units, and $\boldsymbol{\epsilon}$ denotes an error term. A prominent and widely used candidate model is the spatial autoregressive model (see for example LeSage and Pace (2009)), which assumes a known *spatial weight matrix* $\mathbf{W}$ with zero diagonal and $f(\mathbf{y})$ of the form $f(\mathbf{y}) = \rho \mathbf{W}\mathbf{y}$ (or $f(\mathbf{y}) = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta}$ for a model with matrix of covariates $\mathbf{X}$), where ρ is called the spatial correlation coefficient.

Users of these models need to specify the $d \times d$ spatial weight matrix $\mathbf{W}$, which can be a contiguity matrix of 0 and 1, a matrix of inverse distances between two cities/regions, relative amount of import export etc. An obvious shortcoming for practitioners is to specify an “accurate” spatial weight matrix for use, often in the face of too many potential choices. This leads to a series of attempts to estimate the spatial weight matrix itself from data. For instance, see Pinkse et al. (2002) and Sun (2016) for models dealing with cross-sectional data only, both allowing for nonlinear spatial weight matrix estimation. Beenstock and Felsenstein (2012), Bhattacharjee and Jensen-Butler (2013), Lam and Souza (2020) and Higgins and Martellosio (2023) use spatial panel data for spatial weight matrix estimation, with Lam and Souza (2020) and Higgins and Martellosio (2023) allowing for multiple specified spatial weight matrices through a linear combination of them with constant coefficients.

Recent advances in spatial econometrics allows researchers to specify more complex models with an observed panel $\{\mathbf{y}_t\}$. Zhang and Shen (2015) consider partially linear covariate effects and constant spatial interactions using a sieve method to estimate a nonlinear function, while Sun and Malikov (2018) consider varying coefficients in both the spatial correlation coefficient (with underlying variables differ over observed units) and the covariate effects, assuming the nonlinear functions are smooth for kernel estimations. Liang et al. (2022) use kernel estimation on a model with constant spatial interactions but deterministic time-varying coefficient functions for the covariates, while Chang et al. (2024) generalize the model to include an unknown random time trend and deterministic time-varying spatial correlation coefficient, still using kernel estimation. Hong et al. (2024) investigate a model similar to Sun and Malikov (2018), but added dynamic terms involving $\mathbf{y}_{t-1}$.

However, all the above allow for one specified spatial weight matrix only. As mentioned before, practitioners often face with too many potential choices for a spatial weight matrix. Combining the flexibility of allowing for multiple specified spatial weight matrices as input in Lam and Souza (2020) and varying effects in spatial interactions over observed variables or time directly, we propose a model similar to that in Lam and Souza (2020), but with varying coefficients in the linear combination of the spatial weight

matrices. The varying coefficients can be varying over some observed variables (stochastic) or time directly (non-stochastic).

Our contribution in this paper are three-folds. Firstly, using basis representations, we allow for the varying coefficients to be either stochastic or directly time-varying, without the need for any smoothness conditions. Hence the final estimated spatial weight matrix can be either stochastic or deterministic, e.g., directly time dependent. Secondly, our adaptive LASSO estimators are proved to have the oracle properties, so that ill-specified spatial weight matrices which are irrelevant in the end will be dropped with probability going to 1 as the dimension d and the sample size T go to infinity. At the same time, the effects of relevant spatial weight matrices can be seen to be truly varying or not, again with probability going to 1 as $d, T \rightarrow \infty$. This greatly facilitates the interpretability of the spillover effects over time. Last but not least, our framework includes special cases such as spatial autoregressive models with structural changes (Li, 2018) or threshold variables (Deng, 2018; Li and Lin, 2024). Section 5 explores the applications to multiple change points detection in both spatial autoregressive models with structural changes or threshold variables, suggesting an applicable algorithm for consistent change points detection in both cases.

Throughout this paper and unless otherwise specified, we use a lower-case letter a , a bold lower-case letter $\mathbf{a}$, and a bold capital letter $\mathbf{A}$ to denote a scalar, a vector, and a matrix, respectively. We also use $a_i, A_{ij}, \mathbf{A}_i, \mathbf{A}_{\cdot j}$ to denote, respectively, the i -th element of $\mathbf{a}$, the (i, j) -th element of $\mathbf{A}$, the i -th row vector (as a column vector) of $\mathbf{A}$, and the j -th column vector of $\mathbf{A}$. Hereafter, given a positive integer m , define $[m] := \{1, 2, \dots, m\}$, and we use $\otimes$ to represent the Kronecker product.

For a given set, we denote by $|\cdot|$ its cardinality. For a vector $\mathbf{v}$, we denote its L_2 -norm by $\|\mathbf{v}\|$. For a given matrix $\mathbf{A}$, we use $\|\mathbf{A}\|$ to denote its spectral norm, $\|\mathbf{A}\|_F$ its Frobenius norm, $\|\mathbf{A}\|_{\max} := \max_{i,j} |A_{ij}|$ its max norm, $\|\mathbf{A}\|_1 := \max_j \sum_i |A_{ij}|$ its L_1 -norm, and $\|\mathbf{A}\|_{\infty} := \max_i \sum_j |A_{ij}|$ its L_{∞} -norm. For $q > 0$, we define the L_q -norm of a given real-valued random variable x as $\|x\|_q := (\mathbb{E}|x|^q)^{1/q}$. Without loss of generality, we always assume the eigenvalues of a matrix are arranged in descending order, with their eigenvectors ordered correspondingly.

The rest of the paper is organized as follows. Section 2 introduces the dynamic spatial autoregressive model and presents a procedure using adaptive LASSO to estimate the spatial fixed effect, spatial autoregressive parameters in a basis expansion, and the regression coefficients. Section 3 includes the required assumptions and the theoretical guarantees on the parameter estimators. Section 4 covers the algorithm for practical implementations including model selection and covariance matrix estimation for our estimators. Section 5 focuses on multiple change points detection for a spatial autoregressive model with threshold variables or structural changes. Finally, numerical results are presented in Section 6, with a case study of enterprise profits in China. All technical proofs and additional lemmas are deferred to the Appendix.

2 Model and Estimation

2.1 Dynamic spatial autoregressive model

We propose a framework of dynamic spatial autoregressive models with fixed effects such that for each time $t \in [T]$,

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \sum_{j=1}^p \left(\phi_{j,0}^* + \sum_{k=1}^{l_j} \phi_{j,k}^* z_{j,k,t} \right) \mathbf{W}_j \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad (2.1)$$

where $\mathbf{y}_t \in \mathbb{R}^d$ is the observed vector at time t , and $\boldsymbol{\mu}^*$ is a constant vector of spatial fixed effects. Each $\mathbf{W}_j \in \mathbb{R}^{d \times d}$ is a pre-specified spatial weight matrix provided by researchers to feature the spillover effects of cross-sectional units from their neighbours. Each $\mathbf{W}_j$ has zero entries on its main diagonal with no restrictions on the signs of off-diagonal entries, and can be asymmetric. Each term $(\phi_{j,0}^* + \sum_{k=1}^{l_j} \phi_{j,k}^* z_{j,k,t})$ is essentially a spatial correlation coefficient for the spatial weight matrix $\mathbf{W}_j$ (see Lam and Souza (2020) as well), which can be time-varying by being presented as either a basis expansion using some non-random pre-specified set of basis $\{z_{j,k,t}\}$, or an affine combination of random variables $\{z_{j,k,t}\}$. In either case, we call the $\{z_{j,k,t}\}$'s the dynamic variables hereafter. For $j \in [p]$, $k \in [l_j]$, the parameters $\phi_{j,0}^*, \phi_{j,k}^*$ are unknown and need to be estimated. The covariate matrix $\mathbf{X}_t$ has size $d \times r$, with $\boldsymbol{\beta}^*$ the corresponding unknown regression coefficients of length r . Finally, $\boldsymbol{\epsilon}_t$ is the idiosyncratic noise with zero mean.

Without loss of generality, we assume $\mathbf{X}_t$ to have zero mean. Otherwise, we read

$$\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* = (\boldsymbol{\mu}^* + \mathbb{E}[\mathbf{X}_t] \boldsymbol{\beta}^*) + (\mathbf{X}_t - \mathbb{E}[\mathbf{X}_t]) \boldsymbol{\beta}^*,$$

which leads to estimating $(\boldsymbol{\mu}^* + \mathbb{E}[\mathbf{X}_t] \boldsymbol{\beta}^*)$ as the spatial fixed effects instead. We can rewrite (2.1) as a traditional spatial autoregressive model $\mathbf{y}_t = \boldsymbol{\mu}^* + \mathbf{W}_t^* \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t$ by defining the true spatial weight matrix at time t as

$$\mathbf{W}_t^* := \sum_{j=1}^p \left(\phi_{j,0}^* + \sum_{k=1}^{l_j} \phi_{j,k}^* z_{j,k,t} \right) \mathbf{W}_j, \quad \text{with } -1 < \rho_t^* := \sum_{j=1}^p \left(\phi_{j,0}^* + \sum_{k=1}^{l_j} \phi_{j,k}^* z_{j,k,t} \right) < 1. \quad (2.2)$$

The restrictions on the parameter ρ_t^* for all $t \in [T]$ ensure that the model is stationary. See Assumptions (M2) and (M2') for technical details. We define $L := p + \sum_{j=1}^p l_j$ and further reformulate (2.1) as

$$\mathbf{y}_t = \boldsymbol{\mu}^* + (\boldsymbol{\Lambda}_t \boldsymbol{\Phi}^*) \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad \text{where} \quad (2.3)$$

$$\boldsymbol{\Lambda}_t := (\boldsymbol{\Lambda}_{1,t}, \boldsymbol{\Lambda}_{2,t}, \dots, \boldsymbol{\Lambda}_{p,t}) \in \mathbb{R}^{d \times dL}, \quad \text{with } \boldsymbol{\Lambda}_{j,t} := (\mathbf{W}_j, z_{j,1,t} \mathbf{W}_j, \dots, z_{j,l_j,t} \mathbf{W}_j) \in \mathbb{R}^{d \times (d + dl_j)},$$

$$\boldsymbol{\Phi}^* := ((\boldsymbol{\Phi}_1^*)^\top, (\boldsymbol{\Phi}_2^*)^\top, \dots, (\boldsymbol{\Phi}_p^*)^\top)^\top \in \mathbb{R}^{dL \times d}, \quad \text{with } \boldsymbol{\Phi}_j^* := (\phi_{j,0}^* \mathbf{I}_d, \phi_{j,1}^* \mathbf{I}_d, \dots, \phi_{j,l_j}^* \mathbf{I}_d)^\top.$$

Due to the endogeneity in $\mathbf{y}_t$ and potentially $\mathbf{X}_t$, we assume that a set of valid instrumental variables $\mathbf{U}_t$ are available for $t \in [T]$. More specifically, each $\mathbf{U}_t$ is independent of $\boldsymbol{\epsilon}_t$ but is correlated with $\mathbf{y}_t$ and the endogenous $\mathbf{X}_t$. Note that if $\mathbf{X}_t$ is exogenous, we may simply have $\mathbf{U}_t = \mathbf{X}_t$. Following Kelejian and Prucha (1998), we can construct instruments $\mathbf{B}_t$ as a $d \times v$ matrix with $v \geq r$ by interacting each given spatial weight matrix with $\mathbf{U}_t$ such that $\mathbf{B}_t$ is composed of at least a subset of linearly independent columns in¹

$$\left\{ \mathbf{U}_t, \mathbf{W}_1 \mathbf{U}_t, \mathbf{W}_1^2 \mathbf{U}_t, \dots, \mathbf{W}_1 \mathbf{U}_t, \mathbf{W}_1^2 \mathbf{U}_t, \dots, \mathbf{W}_p \mathbf{U}_t, \mathbf{W}_p^2 \mathbf{U}_t, \dots, \mathbf{W}_p \mathbf{U}_t, \mathbf{W}_p^2 \mathbf{U}_t, \dots \right\}. \quad (2.4)$$

To enhance interpretability of the true spatial weight matrix $\mathbf{W}_t^*$, we assume that the vector of coefficients $\boldsymbol{\phi}^* := (\boldsymbol{\phi}_1^{*\top}, \boldsymbol{\phi}_2^{*\top}, \dots, \boldsymbol{\phi}_p^{*\top})^\top$ (with $\boldsymbol{\phi}_j^* := (\phi_{j,0}^*, \phi_{j,1}^*, \dots, \phi_{j,l_j}^*)^\top$) is sparse and that the dynamic feature of model (2.1) is driven only by a few $\{z_{j,k,t}\}$. Using the LASSO (Tibshirani, 1996), an L_1 penalty $\lambda \|\boldsymbol{\phi}\|_1$ can be included in the regression problem to shrink the estimators toward zero and some of them to exactly zero, where $\lambda > 0$ is a tuning parameter. However, this form of regularisation penalizes uniformly on each entry, which may lead to over- or under-penalization. The former induces bias while the latter fails sign-consistency, i.e., zeros are estimated exactly as zeros and non-zeros are estimated with the correct signs.

To ensure the zero-consistency in variable selection, a necessary “irrepresentable condition” is often imposed (Zhao and Yu, 2006). Subsequently, Zou (2006) reweighs the regularization to be $\lambda \mathbf{u}^\top |\boldsymbol{\phi}|$ where $|\cdot|$ is applied entrywise and $\mathbf{u}$ contains the inverse of the initial estimators of $\boldsymbol{\phi}^*$. Now the sign-consistency can be ensured even without the irrepresentable condition if the estimators in $\mathbf{u}$ are $\sqrt{T}$ -consistent. Such a framework adaptively penalizes the magnitude of the estimators and is hence called “adaptive LASSO”. To this end, we start by profiling out $\boldsymbol{\beta}$. To make use of the instruments, define $\bar{\mathbf{B}} := T^{-1} \sum_{t=1}^T \mathbf{B}_t$. If $\boldsymbol{\phi}$ (and hence $\boldsymbol{\Phi}$) is given, by multiplying $(\mathbf{B}_t - \bar{\mathbf{B}})^\top$ and summing over all $t \in [T]$ on both sides of (2.3), we then have

$$\sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}})^\top (\mathbf{I}_d - \boldsymbol{\Lambda}_t \boldsymbol{\Phi}) \mathbf{y}_t = \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{X}_t \boldsymbol{\beta} + \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}})^\top \boldsymbol{\epsilon}_t,$$

where the true values $\boldsymbol{\Phi}^*$ and $\boldsymbol{\beta}^*$ are replaced by $\boldsymbol{\Phi}$ and $\boldsymbol{\beta}$ respectively. Note that the spatial fixed effect $\boldsymbol{\mu}^*$ vanishes since $\sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}})^\top \boldsymbol{\mu}^* = \mathbf{0}$. Therefore, the least squares estimator of $\boldsymbol{\beta}^*$ given $\boldsymbol{\phi}$ can be denoted as

$$\boldsymbol{\beta}(\boldsymbol{\phi}) = \left(\sum_{s=1}^T \mathbf{X}_s^\top (\mathbf{B}_s - \bar{\mathbf{B}}) \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{X}_t \right)^{-1} \sum_{s=1}^T \mathbf{X}_s^\top (\mathbf{B}_s - \bar{\mathbf{B}}) \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}})^\top (\mathbf{I}_d - \boldsymbol{\Lambda}_t \boldsymbol{\Phi}) \mathbf{y}_t. \quad (2.5)$$

In order to facilitate formulating the adaptive LASSO problem by accommodating the instrumental

¹Ideally, we should have $\mathbf{B}_t$ of the form $[\sum_{j=1}^p (\phi_{j,0} + \sum_{k=1}^{l_j} \phi_{j,k} z_{j,k,t}) \mathbf{W}_j]^m \mathbf{U}_t$ for $m = 0, 1, 2, \dots$. However, similar to Lam and Souza (2020), each $\phi_{j,0}$ and $\phi_{j,k}$ is unknown and hence we exclude any cross-terms with more than one $\mathbf{W}_j$.

variables, we write $\gamma := v^{-1}\mathbf{1}_v$ and denote the i -th row of $\mathbf{B}_t$ and $\bar{\mathbf{B}}$ by $\mathbf{B}_{t,i}$ and $\bar{\mathbf{B}}_{i\cdot}$, respectively. We then define the outcome and covariates filtered through instrumental variables as

$$\mathbf{y}_{B,i} := \sum_{t=1}^T (\mathbf{B}_{t,i} - \bar{\mathbf{B}}_{i\cdot})^\top \gamma \mathbf{y}_t, \quad \mathbf{X}_{B,i} := \sum_{t=1}^T (\mathbf{B}_{t,i} - \bar{\mathbf{B}}_{i\cdot})^\top \gamma \mathbf{X}_t, \quad \text{for } i \in [d].$$

The least squares problem is then

$$\tilde{\phi} = \arg \min_{\phi} \frac{1}{2T} \sum_{i=1}^d \left\| (\mathbf{I}_d - \Lambda_t \Phi) \mathbf{y}_{B,i} - \mathbf{X}_{B,i} \beta(\phi) \right\|^2. \quad (2.6)$$

Using this solution as an initial estimator, the adaptive LASSO problem becomes solving for

$$\begin{aligned} \hat{\phi} = \arg \min_{\phi} \frac{1}{2T} \sum_{i=1}^d \left\| (\mathbf{I}_d - \Lambda_t \Phi) \mathbf{y}_{B,i} - \mathbf{X}_{B,i} \beta(\phi) \right\|^2 + \lambda \mathbf{u}^\top |\phi|, \\ \text{subj. to } \|\Lambda_t \Phi\|_\infty < 1, \quad \text{with } |\mathbf{z}_t^\top \phi| < 1, \end{aligned} \quad (2.7)$$

where $\mathbf{z}_t := (\mathbf{z}_{1,t}^\top, \mathbf{z}_{2,t}^\top, \dots, \mathbf{z}_{p,t}^\top)^\top$, $\mathbf{z}_{j,t} := (1, z_{j,1,t}, \dots, z_{j,l_j,t})^\top$, $\mathbf{u} := (|\tilde{\phi}_{1,0}|^{-1}, \dots, |\tilde{\phi}_{p,l_p}|^{-1})^\top$, $|\phi| := (|\phi_{1,0}|, \dots, |\phi_{p,l_p}|)^\top$ and λ is a tuning parameter. With $\hat{\phi}$ (and hence $\hat{\Phi}$), the adaptive LASSO estimators for β^* can be obtained by $\hat{\beta} := \beta(\hat{\phi})$ and the fixed effect estimator by

$$\hat{\mu} := \frac{1}{T} \sum_{t=1}^T \left\{ (\mathbf{I}_d - \Lambda_t \hat{\Phi}) \mathbf{y}_t - \mathbf{X}_t \hat{\beta} \right\}. \quad (2.8)$$

2.2 Full matrix notations

To facilitate both the theoretical results and practical implementation, we present the least squares and the adaptive LASSO problems in matrix notations in this section. First, we introduce

$$\mathbf{B} := T^{-1/2} d^{-a/2} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma) := T^{-1/2} d^{-a/2} \mathbf{I}_d \otimes \{ (\mathbf{I}_T \otimes \gamma^\top) (\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})^\top \}, \quad (2.9)$$

where a is a constant that gauges the correlation between $\mathbf{B}_t$ and $\mathbf{X}_t$ so that a larger a generally means that $\mathbf{B}_t$ is correlated with more covariates in $\mathbf{X}_t$. See Assumption (R4) for more technical details. As in Lam and Souza (2020), in practice we can set $a = 1$ to compute $\mathbf{B}$, which would not change the optimal values of any tuning parameters or estimators in the adaptive LASSO problem below.

Denote by $z_{j,0,t} = 1$ for any $j \in [p], t \in [T]$ for ease of notation. We now rewrite (2.1) as

$$\mathbf{y} = \mu^* \otimes \mathbf{1}_T + \mathbf{V} \phi^* + \mathbf{X}_{\beta^*} \text{vec}(\mathbf{I}_d) + \epsilon, \quad (2.10)$$

where

$$\begin{aligned}\mathbf{y} &:= \text{vec}((\mathbf{y}_1, \dots, \mathbf{y}_T)^\top), \quad \boldsymbol{\epsilon} := \text{vec}((\boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_T)^\top), \quad \mathbf{V} := (\mathbf{V}_1, \dots, \mathbf{V}_p), \\ \mathbf{V}_j &:= [\boldsymbol{\Gamma}_{j,0} \text{vec}(\mathbf{W}_j^\top), \boldsymbol{\Gamma}_{j,1} \text{vec}(\mathbf{W}_j^\top), \dots, \boldsymbol{\Gamma}_{j,l_j} \text{vec}(\mathbf{W}_j^\top)], \\ \boldsymbol{\Gamma}_{j,k} &:= \mathbf{I}_d \otimes (z_{j,k,1} \mathbf{y}_1, \dots, z_{j,k,T} \mathbf{y}_T)^\top, \quad \mathbf{X}_{\boldsymbol{\beta}^*} := \mathbf{I}_d \otimes \{(\mathbf{I}_T \otimes (\boldsymbol{\beta}^*)^\top)(\mathbf{X}_1, \dots, \mathbf{X}_T)^\top\}.\end{aligned}$$

By writing in this way, we observe that the model now has design matrix $\mathbf{V}$ in a classical linear regression setting, except that the endogenous variables $\mathbf{y}_t$ are present in $\mathbf{V}$. We thus obtain the augmented model by left-multiplying both sides of (2.10) by $\mathbf{B}^\top$:

$$\mathbf{B}^\top \mathbf{y} = \mathbf{B}^\top \mathbf{V} \boldsymbol{\phi}^* + \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}^*} \text{vec}(\mathbf{I}_d) + \mathbf{B}^\top \boldsymbol{\epsilon}, \quad (2.11)$$

where the augmented spatial fixed effect vanishes since $\mathbf{B}^\top (\boldsymbol{\mu}^* \otimes \mathbf{1}_T) = T^{-1/2} d^{-a/2} \boldsymbol{\mu}^* \otimes \{(\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})(\mathbf{I}_T \otimes \boldsymbol{\gamma}) \mathbf{1}_T\} = \mathbf{0}$.

For any matrix $\mathbf{C}$, denote by $\mathbf{C}^\otimes := \mathbf{I}_T \otimes \mathbf{C}$, we can read (2.1) as

$$\mathbf{y}^\nu = \mathbf{1}_T \otimes \boldsymbol{\mu}^* + \sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k}^* \mathbf{W}_j^\otimes \mathbf{y}_{j,k}^\nu + \mathbf{X} \boldsymbol{\beta}^* + \boldsymbol{\epsilon}^\nu, \quad \text{where} \quad (2.12)$$

$$\mathbf{y}^\nu := (\mathbf{y}_1^\top, \dots, \mathbf{y}_T^\top)^\top, \quad \mathbf{y}_{j,k}^\nu := (z_{j,k,1} \mathbf{y}_1^\top, \dots, z_{j,k,T} \mathbf{y}_T^\top)^\top, \quad \boldsymbol{\epsilon}^\nu := (\boldsymbol{\epsilon}_1^\top, \dots, \boldsymbol{\epsilon}_T^\top)^\top, \quad \mathbf{X} := (\mathbf{X}_1^\top, \dots, \mathbf{X}_T^\top)^\top,$$

thus with $\mathbf{B}^\nu := (\mathbf{B}_1^\top - \bar{\mathbf{B}}^\top, \dots, \mathbf{B}_T^\top - \bar{\mathbf{B}}^\top)^\top$, we may write (2.5) in matrix form as

$$\boldsymbol{\beta}(\boldsymbol{\phi}) = \left(\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \left(\mathbf{y}^\nu - \sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k} \mathbf{W}_j^\otimes \mathbf{y}_{j,k}^\nu \right). \quad (2.13)$$

Together with (2.11), the least squares problem in (2.6) can be described as

$$\tilde{\boldsymbol{\phi}} = \arg \min_{\boldsymbol{\phi}} \frac{1}{2T} \left\| \mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{V} \boldsymbol{\phi} - \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi})} \text{vec}(\mathbf{I}_d) \right\|^2. \quad (2.14)$$

With the least squares estimator $\tilde{\boldsymbol{\phi}}$, the adaptive LASSO problem in (2.7) in matrix notation is

$$\begin{aligned}\hat{\boldsymbol{\phi}} &= \arg \min_{\boldsymbol{\phi}} \frac{1}{2T} \left\| \mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{V} \boldsymbol{\phi} - \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}(\boldsymbol{\phi})} \text{vec}(\mathbf{I}_d) \right\|^2 + \lambda \mathbf{u}^\top |\boldsymbol{\phi}|, \\ \text{subj. to } &\|\boldsymbol{\Lambda}_t \boldsymbol{\Phi}\|_\infty < 1, \quad \text{with } |\mathbf{z}_t^\top \boldsymbol{\phi}| < 1.\end{aligned} \quad (2.15)$$

Note that the squared error in both (2.14) and (2.15) are still implicit in $\boldsymbol{\phi}$ due to the term $\boldsymbol{\beta}(\boldsymbol{\phi})$. To

this end, define

$$\begin{aligned}\mathbf{Y}_W &:= (\mathbf{W}_1^\otimes \mathbf{y}_{1,0}^\nu, \dots, \mathbf{W}_1^\otimes \mathbf{y}_{1,l_1}^\nu, \dots, \mathbf{W}_p^\otimes \mathbf{y}_{p,0}^\nu, \dots, \mathbf{W}_p^\otimes \mathbf{y}_{p,l_p}^\nu), \\ \Xi &:= T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \right) [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top.\end{aligned}$$

With all lengthy simplification steps relegated to the supplementary materials, we show that our least squares problem in (2.14) can be rewritten as

$$\begin{aligned}\tilde{\phi} &= \arg \min_{\phi} \frac{1}{2T} \left\| \mathbf{B}^\top \mathbf{y} - \Xi \mathbf{y}^\nu - (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W) \phi \right\|^2 \\ &= [(\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)]^{-1} (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{y} - \Xi \mathbf{y}^\nu).\end{aligned}\tag{2.16}$$

The adaptive LASSO problem in (2.15) can be written as

$$\begin{aligned}\hat{\phi} &= \arg \min_{\phi} \frac{1}{2T} \left\| \mathbf{B}^\top \mathbf{y} - \Xi \mathbf{y}^\nu - (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W) \phi \right\|^2 + \lambda \mathbf{u}^\top |\phi|, \\ \text{subj. to } & \|\mathbf{\Lambda}_t \Phi\|_\infty < 1, \quad \text{with } |\mathbf{z}_t^\top \phi| < 1.\end{aligned}\tag{2.17}$$

3 Assumptions and Theoretical Results

We first present some notations involving the measure of serial dependence of all time series variables, which is gauged by the functional dependence measure introduced by Wu (2005). We state all the assumptions used in this paper in Section 3.1. Denote $\{\mathbf{x}_t\} = \{\text{vec}(\mathbf{X}_t)\}$ and $\{\mathbf{b}_t\} = \{\text{vec}(\mathbf{B}_t)\}$ to be the vectorized processes for $\{\mathbf{X}_t\}$ and $\{\mathbf{B}_t\}$ with length dr and dv , respectively. For $t \in [T]$, we assume that

$$\mathbf{x}_t = [f_i(\mathcal{F}_t)]_{i \in [dr]}, \quad \mathbf{b}_t = [g_i(\mathcal{G}_t)]_{i \in [dv]}, \quad \boldsymbol{\epsilon}_t = [h_i(\mathcal{H}_t)]_{i \in [d]},\tag{3.1}$$

where $f_i(\cdot)$'s, $g_i(\cdot)$'s, $h_i(\cdot)$'s are measurable functions defined on the real line, and $\mathcal{F}_t = (\dots, \mathbf{e}_{x,t-1}, \mathbf{e}_{x,t})$, $\mathcal{G}_t = (\dots, \mathbf{e}_{b,t-1}, \mathbf{e}_{b,t})$, $\mathcal{H}_t = (\dots, \mathbf{e}_{\epsilon,t-1}, \mathbf{e}_{\epsilon,t})$ are defined by i.i.d. processes $\{\mathbf{e}_{x,t}\}$, $\{\mathbf{e}_{b,t}\}$ and $\{\mathbf{e}_{\epsilon,t}\}$ respectively, with $\{\mathbf{e}_{b,t}\}$ independent of $\{\mathbf{e}_{\epsilon,t}\}$ but correlated with $\{\mathbf{e}_{x,t}\}$. For $q > 0$, we define

$$\begin{aligned}\theta_{t,q,i}^x &:= \|x_{t,i} - x'_{t,i}\|_q = (\mathbb{E}|x_{t,i} - x'_{t,i}|^q)^{1/q}, \quad i \in [dr], \\ \theta_{t,q,i}^b &:= \|b_{t,i} - b'_{t,i}\|_q = (\mathbb{E}|b_{t,i} - b'_{t,i}|^q)^{1/q}, \quad i \in [dv], \\ \theta_{t,q,i}^\epsilon &:= \|\epsilon_{t,i} - \epsilon'_{t,i}\|_q = (\mathbb{E}|\epsilon_{t,i} - \epsilon'_{t,i}|^q)^{1/q}, \quad i \in [d],\end{aligned}\tag{3.2}$$

where $x'_{t,i} = f_i(\mathcal{F}'_t)$, $\mathcal{F}'_t = (\dots, \mathbf{e}_{x,-1}, \mathbf{e}'_{x,0}, \mathbf{e}_{x,1}, \dots, \mathbf{e}_{x,t})$, with $\mathbf{e}'_{x,0}$ independent of all other $\mathbf{e}_{x,j}$'s. Hence $x'_{t,i}$ is a coupled version of $x_{t,i}$ with $\mathbf{e}_{x,0}$ replaced by its i.i.d. copy $\mathbf{e}'_{x,0}$. We have similar definitions for $b'_{t,i}$ and $\epsilon'_{t,i}$. Intuitively, a large $\theta^x_{t,q,i}$ implies strong serial correlation in $\mathbf{x}_t$ and incorporates some tail conditions of $f_i(\cdot)$'s, i.e., how $f_i(\cdot)$ frames $\mathbf{e}_{x,0}$ at time t and how exaggerated the functional $f_i(\cdot)$ is.

3.1 Assumptions

We present here the set of assumptions for the our model. In summary, (I1) helps to identify the model, assumptions prefixed “M” renders the model framework while those prefixed “R” are more technical.

(I1) (Identification) *The matrix $\mathbf{Q}^\top \mathbf{Q}$ has all its eigenvalues uniformly bounded away from 0, where*

$$\mathbf{Q} = [\mathbb{E}(\mathbf{B}^\top \mathbf{V}), \mathbb{E}(\mathbf{B}^\top \widetilde{\mathbf{X}})], \quad \widetilde{\mathbf{X}} = (\mathbf{x}_{1,1}, \dots, \mathbf{x}_{T,1}, \dots, \mathbf{x}_{1,d}, \dots, \mathbf{x}_{T,d})^\top.$$

(M1) (Time series in $\mathbf{X}_t$, $\mathbf{B}_t$ and ϵ_t) *The processes $\{\mathbf{X}_t\}$, $\{\mathbf{B}_t\}$ and $\{\epsilon_t\}$ are second-order stationary and defined in (3.1), with $\{\mathbf{X}_t\}$ and $\{\epsilon_t\}$ having mean zero. The tail condition $\mathbb{P}(|Z| > z) \leq D_1 \exp(-D_2 z^\ell)$ is satisfied for the variables $B_{t,ij}$, $X_{t,ij}$, $\epsilon_{t,i}$ by the same constants D_1 , D_2 and ℓ .*

With (3.2), define the tail sum

$$\Theta_{m,q}^x = \sum_{t=m}^{\infty} \max_{i \in [dr]} \theta_{t,q,i}^x, \quad \Theta_{m,q}^b = \sum_{t=m}^{\infty} \max_{i \in [dv]} \theta_{t,q,i}^b, \quad \Theta_{m,q}^\epsilon = \sum_{t=m}^{\infty} \max_{i \in [d]} \theta_{t,q,i}^\epsilon.$$

We assume that for some $w > 2$, $\Theta_{m,2w}^x$, $\Theta_{m,2w}^b$, $\Theta_{m,2w}^\epsilon \leq Cm^{-\alpha}$ with $\alpha, C > 0$ being constants that can depend on w .

(M2) (Spatial weight matrix $\mathbf{W}_t^*$ with non-random basis $z_{j,k,t}$) *With the true spatial weight matrix $\mathbf{W}_t^*$ defined in (2.2) using some uniformly bounded non-stochastic basis $\{z_{j,k,t}\}$ for $j \in [p]$, $k \in [l_j]$, there exists a constant $\eta > 0$ such that for all $t \in [T]$, $\|\mathbf{W}_t^*\|_\infty < \eta < 1$ uniformly as $d \rightarrow \infty$. The elements in $\mathbf{W}_t^*$ can be negative, and $\mathbf{W}_t^*$ can be asymmetric. Furthermore, ρ_t^* defined in (2.2) satisfies $|\rho_t^*| < 1$.*

(M2') (Spatial weight matrix $\mathbf{W}_t^*$ with random $z_{j,k,t}$) *Same as Assumption (M2), except that $\{z_{j,k,t}\}$ is a zero mean stochastic process with support $[-1, 1]$, such that $z_{j,k,t} = u_{j,k}(\mathcal{U}_t)$ similar to (3.1), with $\mathbb{E}(z_{j,k,t} \mathbf{X}_t) = \mathbf{0}$, $\mathbb{E}(z_{j,k,t} \epsilon_t) = \mathbf{0}$, and $\Theta_{m,2w}^z \leq Cm^{-\alpha}$ as in Assumption (M1). Furthermore:*

1. $\sum_{j=1}^p \sum_{k=0}^{l_j} \|\phi_{j,k}^* \mathbf{W}_j\|_\infty < \eta < 1$ uniformly as $d \rightarrow \infty$.
2. $\sum_{j=1}^p \sum_{k=0}^{l_j} |\phi_{j,k}^*| < 1$.

(R1) Denote by $\mathbf{D}_W \in \mathbb{R}^{d^2 L \times L}$ the block diagonal matrix with $\mathbf{D}_W := \text{diag}\{\mathbf{I}_{1+l_1} \otimes \text{vec}(\mathbf{W}_1^\top), \dots, \mathbf{I}_{1+l_p} \otimes \text{vec}(\mathbf{W}_p^\top)\}$, then there exists a constant $u > 0$ with $\sigma_L^2(\mathbf{D}_W) \geq du > 0$ uniformly as $d \rightarrow \infty$, where $\sigma_i(\mathbf{A})$ denotes the i -th largest singular value of a matrix $\mathbf{A}$.

Moreover, $\max_j \|\mathbf{W}_j\|_1, \|\mathbf{W}_j\|_\infty \leq c < \infty$ uniformly as $d \rightarrow \infty$ for some constant $c > 0$.

(R2) Write $\boldsymbol{\epsilon}_t = \boldsymbol{\Sigma}_\epsilon^{1/2} \boldsymbol{\epsilon}_t^*$ with $\boldsymbol{\Sigma}_\epsilon$ being the covariance matrix of $\boldsymbol{\epsilon}_t$, then $\|\boldsymbol{\Sigma}_\epsilon\|_{\max} \leq \sigma_{\max}^2 < \infty$ uniformly as $d \rightarrow \infty$. The same applies to the variance of the elements in $\mathbf{B}_t$.

Assume also $\|\boldsymbol{\Sigma}_\epsilon^{1/2}\|_\infty \leq S_\epsilon < \infty$ uniformly as $d \rightarrow \infty$, with $\{\epsilon_{t,i}^*\}_{i \in [d]}$ being a martingale difference with respect to the filtration generated by $\sigma(\epsilon_{t,1}, \dots, \epsilon_{t,i})$. The tail condition and functional dependence in Assumption (M1) is also satisfied by $\{\boldsymbol{\epsilon}_t^*\}_{t \in [T]}$.

(R3) All singular values of $\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)$ are uniformly larger than du for some constant $u > 0$, while the maximum singular value is of order d . Individual entries in the matrix $\mathbb{E}(\mathbf{b}_t \mathbf{x}_t^\top)$ are uniformly bounded away from infinity, with $\mathbf{x}_t$ and $\mathbf{b}_t$ defined in (3.1).

(R4) For $a \in (0, 1)$ the same as in the definition (2.9), define

$$\begin{aligned} \mathbf{G} &:= d^{-a} \mathbf{I}_d \otimes \{\mathbb{E}(\check{\mathbf{G}}) \mathbb{E}(\check{\mathbf{G}})^\top\}, \quad \check{\mathbf{G}} := (\check{\mathbf{G}}_{1,0}, \dots, \check{\mathbf{G}}_{1,l_1}, \dots, \check{\mathbf{G}}_{p,0}, \dots, \check{\mathbf{G}}_{p,l_p}), \\ \check{\mathbf{G}}_{j,k} &:= \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \boldsymbol{\beta}^{*\top} \mathbf{X}_t^\top \boldsymbol{\Pi}_t^{*\top}, \quad \boldsymbol{\Pi}_t^* := (\mathbf{I}_d - \mathbf{W}_t^*)^{-1}. \end{aligned}$$

We assume $\mathbf{G}$ has full rank such that there exists a constant $u > 0$ with $\lambda_{\min}(\mathbf{G}) \geq u > 0$ and $\lambda_{\max}(\mathbf{G}) < \infty$ uniformly as $d \rightarrow \infty$.

(R5) For the same constant a as in Assumption (R4), we have for each d ,

$$\max_{i \in [d]} \sum_{j=1}^d \|\mathbb{E}(\mathbf{B}_{t,i} \mathbf{X}_{t,j}^\top)\|_{\max}, \max_{j \in [d]} \sum_{i=1}^d \|\mathbb{E}(\mathbf{B}_{t,i} \mathbf{X}_{t,j}^\top)\|_{\max} = O(d^a).$$

At the same time, $\mathbb{E}(\mathbf{X}_t \otimes \mathbf{B}_t \boldsymbol{\gamma})$ has all singular values of order d^{1+a} .

(R6) With $\boldsymbol{\Pi}_t^*$ in Assumption (R4), define

$$\ddot{\mathbf{G}} := (\ddot{\mathbf{G}}_{1,0}, \dots, \ddot{\mathbf{G}}_{1,l_1}, \dots, \ddot{\mathbf{G}}_{p,0}, \dots, \ddot{\mathbf{G}}_{p,l_p}), \quad \ddot{\mathbf{G}}_{j,k} := \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top (\mathbf{I}_d \otimes \boldsymbol{\beta}^{*\top} \mathbf{X}_t^\top \boldsymbol{\Pi}_t^{*\top}).$$

We assume $\mathbb{E}(\ddot{\mathbf{G}}) \mathbb{E}(\ddot{\mathbf{G}})^\top$ has full rank such that there exists a constant $u > 0$ with $\lambda_v(\mathbb{E}(\ddot{\mathbf{G}}) \mathbb{E}(\ddot{\mathbf{G}})^\top) \geq u > 0$ and $\lambda_{\max}(\mathbb{E}(\ddot{\mathbf{G}}) \mathbb{E}(\ddot{\mathbf{G}})^\top)$ uniformly as $d \rightarrow \infty$.

(R7) Define the predictive dependence measures

$$P_0^b(B_{t,jk}) := \mathbb{E}(B_{t,jk} \mid \mathcal{G}_0) - \mathbb{E}(B_{t,jk} \mid \mathcal{G}_{-1}), \quad P_0^\epsilon(\epsilon_{t,j}) := \mathbb{E}(\epsilon_{t,j} \mid \mathcal{H}_0) - \mathbb{E}(\epsilon_{t,j} \mid \mathcal{H}_{-1}),$$

with $\mathcal{G}_t, \mathcal{H}_t$ defined after (3.1). Assume

$$\sum_{t \geq 0} \max_{j \in [d]} \max_{k \in [v]} \|P_0^b(B_{t,jk})\|_2 < \infty, \quad \sum_{t \geq 0} \max_{j \in [d]} \|P_0^\epsilon(\epsilon_{t,j})\|_2 < \infty.$$

(R8) For $b \in (0, 1)$, the eigenvalues of $\text{Var}(d^{-b/2} \mathbf{B}_{t,\cdot k})$ and $\text{Var}(\boldsymbol{\epsilon}_t)$ are uniformly bounded away from 0 and infinity, and respectively dominate the singular values of $d^{-b} \mathbb{E}\{[\mathbf{B}_{t,\cdot k} - \mathbb{E}(\mathbf{B}_{t,\cdot k})][\mathbf{B}_{t+\tau,\cdot k} - \mathbb{E}(\mathbf{B}_{t+\tau,\cdot k})]^\top\}$ and $\mathbb{E}(\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_{t+\tau}^\top)$ for any $\tau \neq 0$. The sum of the i -th largest singular values over all lags $\tau \in \mathbb{Z}$ for each $i \in [d]$ is assumed finite for both autocovariance matrices of $\{d^{-b/2} \mathbf{B}_t\}$ and $\{\boldsymbol{\epsilon}_t\}$.

(R9) Define $c_T = gT^{-1/2} \log^{1/2}(T \vee d)$ for some constant $g > 0$. The tuning parameter for the adaptive LASSO problem (2.7) is $\lambda = Cc_T$ for some constant $C > 0$.

(R10) (Rate assumptions) We assume that as $L, d, T \rightarrow \infty$,

$$c_T L^{3/2} d^{1-a}, \quad L d^{-1}, \quad L^2 d^3 T^{2-w}, \quad d^{b+2a+1/w} T^{-1} = o(1), \\ d^{-1/w} \log(T \vee d), \quad d^{b-a-1/w} \log^{-1}(T \vee d) = O(1).$$

3.1.1 Identification of the model

We show that the coefficients $\boldsymbol{\phi}^*$ and $\boldsymbol{\beta}^*$ in model (2.11) are identified under Assumption (I1). Assume that the two sets of parameters $(\check{\boldsymbol{\phi}}, \check{\boldsymbol{\beta}})$ and $(\acute{\boldsymbol{\phi}}, \acute{\boldsymbol{\beta}})$ both satisfy model (2.11), then we have

$$\mathbf{B}^\top \mathbf{V} \check{\boldsymbol{\phi}} + \mathbf{B}^\top \mathbf{X}_{\check{\boldsymbol{\beta}}} \text{vec}(\mathbf{I}_d) = \mathbf{B}^\top \mathbf{V} \acute{\boldsymbol{\phi}} + \mathbf{B}^\top \mathbf{X}_{\acute{\boldsymbol{\beta}}} \text{vec}(\mathbf{I}_d).$$

By noticing that $\mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}} \text{vec}(\mathbf{I}_d) = \mathbf{B}^\top \widetilde{\mathbf{X}} \boldsymbol{\beta}$, we may rearrange the above and arrive at

$$\mathbf{0} = \mathbf{B}^\top \mathbf{V} (\check{\boldsymbol{\phi}} - \acute{\boldsymbol{\phi}}) + \mathbf{B}^\top \widetilde{\mathbf{X}} (\check{\boldsymbol{\beta}} - \acute{\boldsymbol{\beta}}) = \begin{pmatrix} \mathbf{B}^\top \mathbf{V} & \mathbf{B}^\top \widetilde{\mathbf{X}} \end{pmatrix} \begin{pmatrix} \check{\boldsymbol{\phi}} - \acute{\boldsymbol{\phi}} \\ \check{\boldsymbol{\beta}} - \acute{\boldsymbol{\beta}} \end{pmatrix}.$$

Taking expectation and left-multiplying by $(\mathbf{Q}^\top \mathbf{Q})^{-1} \mathbf{Q}^\top$ on both sides, we obtain $\check{\boldsymbol{\phi}} = \acute{\boldsymbol{\phi}}$ and $\check{\boldsymbol{\beta}} = \acute{\boldsymbol{\beta}}$. This implies $\boldsymbol{\phi}^*$ and $\boldsymbol{\beta}^*$ are identified in model (2.1), and hence $\boldsymbol{\mu}^*$ is uniquely identified.

3.1.2 Explanations for the assumptions

The time series components in the model is depicted in Assumption (M1) such that weak serial dependence is allowed. Such a definition of functional or physical dependence of time series is used by various papers such as Shao (2010). An exponential tail behaviour is also assumed to apply the Nagaev-type inequality for functional dependent data by Liu et al. (2013).

Assumptions (M2) and (M2') describe the structure of $\mathbf{W}_t^*$ under two different settings for $\{z_{j,k,t}\}$. The row sum condition for $\mathbf{W}_t^*$ ensures the model (2.1) to be uniformly stationary. (M2) treats $\{z_{j,k,t}\}$ as a constant series while (M2') allows $\{z_{j,k,t}\}$ to be stochastic and specifies its functional form. Note that (M2') ensures stationarity such that $\|\mathbf{W}_t^*\|_\infty < \eta < 1$ and $|\rho_t^*| < 1$ with probability 1, which is sufficient but not necessary. In (M2'), the assumption of zero mean and support $[-1, 1]$ simplifies our discussion as we only require finite mean and bounded support $[-z_{j,k}^*, z_{j,k}^*]$ after demeaning. Otherwise, without loss of generality we may read

$$\phi_{j,k}^* z_{j,k,t} = \phi_{j,k}^* \mathbb{E}(z_{j,k,t}) + \phi_{j,k}^* [z_{j,k,t} - \mathbb{E}(z_{j,k,t})] = \phi_{j,k}^* \mathbb{E}(z_{j,k,t}) + (\phi_{j,k}^* z_{j,k}^*) \left\{ [z_{j,k,t} - \mathbb{E}(z_{j,k,t})] / z_{j,k}^* \right\}.$$

Note that we set $z_{j,0,t} = 1$ for $j \in [p], t \in [T]$. It is justified by the fact that we may allow some $\{z_{j,k,t}\}$ for $j \in \mathcal{P} \subseteq [p], k \in \mathcal{L}_j \subseteq [l_j]$ satisfying (M2) with all other $\{z_{j,k,t}\}$ satisfying (M2'), since the theoretical results are the same for either assumption.

(R1) describes how sparse each spatial weight matrix candidate is. It is worthy to point out that although each $\mathbf{W}_j$ is not necessarily linearly independent with each other by (R1), we actually implicitly impose such linear independence condition from (I1) through the combination of $\{z_{j,k,t}\}$ and $\mathbf{W}_j$ in $\mathbf{B}^\top \mathbf{V}$. (R2) is included for convenience in proofs, as a technical addition to (M2).

Assumptions (R3) to (R6) all draw on the relation between $\mathbf{B}_t$ and $\mathbf{X}_t$. Their dependence structure is non-trivial due to the extra complication from spatial weight matrix and the time-varying components here. Naturally, $\mathbf{B}_t$ needs to be correlated with $\mathbf{X}_t$ to certain extent, captured by an unknown constant a which facilitates the presentation of theoretical results. For instance, as an immediate consequence of (R5), we may derive $\|\mathbb{E}(\mathbf{X}_t \otimes \mathbf{B}_t \boldsymbol{\gamma})\|_1 = O(d^{1+a})$ which is needed to construct the rates of $\|\tilde{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_1$ in Theorem 1.

The predictive dependence measure is defined in (R7) to apply the central limit theorem for data with functional dependence, which is satisfied by, e.g. causal AR processes. Besides further restricting on the serial correlation in the noise, Assumption (R8) introduces another constant b which characterizes how elements in $\mathbf{B}_t$ are contemporarily and temporally correlated with each other. From the comparison of rates derived in the theoretical proofs, we conclude that b is actually upper bounded by $1/w$ in (M1), which

is intuitive since a large w implies weaker temporal correlation and hence a small b .

Lastly, (R9) sets the rate for λ which is related to the rate condition (R10) which characterizes the relation between T , d and potentially the growing L . As an example, (R10) is satisfied if $w = 6$, $a = 1/2$, $L = O(d^{1/3})$ and $T \asymp d^2$.

3.2 Main results

In this section, we formally present the main results for our model estimators.

Theorem 1 *Let all assumptions in Section 3.1 hold ((M2) or (M2')). Given any ϕ as an estimator of ϕ^* , $\beta(\phi)$ according to (2.13) satisfies*

$$\|\beta(\phi) - \beta^*\|_1 = O_p\left(\|\phi - \phi^*\|_1 + c_T d^{-\frac{1}{2} + \frac{1}{2w}}\right),$$

where c_T is defined in Assumption (R9). In particular, the least squares estimator $\tilde{\phi}$ in (2.16) and $\tilde{\beta} := \beta(\tilde{\phi})$ under $L = O(1)$ satisfy

$$\|\tilde{\phi} - \phi^*\|_1 = O_P(c_T d^{-\frac{1}{2} + \frac{1}{2w}}) = \|\tilde{\beta} - \beta^*\|_1.$$

From Theorem 1, the error of our least squares estimator $\beta(\phi)$ might be inflated by the plugged-in estimator for ϕ^* . With $\tilde{\phi}$, we arrive at the rate $c_T d^{-\frac{1}{2} + \frac{1}{2w}}$ under the scenario of dense coefficient vector ϕ^* . The rate depends on w , which is better if we have weaker temporal dependence in the data. Theorem 1 is a foundational step for the results hereafter. For sparse ϕ^* , we next present the sign-consistency of our adaptive LASSO estimator below.

Theorem 2 *(Oracle property for $\hat{\phi}$) Let all assumptions in Section 3.1 hold ((M2) or (M2')), except that (R4) and (R6) are satisfied with $\check{\mathbf{G}}$ and $\ddot{\mathbf{G}}$ respectively replaced by*

$$\check{\mathbf{G}} = (\check{\mathbf{G}}_{j,k})_{i \in H}, \quad \ddot{\mathbf{G}} = (\ddot{\mathbf{G}}_{j,k})_{i \in H}, \quad \text{where } H := \{i : (\phi^*)_i \neq 0\},$$

with the understanding that each index pair (j, k) has a one-to-one correspondence with each $i \in [|\phi^*|] \equiv [p + \sum_{j=1}^p \ell_j]$. Then as $T, d \rightarrow \infty$, with probability 1, we have

$$\text{sign}(\hat{\phi}_H) = \text{sign}(\phi_H^*), \quad \hat{\phi}_{H^c} = \mathbf{0}.$$

If we also assume $\mathbf{R}_H \mathbf{S}_\gamma \mathbf{S}_\gamma^\top \mathbf{R}_H^\top$ has the smallest eigenvalue of constant order where $\mathbf{R}_H, \mathbf{S}_\gamma$ are defined

below, we have $\widehat{\phi}_H$ asymptotically normal with rate $T^{-1/2}d^{-(1-b)/2}$ such that

$$T^{1/2}(\mathbf{R}_H \mathbf{S}_\gamma \mathbf{R}_\beta \Sigma_\beta \mathbf{R}_\beta^\top \mathbf{S}_\gamma^\top \mathbf{R}_H^\top)^{-1/2}(\widehat{\phi}_H - \phi_H^*) \xrightarrow{D} \mathcal{N}(\mathbf{0}, \mathbf{I}_{|H|}),$$

where $\mathbf{R}_\beta$, Σ_β , $\mathbf{S}_\gamma$ are defined as

$$\begin{aligned} \mathbf{R}_\beta &= [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} \mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t), \quad \Sigma_\beta = \sum_{\tau} \mathbb{E}\{(\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top \boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_{t+\tau}^\top (\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_{t+\tau}))\}, \\ \mathbf{S}_\gamma &= \left(\mathbb{E}[\mathbf{X}_{t,1\cdot}(\mathbf{B}_{t,1\cdot} - \mathbb{E}(\mathbf{B}_{t,1\cdot}))^\top] \gamma, \dots, \mathbb{E}[\mathbf{X}_{t,1\cdot}(\mathbf{B}_{t,d\cdot} - \mathbb{E}(\mathbf{B}_{t,d\cdot}))^\top] \gamma, \right. \\ &\quad \left. \dots, \mathbb{E}[\mathbf{X}_{t,d\cdot}(\mathbf{B}_{t,1\cdot} - \mathbb{E}(\mathbf{B}_{t,1\cdot}))^\top] \gamma, \dots, \mathbb{E}[\mathbf{X}_{t,d\cdot}(\mathbf{B}_{t,d\cdot} - \mathbb{E}(\mathbf{B}_{t,d\cdot}))^\top] \gamma \right)^\top. \end{aligned}$$

The term $\mathbf{R}_H$ is constructed by denoting $(\cdot)_H$ the corresponding matrix with columns restricted on the set H , and define for any given $n \times d$ matrix $\mathbf{H} = (\mathbf{h}_1, \dots, \mathbf{h}_n)^\top$ that

$$\mathbf{V}_{\mathbf{H},K} = \begin{pmatrix} \mathbf{I}_K \otimes \mathbf{h}_1 \\ \vdots \\ \mathbf{I}_K \otimes \mathbf{h}_n \end{pmatrix},$$

then $\mathbf{R}_H = [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} (\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top$ with

$$\begin{aligned} \mathbf{U}_{\mathbf{x},j,k} &= \frac{1}{T} \sum_{t=1}^T z_{j,k,t} \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \mathbf{x}_t^\top, \quad \mathbf{U}_{\boldsymbol{\mu},j,k} = \frac{1}{T} \sum_{t=1}^T z_{j,k,t} \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\mu}^{*\top}, \\ \mathbf{H}_{10} &= \left\{ \left(\mathbf{I}_d \otimes (\gamma^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\mathbf{x},1,0}(\boldsymbol{\beta}^* \otimes \mathbf{I}_d) \boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_1^\top) + \left(\mathbf{I}_d \otimes (\gamma^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},1,0} \boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_1^\top), \right. \\ &\quad \left. \dots, \left(\mathbf{I}_d \otimes (\gamma^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\mathbf{x},p,l_p}(\boldsymbol{\beta}^* \otimes \mathbf{I}_d) \boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_p^\top) + \left(\mathbf{I}_d \otimes (\gamma^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},p,l_p} \boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_p^\top) \right\}, \\ \mathbf{H}_{20} &= \mathbb{E}(\mathbf{X}_t \otimes \mathbf{B}_t \gamma) [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} \mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \\ &\quad \cdot \left\{ \mathbf{V}_{\mathbf{W}_1^\top, v}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\mathbf{x},1,0}) \mathbf{V}_{\boldsymbol{\Pi}_t^*, r}] \boldsymbol{\beta}^* + \mathbf{V}_{\mathbf{W}_1^\top, v}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\mu},1,0}) \text{vec}(\boldsymbol{\Pi}_t^{*\top})], \right. \\ &\quad \left. \dots, \mathbf{V}_{\mathbf{W}_p^\top, v}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\mathbf{x},p,l_p}) \mathbf{V}_{\boldsymbol{\Pi}_t^*, r}] \boldsymbol{\beta}^* + \mathbf{V}_{\mathbf{W}_p^\top, v}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\mu},p,l_p}) \text{vec}(\boldsymbol{\Pi}_t^{*\top})] \right\}. \end{aligned}$$

Regarding the new assumptions stated in Theorem 2, the term $\check{\mathbf{G}}$ (resp. $\ddot{\mathbf{G}}$) in (R4) (resp. (R6)) is simply replaced by its H -restricted version. Moreover, the largest eigenvalue of $\mathbf{R}_H \mathbf{S}_\gamma \mathbf{S}_\gamma^\top \mathbf{R}_H^\top$ has constant order, and the same order is required to obtain the exact rate for the asymptotic normality result.

Despite the inevitable complications in Theorem 2, two key results are obtained: $\widehat{\phi}$ consistently estimate the zeros in ϕ^* as exactly zeros, and are asymptotically normal on the non-zero entries in ϕ^* . The convergence rate is $T^{-1/2}d^{-(1-b)/2}$, which is worse off if more variables in $\mathbf{B}_t$ are correlated. In practice, we may perform inferences on $\widehat{\phi}_H$, with the covariance matrix estimated by plugging in the moment

estimator for each component.

If the $\{z_{j,k,t}\}$'s are non-stochastic, Theorem 2 implies that we can make inference on ρ_t^* using $\hat{\rho}_t := (\mathbf{z}_t)_H^\top \hat{\boldsymbol{\phi}}_H$, since

$$T^{1/2}((\mathbf{z}_t)_H^\top \mathbf{R}_H \mathbf{S}_\gamma \mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top \mathbf{S}_\gamma^\top \mathbf{R}_H^\top (\mathbf{z}_t)_H)^{-1/2}(\hat{\rho}_t - \rho_t^*) \xrightarrow{D} \mathcal{N}(0, 1).$$

Lastly, we present below the consistency of the spatial weight matrix estimator and the spatial fixed effect estimator. Note also the theorem below implies that the spectral norm has error $\|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\| = O_P(T^{-1/2}d^{-(1-b)/2})$.

Theorem 3 *Let assumptions in Theorems 1 and 2 hold. For $\widehat{\mathbf{W}}_t := \sum_{j=1}^p \left(\hat{\phi}_{j,0} + \sum_{k=1}^{l_j} \hat{\phi}_{j,k} z_{j,k,t} \right) \mathbf{W}_j$ and the spatial fixed effect estimator $\hat{\boldsymbol{\mu}}$ defined in (2.8), we have*

$$\|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\|_\infty = O_P(T^{-1/2}d^{-(1-b)/2}) = \|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\|_1, \quad \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\|_{\max} = O_P(c_T).$$

4 Practical Implementation

With Section 2.2, we may estimate $(\boldsymbol{\mu}^*, \boldsymbol{\phi}^*, \boldsymbol{\beta}^*)$ by obtaining a penalized estimator for $\boldsymbol{\phi}^*$ first, followed by the least squares estimator for $\boldsymbol{\beta}^*$ and $\boldsymbol{\mu}^*$. The detailed algorithm is presented below.

Algorithm for $(\boldsymbol{\mu}^*, \boldsymbol{\phi}^*, \boldsymbol{\beta}^*)$ Estimation

1. Compute the least squares estimator stated in (2.16) and denote it as $\tilde{\boldsymbol{\phi}}$.
2. Construct $\mathbf{u}$ using $\tilde{\boldsymbol{\phi}}$. Using the Least Angle Regressions (LARS) (Efron et al., 2004), solve the adaptive LASSO problem stated in (2.17) subject to its corresponding constraints, and denote the solution by $\hat{\boldsymbol{\phi}}$.
3. Using (2.13), obtain the least squares estimator for $\boldsymbol{\beta}^*$ as $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}(\hat{\boldsymbol{\phi}})$.
4. According to (2.3), construct $\hat{\boldsymbol{\Phi}}$ using $\hat{\boldsymbol{\phi}}$ and obtain the least squares estimator for $\boldsymbol{\mu}^*$ as

$$\hat{\boldsymbol{\mu}} = \frac{1}{T} \sum_{t=1}^T \left\{ (\mathbf{I}_d - \boldsymbol{\Lambda}_t \hat{\boldsymbol{\Phi}}) \mathbf{y}_t - \mathbf{X}_t \hat{\boldsymbol{\beta}} \right\}.$$

The tuning parameter λ involved in step 2 is obtained by minimising the following BIC which is inspired by Wang et al. (2009):

$$\text{BIC}(\lambda) = \log \left(T^{-1} \left\| \mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{V} \hat{\boldsymbol{\phi}} - \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}(\hat{\boldsymbol{\phi}})} \text{vec}(\mathbf{I}_d) \right\|^2 \right) + |\widehat{H}| \frac{\log(T)}{T} \log(\log(L)), \quad (4.1)$$

where $\widehat{\boldsymbol{\phi}}$ is the adaptive LASSO solution with parameter λ and $\widehat{H}$ is the set of indices for which $\widehat{\phi}_j$ is non-zero for $j \in \widehat{H}$. Note that although $\mathbf{B}$ contains the unknown constant a , the optimal λ is independent of a . A procedure to assess the goodness of fit of a given set of dynamic variables $\{z_{j,k,t}\}$ can also be facilitated by (4.1). In details, we hope to compare (2.1) with the null model $\mathcal{H}_0 : \mathbf{y}_t = \boldsymbol{\mu}^* + \sum_{j=1}^p \phi_{j,0}^* \mathbf{W}_j \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t$, where their BIC are respectively computed according to (4.1) and the below inspired by Wang and Leng (2007):

$$\text{BIC}(\mathcal{H}_0) = \log \left(T^{-1} \left\| \mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{V} \widehat{\boldsymbol{\phi}} - \mathbf{B}^\top \mathbf{X}_{\boldsymbol{\beta}(\widehat{\boldsymbol{\phi}})} \text{vec}(\mathbf{I}_d) \right\|^2 \right) + |\widehat{H}| \frac{\log(T)}{T}. \quad (4.2)$$

All the variables above are constructed using the null model and an adaptive LASSO estimation is performed similarly. Lastly, to utilize the asymptotic normality result in Theorem 2, we propose a feasible covariance estimator in the succeeding section.

4.1 Feasible inference

To leverage the asymptotic normality for $\widehat{\boldsymbol{\phi}}_H$ in Theorem 2, we need appropriate estimators for $\mathbf{R}_H$, $\mathbf{R}_\beta$, $\boldsymbol{\Sigma}_\beta$ and $\mathbf{S}_\gamma$. According to Theorem 1, 2 and 3, it is reasonable to use the plug-in estimators, denoted by $\widehat{\mathbf{R}}_H$, $\widehat{\mathbf{R}}_\beta$ and $\widehat{\mathbf{S}}_\gamma$, based on the consistent estimators $(\boldsymbol{\beta}(\widehat{\boldsymbol{\phi}}), \widehat{\boldsymbol{\mu}})$. In particular, the set H is directly estimated from $\widehat{H} = \{i : (\widehat{\boldsymbol{\phi}})_i \neq 0\}$. All expected values may be replaced by their sample estimators.

On estimating $\boldsymbol{\Sigma}_\beta$, we use the following estimated error which is consistent due to the consistency of all model parameters,

$$\widehat{\boldsymbol{\epsilon}}_t := \mathbf{y}_t - \widehat{\boldsymbol{\mu}} - \widehat{\mathbf{W}}_t \mathbf{y}_t - \mathbf{X}_t \boldsymbol{\beta}(\widehat{\boldsymbol{\phi}}).$$

Moreover, since $\boldsymbol{\Sigma}_\beta$ involves an infinite sum, we sum up to a cut-off τ after which the sum changes little, and denote the constructed estimator $\widehat{\boldsymbol{\Sigma}}_\beta$. Lastly, the covariance matrix for the asymptotic normality in Theorem 2 can be computed by $T^{-1} \widehat{\mathbf{R}}_{\widehat{H}} \widehat{\mathbf{S}}_\gamma \widehat{\mathbf{R}}_\beta \widehat{\boldsymbol{\Sigma}}_\beta \widehat{\mathbf{R}}_\beta^\top \widehat{\mathbf{S}}_\gamma^\top \widehat{\mathbf{R}}_{\widehat{H}}^\top$.

5 Change Point Detection and Estimation in Dynamic Spatial Autoregressive Models

5.1 Threshold spatial autoregressive models

The early work by Tong (1978) proposes a regime switching mechanism by introducing threshold autoregressive models which have been well studied in the past few decades. More recently, such threshold structure is used by researchers in spatial econometrics. For example, see Deng (2018) and Li (2022). One benefit of the framework proposed in this paper is that threshold variables can be directly adapted in (2.1). As a simple example, we consider a spatial autoregressive model with regime switching on the

spatial weight matrix such that

$$\mathbf{y}_t = \begin{cases} \boldsymbol{\mu}^* + \phi_1^* \mathbf{W}_1 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & q_t \leq \gamma^*, \\ \boldsymbol{\mu}^* + \phi_2^* \mathbf{W}_2 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & q_t > \gamma^*, \end{cases} \quad (5.1)$$

where q_t is some observed threshold variable with an unknown threshold value γ^* , and other parameters are as specified in (2.1) by rewriting (5.1) compactly as

$$\mathbf{y}_t = \boldsymbol{\mu}^* + z_{1,1,t} \phi_1^* \mathbf{W}_1 \mathbf{y}_t + z_{2,1,t} \phi_2^* \mathbf{W}_2 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad (5.2)$$

where $z_{1,1,t} := \mathbb{1}_{\{q_t \leq \gamma^*\}}$ and $z_{2,1,t} := \mathbb{1}_{\{q_t > \gamma^*\}}$.

Consider the estimation of the threshold value γ^* and suppose there is a domain of possible threshold values $\Gamma = [\gamma_{\min}, \gamma_{\max}]$. A standard approach in threshold models is to search the minimum regression error over the intersection $\Gamma \cap \{q_1, \dots, q_T\}$, see Deng (2018) for example. Our framework provides another approach with $L \equiv |\Gamma \cap \{q_1, \dots, q_T\}|$. Denote the elements in the intersection by $\gamma_1 \leq \gamma_2 \leq \dots \leq \gamma_L$, and let γ^* be identified to be one of them. Let $z_{1,l,t} := \mathbb{1}_{\{q_t \leq \gamma_l\}}$ and $z_{2,l,t} := \mathbb{1}_{\{q_t > \gamma_l\}}$ for all $l \in [L]$. Then we can consider a spatial autoregressive model such that

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \sum_{l=1}^L z_{1,l,t} \phi_{1,l}^* \mathbf{W}_1 \mathbf{y}_t + \sum_{l=1}^L z_{2,l,t} \phi_{2,l}^* \mathbf{W}_2 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t. \quad (5.3)$$

Since $\phi_{1,l}^*$ and $\phi_{2,l}^*$ would only be non-zero² for $\gamma_l = \gamma^*$, the threshold value can be selected consistently in one step by the oracle property of our adaptive LASSO estimator in Theorem 2. This is presented in Corollary 1. Given the sparse solution of (5.3), we can use only the non-zero pairs $(\hat{\phi}_{1,l}, \hat{\phi}_{2,l})$ to perform model estimations again. Note that such an approach remains applicable for L growing with T . In practice, the order of L might not fulfil Assumption (R10), but we may circumvent this issue by a sequential procedure. See Remark 1 in Section 5.2 for details.

Our framework also allows us to consider a spatial autoregressive model with regime switching on the spatial correlation coefficients similar to Li (2022):

$$\mathbf{y}_t = \begin{cases} \boldsymbol{\mu}^* + \phi_1^* \mathbf{W}_1 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & q_t \leq \gamma^*, \\ \boldsymbol{\mu}^* + \phi_2^* \mathbf{W}_2 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & q_t > \gamma^*, \end{cases} \quad (5.4)$$

where the variables are defined in the same way as in (5.1). To estimate the parameters in (5.4), Li (2022)

² In practice, we often have no prior information on which spatial weight matrix corresponding to the regime $q_t \leq \gamma^*$. This can be resolved in (5.3) by writing $(\sum_{l=1}^L z_{1,l,t} \phi_{1,l}^*)$ and $(\sum_{l=1}^L z_{2,l,t} \phi_{2,l}^*)$ as $(\phi_{1,0}^* + \sum_{l=1}^L z_{1,l,t} \phi_{1,l}^*)$ and $(\phi_{2,0}^* + \sum_{l=1}^L z_{2,l,t} \phi_{2,l}^*)$, respectively.

considers using quasi maximum likelihood (QML) estimators and traverse over a finite parameter space for the threshold value γ^* . In contrast, a one-step estimation is again feasible by our framework. To this end, we may read (5.4) in the form,

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \left(\phi_{1,0}^* + \sum_{l=1}^L z_{1,l,t} \phi_{1,l}^* \right) \mathbf{W}_1 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad (5.5)$$

where $z_{1,l,t}$ for $l \in [L]$ is as previously defined. With our adaptive LASSO estimators, only $\hat{\phi}_{1,0}$ and one $\hat{\phi}_{1,l}$ such that $\gamma_l = \gamma^*$ are expected to be non-zero. The consistency of such γ^* estimator is included in Corollary 1. The results are then interpreted as $\hat{\phi}_1 \equiv \hat{\phi}_{1,0} + \hat{\phi}_{1,l}$ and $\hat{\phi}_2 \equiv \hat{\phi}_{1,0}$.

As we often have limited prior knowledge on the parameter space in practice, we recommend using our framework in an exploratory way. This should help researchers to discover more reasonable threshold structures in the data.

Corollary 1 (*Consistency on threshold value estimation*) *Let all assumptions in Theorem 2 hold.*

(a) *Consider model (5.1). The threshold value γ^* can be consistently estimated by the set of estimators $\{\gamma_l : \hat{\phi}_{1,l} \neq 0, \hat{\phi}_{2,l} \neq 0\}$, where $\{\hat{\phi}_{1,l}, \hat{\phi}_{2,l}\}_{l \in [L]}$ denotes the adaptive LASSO solution for $\{\phi_{1,l}^*, \phi_{2,l}^*\}_{l \in [L]}$ in (5.3).*

(b) *Consider model (5.4). The threshold value γ^* can be consistently estimated by the set of estimators $\{\gamma_l : \hat{\phi}_{1,l} \neq 0\}$, where $\{\hat{\phi}_{1,l}\}_{l \in [L]}$ denotes the adaptive LASSO solution for $\{\phi_{1,l}^*\}_{l \in [L]}$ in (5.5).*

In fact, our framework (2.1) can be applied to spatial autoregressive models with more complicated threshold structures in the spatial weight matrix $\mathbf{W}_t^*$, i.e., regimes from multiple threshold variables with multiple threshold values. For illustrations, consider the following spatial autoregressive model with $(k+1)$ regimes switching on the spatial correlation coefficients:

$$\mathbf{y}_t = \begin{cases} \boldsymbol{\mu}^* + \phi_1^* \mathbf{W}_1 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & q_t \leq \gamma_1^*, \\ \boldsymbol{\mu}^* + \phi_2^* \mathbf{W}_1 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & \gamma_1^* < q_t \leq \gamma_2^*, \\ \dots & \\ \boldsymbol{\mu}^* + \phi_k^* \mathbf{W}_1 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & q_t > \gamma_k^*. \end{cases} \quad (5.6)$$

The consistency of variable selection by adaptive LASSO facilitates consistent estimation of k if it is unknown a priori. Model (5.6) can again be written as in (5.5), and the theoretical guarantee is stated in Corollary 2.

Corollary 2 (*Consistency on the number of threshold regimes and multiple threshold values estimation*) *Let all assumptions in Theorem 2 hold. Consider model (5.6). Let $\{\hat{\phi}_{1,l}\}_{l \in [L]}$ denote the adaptive LASSO*

solution for $\{\phi_{1,l}^*\}_{l \in [L]}$ in (5.5), then $\widehat{k} := |\{\gamma_l : \widehat{\phi}_{1,l} \neq 0\}|$ estimates k consistently. Moreover, the i -th smallest element in $\{\gamma_l : \widehat{\phi}_{1,l} \neq 0\}$ estimates γ_i^* consistently for $i \in [\widehat{k}]$.

5.2 Spatial autoregressive models with structural change points

Structural changes in the relationship of variables in econometric models have been studied extensively in the literature. For example, see Sengupta (2017) and Barigozzi and Trapani (2020). As a second example demonstrating the applicability of our framework, we consider the spatial autoregressive model with a structural change, which can be written as

$$\mathbf{y}_t = \begin{cases} \boldsymbol{\mu}^* + \phi_1^* \mathbf{W}_1 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & t \leq t^*, \\ \boldsymbol{\mu}^* + \phi_2^* \mathbf{W}_2 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & t > t^*, \end{cases} \quad (5.7)$$

where t^* is some unknown change location. Similar to the threshold model example, this can also be expressed in the form of (5.2), now with $z_{1,1,t} := \mathbb{1}_{\{t \leq t^*\}}$ and $z_{2,1,t} := \mathbb{1}_{\{t > t^*\}}$. It is important to note that, despite the models taking the same form, the dynamic variables $z_{1,1,t}$ and $z_{2,1,t}$ are random in the threshold model but non-random in the change point model here. Recall that our main results hold for both types of $\{z_{j,k,t}\}$; see Section 3.1.2 for a more detailed discussion.

To estimate the change location t^* , Li (2018) calculates the maximum (quasi) likelihood for each possible location and sets the maximizer as the estimator. When the set of possible change locations is large, this approach requires a significant number of model fittings. Using our framework, it is again possible to estimate t^* in one go. We rewrite model (5.7) as

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \sum_{l=1}^{|\mathcal{T}|} z_{1,l,t} \phi_{1,l}^* \mathbf{W}_1 \mathbf{y}_t + \sum_{l=1}^{|\mathcal{T}|} z_{2,l,t} \phi_{2,l}^* \mathbf{W}_2 \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad (5.8)$$

where $\mathcal{T} = (\mathcal{T}_l)$ is the set of all candidate change point locations, with $z_{1,l,t} := \mathbb{1}_{\{t \leq \mathcal{T}_l\}}$ and $z_{2,l,t} := \mathbb{1}_{\{t > \mathcal{T}_l\}}$. We then follow the same argument used in the threshold model below (5.3), with the consistency of our change location estimate guaranteed.

Corollary 3 (*Consistency on change location estimation*) *Let all assumptions in Theorem 2 hold, with L replaced by $|\mathcal{T}|$. Consider model (5.7) and assume that $t^* \in \mathcal{T}$. Then, the change location t^* can be consistently estimated by the set of estimators $\{l \in [|\mathcal{T}|] : \widehat{\phi}_{1,l} \neq 0, \widehat{\phi}_{2,l} \neq 0\}$, where $\{\widehat{\phi}_{1,l}, \widehat{\phi}_{2,l}\}_{l \in [|\mathcal{T}|]}$ denotes the adaptive LASSO solution for $\{\phi_{1,l}^*, \phi_{2,l}^*\}_{l \in [|\mathcal{T}|]}$ in (5.8).*

Similar to (5.6), we may allow for k change points in $\mathbf{W}_t^*$ consisting of p spatial weight candidates with

the number of change points unknown. In other words, consider

$$\mathbf{y}_t = \begin{cases} \boldsymbol{\mu}^* + \sum_{j=1}^p \phi_{j,1}^* \mathbf{W}_j \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & t \leq t_1^*, \\ \boldsymbol{\mu}^* + \sum_{j=1}^p \phi_{j,2}^* \mathbf{W}_j \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & t_1^* < t \leq t_2^*, \\ \dots & \\ \boldsymbol{\mu}^* + \sum_{j=1}^p \phi_{j,k}^* \mathbf{W}_j \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, & t > t_k^*. \end{cases} \quad (5.9)$$

For the estimation and consistency guarantee, see Corollary 4.

Corollary 4 (*Consistency on the number of changes and multiple change locations estimation*) Given a set of all candidate change point locations $\mathcal{T}$, let all assumptions in Theorem 2 hold, with L replaced by $|\mathcal{T}|$. Consider model (5.9) and assume that $t_i^* \in \mathcal{T}$ for $i \in [k]$. Let $\{\hat{\phi}_{j,l}\}_{j \in [p], l \in [|\mathcal{T}|]}$ denote the adaptive LASSO solution for $\{\phi_{j,l}^*\}_{j \in [p], l \in [|\mathcal{T}|]}$ in

$$\mathbf{y}_t = \boldsymbol{\mu}^* + \sum_{j=1}^p \left(\phi_{j,0}^* + \sum_{l=1}^{|\mathcal{T}|} z_{j,l,t} \phi_{j,l}^* \right) \mathbf{W}_j \mathbf{y}_t + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t,$$

where $z_{j,l,t} := \mathbb{1}_{\{t \leq \tau_l\}}$ for $j \in [p], l \in [|\mathcal{T}|]$. Then $\hat{k} := |\hat{\mathcal{T}}| := |\{l \neq 0 : \hat{\phi}_{j,l} \neq 0 \text{ for some } j \in [p]\}|$ estimates k consistently. Moreover, the i -th smallest element in $\hat{\mathcal{T}}$ estimates t_i^* consistently for $i \in [\hat{k}]$.

Remark 1 Throughout Section 5.2, the size of $\mathcal{T}$ is restricted by Assumption (R10). Specifying appropriate $\mathcal{T}$ requires prior information which might be infeasible in practice. Suppose $\mathcal{T}$ is given but $|\mathcal{T}|$ can be too large and fails (R10) with $L \asymp |\mathcal{T}|$. The order of L in Section 5.1 raises a similar concern. For practical implementation, we might resort to a sequential search in the following manner.

Consider model (5.7) for instance. We first partition $\mathcal{T}$ into subsets $\mathcal{T} = \cup_j \mathcal{T}_j$ so that each $\mathcal{T}_j$ satisfies (R10) (with L replaced by $|\mathcal{T}_j|$). We perform estimation on those subsets and obtain all time points identified as change locations, i.e. $\{l \in \mathcal{T}_j : \hat{\phi}_{1,l} \neq 0, \hat{\phi}_{2,l} \neq 0\}$, where $\{\hat{\phi}_{1,l}, \hat{\phi}_{2,l}\}_{l \in \mathcal{T}}$ denotes the adaptive LASSO solution for $\{\phi_{1,l}^*, \phi_{2,l}^*\}_{l \in \mathcal{T}_j}$ in (5.8) with $\mathcal{T}$ replaced by $\mathcal{T}_j$. Aggregate those identified change locations as a set $\tilde{\mathcal{T}}$ which shall satisfy (R10) (with L replaced by $|\tilde{\mathcal{T}}|$) by iteratively applying Corollary 3 on $\mathcal{T}_j$. Eventually, estimating (5.8) with $\mathcal{T}$ replaced by $\tilde{\mathcal{T}}$ locates the change points consistently.

6 Numerical Studies

6.1 Simulations

We run Monte-Carlo simulations to demonstrate the performance of our estimators in this section. For the general setting, the data is generated according to

$$\mathbf{y}_t = \left\{ \mathbf{I}_d - (0.2 + 0.2 z_{1,1,t} + 0 z_{1,2,t}) \mathbf{W}_1 - (0 + 0 z_{2,1,t} + 0.3 z_{2,2,t}) \mathbf{W}_2 \right\}^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t), \quad (6.1)$$

where $\boldsymbol{\mu}^*$ and $\boldsymbol{\beta}^*$ are vectors of 1's, $\mathbf{W}_1$ is generated to have two neighbours ahead and two behind as in Kelejian and Prucha (1998), $\mathbf{W}_2$ is a contiguity matrix with each $(\mathbf{W}_2)_{i,j}$ i.i.d. Bernoulli (0.2) for $i \neq j$. The parameter is such that $\boldsymbol{\phi}^* = (0.2, 0.2, 0, 0, 0, 0.3)^\top$. The disturbance $\boldsymbol{\epsilon}_t$ is jointly Gaussian with its variance-covariance matrix having 1 at the diagonal and 0.1 above its diagonal with probability 0.2 (then replicated to the entries below the diagonal). For the rows of $\mathbf{W}_1$ or $\mathbf{W}_2$ with row sums exceeding one, we divide by the L_1 norm of the row. We use independent standard normal random variables for the dynamic variables $\{z_{1,1,t}\}$, $\{z_{1,2,t}\}$, $\{z_{2,1,t}\}$ and $\{z_{2,2,t}\}$. The covariate matrix $\mathbf{X}_t$ has three columns, with each element generated as independent standard normal, except that the third is endogenous by adding a half of ϵ_t . Let $\mathbf{X}_{\text{exo},t}$ be $\mathbf{X}_t$ with the disturbance part removed, then instruments can be set as $\mathbf{B}_t = [\mathbf{X}_{\text{exo},t}, \mathbf{W}_1 \mathbf{X}_{\text{exo},t}, \mathbf{W}_2 \mathbf{X}_{\text{exo},t}]$. Tuning parameters for the adaptive LASSO is selected by minimising the BIC in (4.1).

We experiment $d = 25, 50, 75$ and $T = 50, 100, 150$, with each setting repeated 1000 times. Results are presented in Table 1. In there, specificity is the proportion of true zeros estimated as zeros, and sensitivity is the proportion of non-zeros estimated as non-zeros, while MSE is the mean squared error. The MSE results corroborate our Theorem 1, while the specificity and sensitivity results corroborate the sparsity consistency in Theorem 2. Zeros in $\boldsymbol{\phi}^*$ can be selected with high accuracy, yet the sensitivity results suggest over-identification of zeros. Both increasing the spatial dimension d and time span T improve the performance of our estimators in general, except that when T increases from 100 to 150 and $d = 25$, all measures get a bit worse, which is similarly seen in Table 1 of Lam and Souza (2020). This might hint on the issue of a data set with unbalanced dimensions in practice.

To better illustrate the asymptotic normality for $\hat{\boldsymbol{\phi}}$ in Theorem 2, the data is generated according to (6.1) with $(T, d) = (200, 50)$, except that $\boldsymbol{\phi}^* = (0, -0.5, 0.5, 0, 0, 0)^\top$, $\mathbf{X}_t$ is exogenous and $\boldsymbol{\epsilon}_t$ has diagonal covariance matrix. For ease of presentation, we fix $\hat{H} = \{2, 3\}$ which is the index set of true non-zero parameters, while the remaining components of the covariance matrix is estimated according to Section 4.1. Fig 1 displays the histogram of $T^{1/2}(\hat{\mathbf{R}}_{\hat{H}} \hat{\mathbf{S}}_{\boldsymbol{\gamma}} \hat{\mathbf{R}}_{\hat{\beta}} \hat{\boldsymbol{\Sigma}}_{\beta} \hat{\mathbf{R}}_{\hat{\beta}}^\top \hat{\mathbf{S}}_{\boldsymbol{\gamma}}^\top \hat{\mathbf{R}}_{\hat{H}}^\top)^{-1/2}(\hat{\boldsymbol{\phi}}_{\hat{H}} - \boldsymbol{\phi}_{\hat{H}}^*)$. The plots show good approximations for our asymptotic normality result in Theorem 2, apart from some tail issues,

	$T = 50$			$T = 100$			$T = 150$		
	$d = 25$	$d = 50$	$d = 75$	$d = 25$	$d = 50$	$d = 75$	$d = 25$	$d = 50$	$d = 75$
$\hat{\phi}$ MSE	.002 (.008)	.001 (.001)	.002 (.001)	.000 (.000)	.000 (.000)	.000 (.000)	.003 (.007)	.001 (.000)	.000 (.000)
$\hat{\beta}$ MSE	.087 (.434)	.029 (.011)	.080 (.025)	.005 (.004)	.008 (.002)	.001 (.001)	.088 (.278)	.013 (.005)	.005 (.002)
$\hat{\mu}$ MSE	.039 (.124)	.038 (.010)	.044 (.010)	.013 (.006)	.010 (.002)	.011 (.002)	.036 (.083)	.010 (.003)	.008 (.002)
$\hat{\phi}$ Specificity	.992 (.064)	1.000 (.000)	1.000 (.000)	1.000 (.000)	1.000 (.000)	1.000 (.000)	.994 (.048)	1.000 (.000)	1.000 (.000)
$\hat{\phi}$ Sensitivity	.921 (.133)	.983 (.069)	.992 (.048)	.910 (.132)	.961 (.099)	.991 (.051)	.873 (.148)	.983 (.070)	.956 (.103)

Table 1: Simulation results for the general setting (6.1). Mean and standard deviation (in brackets) of the corresponding error measures over the 1000 repetitions are presented.

especially on the left tail, as our covariance estimator suffers from insufficient dimension. We leave potential improvements to the future study.

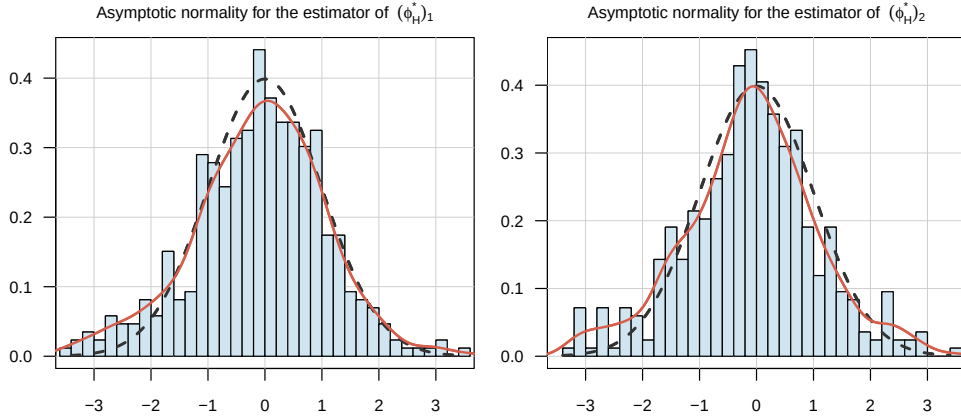


Figure 1: Histogram of the first (left panel) and second (right panel) entries in $T^{1/2}(\hat{\mathbf{R}}_{\hat{H}}\hat{\mathbf{S}}_{\gamma}\hat{\mathbf{R}}_{\beta}\hat{\Sigma}_{\beta}\hat{\mathbf{R}}_{\beta}^{\top}\hat{\mathbf{S}}_{\gamma}^{\top}\hat{\mathbf{R}}_{\hat{H}}^{\top})^{-1/2}(\hat{\phi}_{\hat{H}} - \phi_{\hat{H}}^*)$ for $(T, d) = (200, 50)$. The curves (in red) are the empirical density, and the density curves for $\mathcal{N}(0, 1)$ (in black, dotted) are superimposed.

Next, we consider the threshold spatial autoregressive model in Section 5.1. The data is generated according to

$$\mathbf{y}_t = (\mathbf{I}_d - 0.3 z_{1,t} \mathbf{W}_1 - 0.8 z_{2,t} \mathbf{W}_2)^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t), \quad (6.2)$$

where $\mathbf{W}_1$, $\mathbf{W}_2$, $\boldsymbol{\mu}^*$, $\mathbf{X}_t$, $\boldsymbol{\beta}^*$ and $\boldsymbol{\epsilon}_t$ are constructed in the same way as those in (6.1), $z_{1,t} = \mathbb{1}_{\{q_t \leq \gamma^*\}}$ and $z_{2,t} = \mathbb{1}_{\{q_t > \gamma^*\}}$ for some threshold variables and values q_t and γ^* . Hence (6.2) represents a threshold spatial autoregressive model with changes in both coefficients and spatial weight matrices. To estimate the

parameters and the threshold value simultaneously, we fit a model of the form

$$\mathbf{y}_t = \left(\mathbf{I}_d - \sum_{l=1}^{19} z_{1,l,t} \phi_{1,l}^* \mathbf{W}_1 - \sum_{l=1}^{19} z_{2,l,t} \phi_{2,l}^* \mathbf{W}_2 \right)^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t), \quad \text{where} \quad (6.3)$$

$$z_{1,l,t} = \mathbb{1}_{\{q_t \leq \hat{\gamma}_l\}}, \quad z_{2,l,t} = \mathbb{1}_{\{q_t > \hat{\gamma}_l\}}, \quad \hat{\gamma}_l = (5\% \cdot l)\text{-th empirical quantile of } \{q_t\}_{t \in [T]}.$$

As discussed in Section 5.1, we expect all entries are zero in $(\phi_{1,1}^*, \dots, \phi_{1,19}^*)$ but one corresponding to $z_{1,l,t}$ such that $\hat{\gamma}_l$ is the nearest to γ^* . Denote the index of such $\hat{\gamma}_l$ as l^* . Moreover, entries in $(\phi_{2,1}^*, \dots, \phi_{2,19}^*)$ should hold the same pattern with the same index of l^* being non-zero. It should also hold $(\phi_{1,l^*}^*, \phi_{2,l^*}^*) \approx (0.3, 0.8)$.

We experiment two types of threshold variables: 1) (*AR(5)*) q_t is AR(5) with i.i.d. standard normal innovations, with $\gamma^* = 0.3$; 2) (*Self-exciting on mean*) $q_t = \mathbf{1}^\top \mathbf{y}_{t-1}/d$, i.e., the regime changes in a self-exciting manner on the mean of the previous data point, with $\gamma^* = 1.5$. From the above discussion, results for $d = 50, 75$ and $T = 100, 150$ are presented in Table 2 with $\hat{\phi}$ and $\hat{\gamma}_l$ estimated from (6.3), where

$$\hat{\phi} \text{ MSE} := \text{MSE of } \hat{\phi} \text{ with } \phi \text{ all zero except } (\phi_{1,l^*}^*, \phi_{2,l^*}^*) \text{ set as } (0.3, 0.8),$$

$$\hat{\phi} \text{ True-Unique} := \mathbb{1}_{\{\hat{\phi}_{1,l^*} \text{ and } \hat{\phi}_{2,l^*} \text{ are both nonzero in } \hat{\phi} \text{ uniquely}\}},$$

$$\hat{\gamma}_l \text{ MSE} := \text{MSE of } \hat{\gamma}_l \text{ with true threshold value } \gamma^*.$$

$\hat{\phi}$ True-Unique is the key to demonstrating the validity of our algorithm as it relates to both specificity and sensitivity. More importantly, it measures if the estimated threshold value is unique. On computing $\hat{\gamma}_l$ -MSE with multiple $\hat{\gamma}_l$ values, i.e., the intersecting index set is not a singleton, we choose l corresponding to the largest $\hat{\phi}_{1,l^*}$. Table 2 confirms that our procedure is capable of estimating the threshold value and other model parameters in one go. Although the estimator of γ^* is coarse up by the 5% empirical quantile of the threshold variable, Table 2 shows the increase in data dimensions enhances the performance of $\hat{\gamma}_l$. In practice, re-estimation using a finer grid based on such initial threshold estimator could be performed.

q_t setting	AR(5)				Self-exciting on mean			
(T, d)	(100, 50)	(100, 75)	(150, 50)	(150, 75)	(100, 50)	(100, 75)	(150, 50)	(150, 75)
$\hat{\phi}$ MSE	.002 (.002)	.003 (.003)	.017 (.013)	.020 (.011)	.009 (.011)	.007 (.010)	.003 (.006)	.001 (.003)
$\hat{\phi}$ True-Unique	.562 (.497)	.439 (.500)	.706 (.456)	.844 (.363)	.537 (.499)	.621 (.485)	.797 (.403)	.938 (.242)
$\hat{\gamma}_l$ MSE	.343 (.870)	.066 (.309)	.207 (.707)	.025 (.193)	.128 (.881)	.031 (.126)	.080 (.802)	.005 (.023)

Table 2: Simulation results for the threshold model (6.3). Mean and standard deviation (in brackets) of the corresponding error measures over the 500 repetitions are presented.

6.2 Change point analysis

In this section, we demonstrate the performance of our dynamic framework on models with structural changes as described in Section 5.2. We first consider a spatial autoregressive model with a single change such that

$$\mathbf{y}_t = (\mathbf{I}_d - 0.3 \mathbb{1}_{\{t \leq 30\}} \mathbf{W}_1 - 0.3 \mathbb{1}_{\{t > 30\}} \mathbf{W}_2)^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t), \quad (6.4)$$

where $\mathbf{W}_1$, $\mathbf{W}_2$, $\boldsymbol{\mu}^*$, $\mathbf{X}_t$ and $\boldsymbol{\beta}^*$ are constructed in the same way as those in (6.1). The model (6.4) represents a change on the true spatial weight matrix at $t = 30$ from $0.3 \mathbf{W}_1$ to $0.3 \mathbf{W}_2$. Akin to (6.3), we consider

$$\mathbf{y}_t = \left(\mathbf{I}_d - \sum_{l=1}^{\lfloor T/\Delta \rfloor - 1} z_{1,l,t} \phi_{1,l}^* \mathbf{W}_1 - \sum_{l=1}^{\lfloor T/\Delta \rfloor - 1} z_{2,l,t} \phi_{2,l}^* \mathbf{W}_2 \right)^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t), \quad \text{where} \quad (6.5)$$

$$z_{1,l,t} = \mathbb{1}_{\{t \leq t_l\}}, \quad z_{2,l,t} = \mathbb{1}_{\{t > t_l\}}, \quad t_l = \Delta \cdot l.$$

We consider a grid of candidate change locations, spaced at intervals of $\Delta = 5$. From (6.4), every $(\phi_{1,l}^*, \phi_{2,l}^*) = (0, 0)$ except the one l such that $t_l = 30$. Similarly to Table 2, the error measure $\hat{\phi}$ True-Unique is hence defined to be 1 if the only non-zero pair $(\phi_{1,l}^*, \phi_{2,l}^*)$ corresponds to $t_l = 30$. To showcase the robustness under heavy-tailed noise, we compare the results with $\boldsymbol{\epsilon}_t$ generated by i.i.d. $\mathcal{N}(0, 1)$ and t_6 , respectively. Table 3 displays the results with each (T, d) setting specified, and Table 4 shows the results under a stronger change signal such that the true spatial weight matrix changes from $0.5 \mathbf{W}_1$ to $0.5 \mathbf{W}_2$.

Results for both weak and strong change signals display similar patterns. Unsurprisingly, the change detection slightly suffers from fat-tailed noise and weaker signals. Increasing spatial dimension clearly rescues the accuracy of detection. Although a larger T seems to undermine the $\hat{\phi}$ True-Unique measure, it is essentially due to more dynamic variables ($z_{1,l,t}$ and $z_{2,l,t}$) used in the setting.

$\boldsymbol{\epsilon}_t$	i.i.d. $\mathcal{N}(0, 1)$				i.i.d. t_6			
(T, d)	(50, 25)	(50, 50)	(100, 50)	(100, 75)	(50, 25)	(50, 50)	(100, 50)	(100, 75)
$\hat{\phi}$ MSE	.008 (.007)	.004 (.003)	.004 (.002)	.003 (.002)	.010 (.009)	.004 (.003)	.004 (.003)	.003 (.002)
$\hat{\phi}$ True-Unique	.611 (.488)	.965 (.184)	.668 (.472)	.938 (.242)	.559 (.497)	.944 (.231)	.647 (.479)	.898 (.303)

Table 3: Simulation results for the change point model (6.5) with a weak change signal. Mean and standard deviation (in brackets) of the corresponding error measures over the 500 repetitions are presented.

We also use one example to demonstrate the numerical performance of the iterative searching scheme depicted in Remark 1. Under $(T, d) = (100, 75)$, consider an extension of (6.4) with two change points such

ϵ_t	i.i.d. $\mathcal{N}(0, 1)$				i.i.d. t_6			
(T, d)	(50, 25)	(50, 50)	(100, 50)	(100, 75)	(50, 25)	(50, 50)	(100, 50)	(100, 75)
$\hat{\phi}$ MSE	.006 (.011)	.001 (.001)	.003 (.006)	.001 (.001)	.007 (.012)	.001 (.003)	.004 (.007)	.001 (.002)
$\hat{\phi}$ True-Unique	.826 (.379)	.994 (.077)	.846 (.361)	.990 (.100)	.759 (.428)	.984 (.126)	.811 (.392)	.972 (.166)

Table 4: Simulation results for the change point model (6.5) with a strong change signal. Mean and standard deviation (in brackets) of the corresponding error measures over the 500 repetitions are presented.

that

$$\mathbf{y}_t = (\mathbf{I}_d - 0.8 \mathbb{1}_{\{t \leq 30\}} \mathbf{W}_1 + 0.9 \mathbb{1}_{\{t \leq 60\}} \mathbf{W}_1 + 0.9 \mathbb{1}_{\{t > 60\}} \mathbf{W}_2)^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t). \quad (6.6)$$

That is, the spatial weight matrix changes from $-0.1\mathbf{W}_1$ to $-0.9\mathbf{W}_1$ at $t = 30$, followed by a change from $-0.9\mathbf{W}_1$ to $-0.9\mathbf{W}_2$ at $t = 60$. Suppose it is only known a priori that on $\mathcal{T} = \{2, 4, \dots, 98, 100\}^3$, the spatial weight matrix might change from $\mathbf{W}_1$ to $\mathbf{W}_2$ and potentially the spatial correlation coefficients change as well. We hope to estimate the number of changes and the change locations. Following Remark 1, we construct the ordered sets $\mathcal{T}_1 = \{2, 4, \dots, 20\}$, $\mathcal{T}_2 = \{20, 22, \dots, 40\}$, $\mathcal{T}_3 = \{40, 42, \dots, 60\}$, $\mathcal{T}_4 = \{60, 62, \dots, 80\}$ and $\mathcal{T}_5 = \{80, 82, \dots, 100\}$. Note that we add “ties” between adjacent sets to circumvent falsely identifying change candidates at the margin. Now the size $|\mathcal{T}_j| \leq 11$ for $j \in [5]$ should be reasonable according to Tables 3 and 4. Then for each $j \in [5]$, consider⁴,

$$\mathbf{y}_t = \left(\mathbf{I}_d - \sum_{l=1}^{|\mathcal{T}_j|} z_{j,1,l,t} \phi_{j,1,l}^* \mathbf{W}_1 - \sum_{l=1}^{|\mathcal{T}_j|} z_{j,2,l,t} \phi_{j,2,l}^* \mathbf{W}_2 \right)^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t), \quad \text{where} \quad (6.7)$$

$$z_{j,1,l,t} := \mathbb{1}_{\{t \leq (\mathcal{T}_j)_l\}}, \quad z_{j,2,l,t} := \mathbb{1}_{\{t > (\mathcal{T}_j)_l\}}.$$

As in Remark 1, all time points corresponding to nonzero estimates of $\phi_{j,1,l}^*$ or $\phi_{j,2,l}^*$ for $j \in [5], l \in [|\mathcal{T}_j|]$ are identified and collected to form a refined candidate set $\tilde{\mathcal{T}}$, where marginal time points, i.e., $t \in \{20, 40, 60, 80\}$, are discarded if they are not identified in all $\mathcal{T}_j$ containing them. Finally, consider

$$\mathbf{y}_t = \left(\mathbf{I}_d - \sum_{l=1}^{|\tilde{\mathcal{T}}|} z_{1,l,t} \phi_{1,l}^* \mathbf{W}_1 - \sum_{l=1}^{|\tilde{\mathcal{T}}|} z_{2,l,t} \phi_{2,l}^* \mathbf{W}_2 \right)^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t), \quad \text{where} \quad (6.8)$$

$$z_{1,l,t} = \mathbb{1}_{\{t \leq \tilde{\mathcal{T}}_l\}}, \quad z_{2,l,t} = \mathbb{1}_{\{t > \tilde{\mathcal{T}}_l\}}.$$

Then change points are estimated as the times points corresponding to nonzero estimates for $\phi_{1,l}^*$ or $\phi_{2,l}^*$. The experiment is repeated 500 times with estimated change locations shown in the first histogram of Figure 2. Although we find that only 64.2% of the experiments estimate the change location to be exactly

³Change point identified at the last observed time point $T = 100$ represents no change in the structure.

⁴Note that the last dynamic variable in $\mathcal{T}_5$ for $\mathbf{W}_2$, $z_{5,2,11,t} = 0$ for all t , so we directly specify $\phi_{5,2,11}^*$ as 0.

$\{30, 60\}$, the quality of the search is guaranteed by the adjusted Rand index (ARI) between the estimated time segmentation and the truth⁵. ARI is a similarity measure between set partitions and is often used by change point researchers, see Wang and Samworth (2017) and the references therein. The computed ARI is 0.901 on average, where zero ARI implies no alignment between the compared partitions and ARI at the maximum 1 implies exactly same partitions. Hence our iterative scheme is performing well according to the histogram and ARI.

We also consider (6.6) under no changes or equivalently, one change at $t = 100$, such that

$$\mathbf{y}_t = (\mathbf{I}_d + 0.9 \mathbb{1}_{\{t \leq T\}} \mathbf{W}_1)^{-1} (\boldsymbol{\mu}^* + \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t). \quad (6.9)$$

We follow exactly the same procedure in estimating (6.6), and the estimated change points over 500 runs is shown in the right-hand histogram of Figure 2. Among 98% of the experiments, exactly $T = 100$ is identified, i.e., no change is detected, implying a 2% of false change discovery. Moreover, the computed ARI⁶ is on average 0.980.

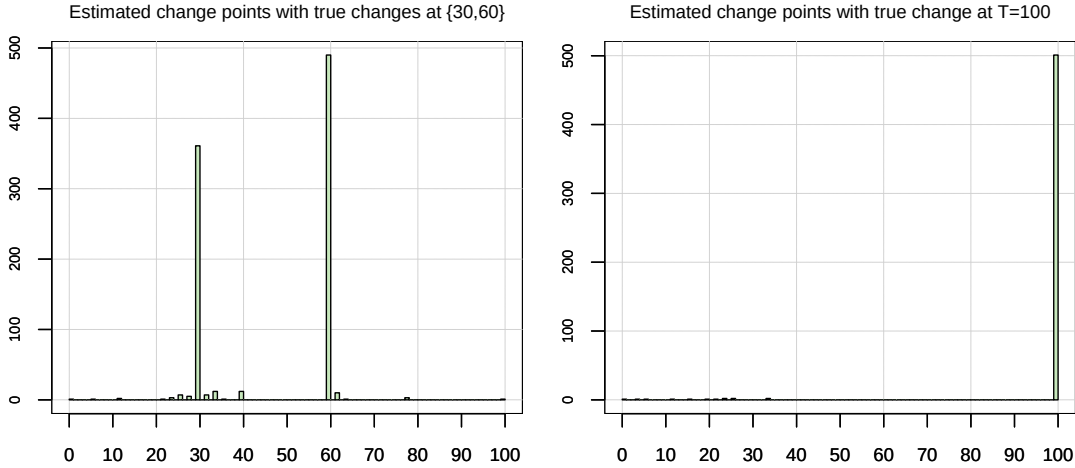


Figure 2: Histograms of estimated change locations under true models (6.6) (left panel) and (6.9) (right panel). Both experiments are repeated 500 times.

⁵The estimated time segmentation attributes the same labels for time points in between estimated change points, and labels are different after each change. The true time partition for changes at $\{30, 60\}$ is such that $\{1, 2, \dots, 30\}$ are labelled 1, $\{31, 32, \dots, 60\}$ are labelled 2, and $\{61, 62, \dots, 100\}$ are labelled 3.

⁶The true time partition when there are no changes has all time points labelled as 1.

6.3 A real data example

In this case study, we use our proposed model to analyse the total profits of enterprises for a selection of provincial regions in China. Our panel data covers $T = 86$ monthly periods from March 2016 to August 2024 and 25 provinces and 4 direct-administered municipalities (i.e., $d = 29$)⁷. A snippet⁸ of the total profits for August 2024 is shown in Figure 3. The set of covariates (all standardized) consists of Consumer Price Index (CPI), Purchasing Price Index for industrial producers (PPI) and output of electricity ($elec$). The data is available at the National Bureau of Statistics of China: <https://data.stats.gov.cn/english/>.

We consider three spatial weight matrix candidates specified as the following, with each row standardized by its L_1 norm if the row sum exceeds one: inverse distance matrix using inverse of geographical distances between locations computed by the Geodesic WGS-84 System ($\mathbf{W}_1$), contiguity matrix ($\mathbf{W}_2$), municipality matrix such that all direct-administered municipalities are neighbours ($\mathbf{W}_3$).

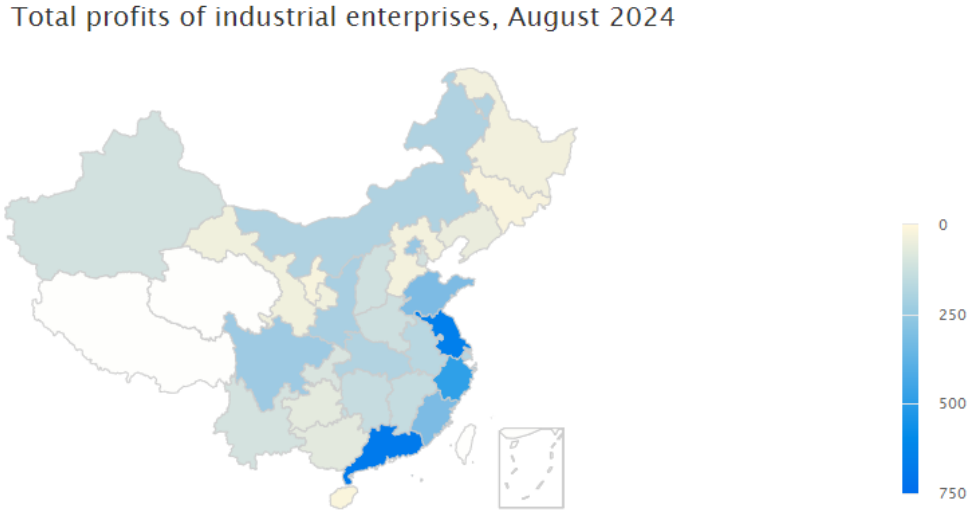


Figure 3: Illustration of the total profits of industrial enterprises within considered Chinese provinces and direct-administered municipalities, in 100 million yuans.

We treat the covariates as exogenous for two reasons: CPI and PPI are largely independent of the internal economic activities specific to enterprises within each province or municipality, and electricity supply as a public utility is often price inelastic. Using the aforementioned covariates and spatial weight matrices, we first specify a time-invariant spatial autoregressive model as our null model:

$$profit_t = \boldsymbol{\mu}^* + (\phi_{1,0}^* \mathbf{W}_1 + \phi_{2,0}^* \mathbf{W}_2 + \phi_{3,0}^* \mathbf{W}_3) profit_t + (CPI_t, PPI_t, elec_t) \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t. \quad (6.10)$$

⁷Due to missingness, we exclude January and February data. The 25 provinces included in the data are Hebei, Shanxi, Inner Mongolia, Liaoning, Jilin, Heilongjiang, Jiangsu, Zhejiang, Anhui, Fujian, Jiangxi, Shandong, Henan, Hubei, Hunan, Guangdong, Guangxi, Hainan, Sichuan, Guizhou, Yunnan, Shaanxi, Gansu, Ningxia, Xinjiang, and the 4 direct-administered municipalities are Beijing, Tianjin, Shanghai and Chongqing.

⁸The map is produced using the R package `hchinamap`.

	$\hat{\phi}_{1,0}$	$\hat{\phi}_{2,0}$	$\hat{\phi}_{3,0}$	$\hat{\beta}_{CPI}$	$\hat{\beta}_{PPI}$	$\hat{\beta}_{elec}$	BIC
Null model	15.184 (3.725)	.000	-.285 (.066)	.021	.053	.394	2.790

Table 5: Estimated coefficients for model (6.10), with standard error (in brackets) computed according to Section 4.1. The computed BIC is displayed in the last column, and $\hat{\beta}_{CPI}$, $\hat{\beta}_{PPI}$ and $\hat{\beta}_{elec}$ denote the estimates of β^* corresponding to CPI_t , PPI_t and $elec_t$, respectively.

We estimate the parameters in (6.10) by our adaptive LASSO estimators. It is revealed from $\hat{\mu}$ that Guangdong, Beijing, Jiangsu and Shanghai have significantly larger spatial fixed effects than other provinces or municipalities. The estimated coefficients $\{\hat{\phi}_{1,0}, \hat{\phi}_{2,0}, \hat{\phi}_{3,0}, \hat{\beta}\}$ are presented in Table 5, together with the BIC computed according to (4.2). The table also shows the standard error of $(\hat{\phi}_{1,0}, \hat{\phi}_{3,0})$ based on Theorem 2, and the p-values on testing zero coefficients are both less than 0.0001. This reveals the spillovers among the neighbours of provinces and municipalities. Interestingly, $\hat{\phi}_{3,0}$ suggests a negative spillover effect among the four direct-administered municipalities, which might be explained by enterprises within municipalities are main competitors in the market.

Hereafter, we refer to (6.10) as the null model. The rest of the analysis is performed in an exploratory fashion such that we consider spatial autoregressive models of the form (2.1) with some l_j and dynamic variables $\{z_{j,k,t}\}$. Some of our findings comparable to the null model are:

$$\begin{aligned}
\text{Model 1 : } profit_t &= \mu^* + \sum_{j=1}^3 \left(\phi_{j,0}^* + \sum_{k=1}^{15} \phi_{j,k}^* \mathbb{1}\{t \leq 5 + 5k\} \right) \mathbf{W}_j profit_t + (CPI_t, PPI_t, elec_t) \beta^* + \epsilon_t; \\
\text{Model 2 : } profit_t &= \mu^* + \sum_{j=1}^3 \left(\phi_{j,0}^* + \sum_{k=1}^9 \phi_{j,k}^* \mathbb{1}\{\text{sd}(profit_{t-5}) \leq \gamma_k\} \right) \mathbf{W}_j profit_t + (CPI_t, PPI_t, elec_t) \beta^* + \epsilon_t, \\
&\text{where } (\gamma_1, \dots, \gamma_9) = (.177, .193, .202, .207, .217, .219, .231, .250, .302); \\
\text{Model 3 : } profit_t &= \mu^* + \sum_{j=1}^3 \left(\phi_{j,0}^* + \sum_{k=1}^5 \phi_{j,k}^* \mathbb{1}\{t \text{ divides } 2k\} \right) \mathbf{W}_j profit_t + (CPI_t, PPI_t, elec_t) \beta^* + \epsilon_t.
\end{aligned} \tag{6.11}$$

Model 1 represents a spatial autoregressive model with the spatial weight matrix potentially changing at (10, 15, 20, ..., 75, 80), Model 2 is a self-exciting threshold spatial autoregressive model with the standard deviation of $profit_{t-5}$ as the threshold variable. The sequence of threshold value is in fact the empirical quantile, from 10% to 90%, of $\text{sd}(profit_{t-5})$. Model 3 is similar to the null model but accounts for monthly spillovers for lags of two, four, six, eight and ten months.

In our analyses above, the specification of the spatial weight matrices can be inappropriate. Our framework can be adapted to spatial autoregressive models with constant and nonlinear spatial weight

matrices. For instance, we consider

$$\text{Model 4 : } profit_t = \boldsymbol{\mu}^* + \left(\phi_{1,0}^* \alpha_{3,1}(\mathbf{W}_1) + \phi_{2,0}^* \alpha_{3,2}(\mathbf{W}_1) + \phi_{3,0}^* \alpha_{3,3}(\mathbf{W}_1) \right) profit_t + (CPI_t, PPI_t, elec_t) \boldsymbol{\beta}^* + \boldsymbol{\epsilon}_t, \quad (6.12)$$

where $\alpha_{3,1}(\mathbf{W}_1)$, $\alpha_{3,2}(\mathbf{W}_1)$ and $\alpha_{3,3}(\mathbf{W}_1)$ denote the matrix formed by series expansion using the order-3 normalized Laguerre functions⁹ based on $\{(\mathbf{W}_1)_{ij}^{-1}\}_{i,j \in [29]}$.

Table 6 reports the estimated parameters and BIC for each model. From the table, Model 1 has non-zero $\hat{\phi}_{1,9}$ corresponding to a change in the spillovers featured by $\mathbf{W}_1$ in March 2020. Although its p-value of 0.185 is not satisfying, this could hint on the inactive economic activities due to COVID-19 starting at the beginning of 2020.

Model 2 has the best BIC among all models shown here, including the null model. From Table 6, we may identify four threshold regions such that

$$\widehat{\mathbf{W}}_t = \begin{cases} 28.718 \mathbf{W}_1 - 0.522 \mathbf{W}_3, & \text{sd}(profit_{t-5}) \leq 0.202; \\ 18.997 \mathbf{W}_1 - 0.522 \mathbf{W}_3, & 0.202 < \text{sd}(profit_{t-5}) \leq 0.231; \\ 18.997 \mathbf{W}_1, & 0.231 < \text{sd}(profit_{t-5}) \leq 0.302; \\ 0, & \text{sd}(profit_{t-5}) > 0.302; \end{cases}$$

where $\widehat{\mathbf{W}}_t = \sum_{j=1}^3 (\hat{\phi}_{j,0}^* + \sum_{k=1}^9 \hat{\phi}_{j,k}^* \mathbb{1}\{\text{sd}(profit_{t-5}) \leq \gamma_k\}) \mathbf{W}_j$. For better illustration, the series of $\widehat{\mathbf{W}}_t$ among Beijing, Shanghai and Guangdong are plotted in Figure 4. We see that the spillovers between Beijing and Shanghai is more significant than their respective spillovers with Guangdong. Other lags of $profit_t$ or the mean on the lagged data as the threshold variable are experimented, and the presented Model 2 best explained our data among them.

On Model 3, Table 6 suggests that the effect of $\mathbf{W}_1$ (representing domestic spillovers) remains constant, that of $\mathbf{W}_2$ (representing more local spillovers) persists every two months but roughly cancels out every half year, and the “municipality spillover” by $\mathbf{W}_3$ occurs every December. It is interesting to see different spillover patterns featured by the expert spatial weight matrices, which certainly merits further study.

In contrast to Models 1 to 3, Model 4 considers a time-invariant spillover effect. An example of the estimated spatial weight matrix is displayed in Figure 5. It depicts how the spillovers diminish with the geographical distance, albeit the model BIC is worse than the null model. Lastly, the analysis on the total profits data is for demonstrating our dynamic spatial autoregressive framework, and more sophisticated investigations are required to further understand the spatial relations among industrial enterprises in Chinese provinces and direct-administered municipalities.

⁹The basis function selected here is inspired by Sun (2016).

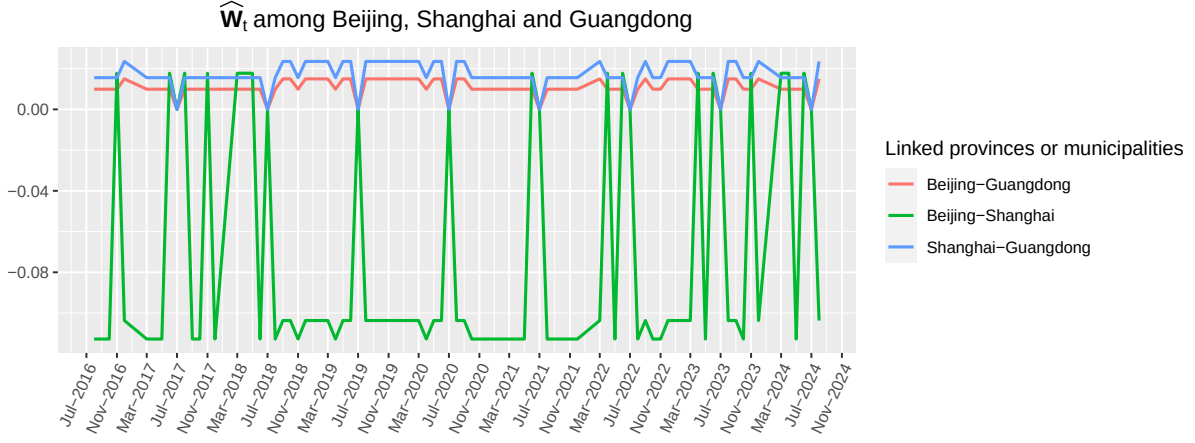


Figure 4: Illustration of $\widehat{\mathbf{W}}_t$ of Model 2 in (6.11) among Beijing, Shanghai and Guangdong, from August 2016 to August 2024.

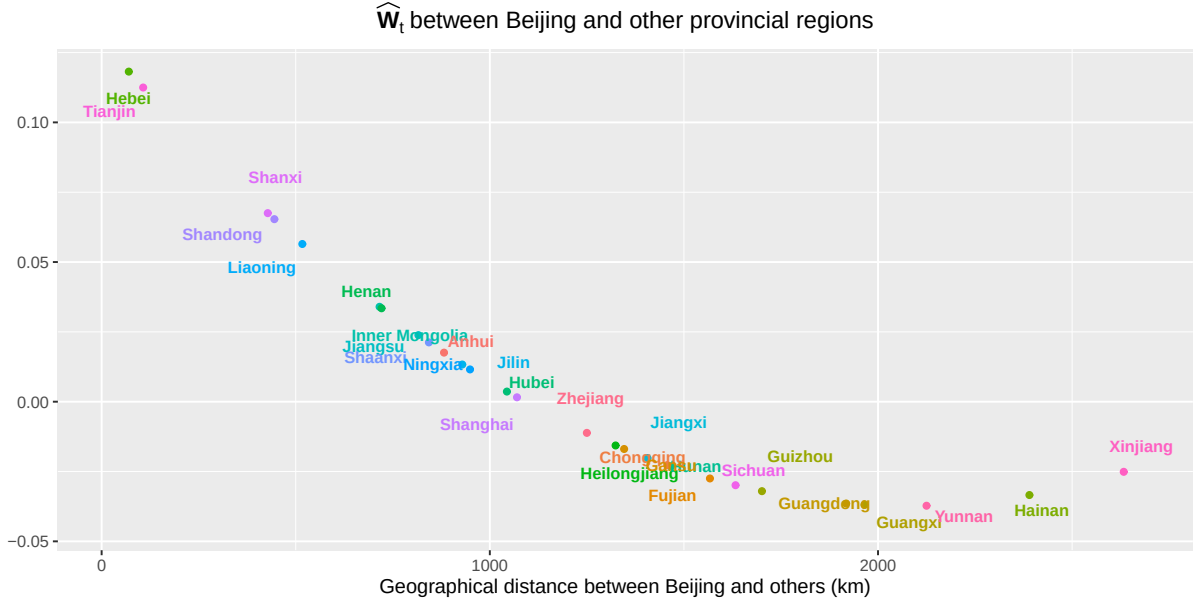


Figure 5: Illustration of (time-invariant) $\widehat{\mathbf{W}}_t$ of Model 4 in (6.12) between Beijing and other provincial regions, against their geographical distances.

	Non-zero $\hat{\phi}_{j,k}$				$\hat{\beta}_{CPI}$	$\hat{\beta}_{PPI}$	$\hat{\beta}_{elec}$	BIC
Model 1	$\hat{\phi}_{1,9} = 18.030$ (13.601)				.044	.038	.309	2.978
Model 2	$\hat{\phi}_{1,3} = 9.721$ (3.487)	$\hat{\phi}_{1,9} = 18.997$ (.717)	$\hat{\phi}_{3,7} = -.522$ (.001)		.030	.038	.368	2.744
Model 3	$\hat{\phi}_{1,0} = 15.424$ (.000)	$\hat{\phi}_{2,1} = .335$ (.000)	$\hat{\phi}_{2,3} = -.331$ (.000)	$\hat{\phi}_{3,5} = -.401$ (.000)	.023	.052	.398	2.747
Model 4	$\hat{\phi}_{1,0} = .048$ (.001)	$\hat{\phi}_{2,0} = -.034$ (.003)	$\hat{\phi}_{3,0} = .114$ (.001)		.043	.056	.512	2.848

Table 6: Estimated coefficients for different models specified in (6.11), with standard errors (in brackets) computed according to Section 4.1. The computed BIC is displayed in the last column. Refer to Table 5 for the definitions of $\hat{\beta}_{CPI}$, $\hat{\beta}_{PPI}$ and $\hat{\beta}_{elec}$.

7 Appendix

Proofs of all the theorems in this paper can be found in the supplement.

A Appendix: Additional explanations of Section 2.2

Note from (2.13) and the definition of $\mathbf{B}$ in (2.9), we have

$$\begin{aligned}
& \mathbf{B}^\top \left(\mathbf{I}_d \otimes \left\{ \left(\mathbf{I}_T \otimes \left\{ (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \right\} \right) (\mathbf{X}_1, \dots, \mathbf{X}_T)^\top \right\} \right) \text{vec}(\mathbf{I}_d) \\
&= T^{-1/2} d^{-a/2} \left(\mathbf{I}_d \otimes \left\{ (\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}}) (\mathbf{I}_T \otimes \gamma) \right. \right. \\
&\quad \left. \left. \left(\mathbf{I}_T \otimes \left\{ (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \right\} \right) (\mathbf{X}_1, \dots, \mathbf{X}_T)^\top \right\} \right) \text{vec}(\mathbf{I}_d) \\
&= T^{-1/2} d^{-a/2} \left(\mathbf{I}_d \otimes \left\{ \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}_t^\top \right\} \right) \text{vec}(\mathbf{I}_d) \\
&= T^{-1/2} d^{-a/2} \text{vec} \left(\sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}_t^\top \right) \\
&= T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \right) \text{vec} \left((\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \right) \\
&= T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \right) [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{y}^\nu.
\end{aligned}$$

Similarly, by the definition of $\mathbf{Y}_W$,

$$\begin{aligned}
& \mathbf{B}^\top \left(\mathbf{I}_d \otimes \left\{ \left(\mathbf{I}_T \otimes \left\{ \left(\sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k} (\mathbf{y}_{j,k}^\nu)^\top (\mathbf{W}_j^\otimes)^\top \right) \right. \right. \right. \right. \\
&\quad \left. \left. \left. \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \right\} \right) (\mathbf{X}_1, \dots, \mathbf{X}_T)^\top \right\} \right) \text{vec}(\mathbf{I}_d) \\
&= T^{-1/2} d^{-a/2} \left(\mathbf{I}_d \otimes \left\{ (\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}}) (\mathbf{I}_T \otimes \gamma) \right. \right. \\
&\quad \left. \left. \left(\mathbf{I}_T \otimes \left\{ \left(\sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k} (\mathbf{y}_{j,k}^\nu)^\top (\mathbf{W}_j^\otimes)^\top \right) \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \right\} \right) (\mathbf{X}_1, \dots, \mathbf{X}_T)^\top \right\} \right) \text{vec}(\mathbf{I}_d) \\
&= T^{-1/2} d^{-a/2} \left(\mathbf{I}_d \otimes \left\{ \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \left(\sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k} (\mathbf{y}_{j,k}^\nu)^\top (\mathbf{W}_j^\otimes)^\top \right) \right. \right. \\
&\quad \left. \left. \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}_t^\top \right\} \right) \text{vec}(\mathbf{I}_d) \\
&= T^{-1/2} d^{-a/2} \text{vec} \left(\sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma (\phi^\top \mathbf{Y}_W^\top) \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}_t^\top \right) \\
&= T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \right) [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{Y}_W \phi.
\end{aligned}$$

With (2.13), the term inside the squared Euclidean norm of (2.14) can hence be further written as

$$\begin{aligned}
& \mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{V} \phi - \mathbf{B}^\top \mathbf{X}_{\beta(\phi)} \text{vec}(\mathbf{I}_d) \\
&= \mathbf{B}^\top \mathbf{y} - \mathbf{B}^\top \mathbf{V} \phi - \mathbf{B}^\top \left(\mathbf{I}_d \otimes \left\{ \left(\mathbf{I}_T \otimes \left\{ (\mathbf{y}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \right\} \right) (\mathbf{X}_1, \dots, \mathbf{X}_T)^\top \right\} \right) \text{vec}(\mathbf{I}_d) \\
&\quad + \mathbf{B}^\top \left(\mathbf{I}_d \otimes \left\{ \left(\mathbf{I}_T \otimes \left\{ \left(\sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k} (\mathbf{y}_{j,k}^\nu)^\top (\mathbf{W}_j^\otimes)^\top \right) \right. \right. \right. \right. \\
&\quad \left. \left. \left. \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \right\} \right) (\mathbf{X}_1, \dots, \mathbf{X}_T)^\top \right\} \right) \text{vec}(\mathbf{I}_d) \\
&= \mathbf{B}^\top \mathbf{y} - T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \right) [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{y}^\nu \\
&\quad - \left\{ \mathbf{B}^\top \mathbf{V} - T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \right) [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{Y}_W \right\} \phi \\
&= \mathbf{B}^\top \mathbf{y} - \Xi \mathbf{y}^\nu - (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W) \phi.
\end{aligned} \tag{A.1}$$

B Appendix: Proof of all theorems, lemmas and corollaries

B.1 Lemmas and proofs

To prove our main theorems, recall first $B_{t,ij}$ and $X_{t,ij}$ represent the (i, j) entry of $\mathbf{B}_t$ and $\mathbf{X}_t$ respectively.

Define $\mathcal{M} = \bigcap_{i=1}^{13} \mathcal{A}_i$, where

$$\begin{aligned}
\mathcal{A}_1 &= \left\{ \max_{i,q \in [d]} \max_{j,l \in [r]} \left| \frac{1}{T} \sum_{t=1}^T [B_{t,ij} X_{t,ql} - \mathbb{E}(B_{t,ij} X_{t,ql})] \right| < c_T \right\}, \\
\mathcal{A}_2 &= \left\{ \max_{i,q \in [d]} \max_{j \in [r]} \left| \frac{1}{T} \sum_{t=1}^T B_{t,ij} \epsilon_{t,q} \right| < c_T \right\}, \\
\mathcal{A}_3 &= \left\{ \max_{j \in [r]} \left| \frac{1}{T} \sum_{t=1}^T \sum_{q=1}^d B_{t,qj} \epsilon_{t,q} \right| < c_T d^{\frac{1}{2} + \frac{1}{2w}} \right\}, \\
\mathcal{A}_4 &= \left\{ \max_{i \in [d]} \max_{j \in [r]} |\bar{B}_{\cdot,ij} - \mathbb{E}[B_{t,ij}]| < c_T \right\}, \\
\mathcal{A}_5 &= \left\{ \max_{q \in [d]} |\bar{\epsilon}_{\cdot,q}| < c_T \right\}, \\
\mathcal{A}_6 &= \left\{ \max_{i \in [d]} \max_{j \in [r]} |\bar{X}_{\cdot,ij}| < c_T \right\}, \\
\mathcal{A}_7 &= \left\{ \max_{j \in [r]} \left| \sum_{q=1}^d \bar{B}_{\cdot,qj} \bar{\epsilon}_{\cdot,q} \right| < 2^{1/2} c_T d^{1/2} \log^{1/2}(T \vee d) S_\epsilon(\mu_{b,\max} + c_T) \right\}, \\
\mathcal{A}_8 &= \left\{ \max_{m \in [p]} \max_{n \in [l_m] \cup \{0\}} \max_{i,q \in [d]} \max_{j \in [v]} \max_{l \in [r]} \left| \frac{1}{T} \sum_{t=1}^T [z_{m,n,t} B_{t,ij} X_{t,ql} - \mathbb{E}(z_{m,n,t} B_{t,ij} X_{t,ql})] \right| < c_T \right\}, \\
\mathcal{A}_9 &= \left\{ \max_{m \in [p]} \max_{n \in [l_m] \cup \{0\}} \max_{i \in [d]} \max_{j \in [r]} \left| \frac{1}{T} \sum_{t=1}^T z_{m,n,t} X_{t,ij} \right| < c_T \right\}, \\
\mathcal{A}_{10} &= \left\{ \max_{m \in [p]} \max_{n \in [l_m] \cup \{0\}} \max_{i,q \in [d]} \max_{j \in [v]} \left| \frac{1}{T} \sum_{t=1}^T z_{m,n,t} B_{t,ij} \epsilon_{t,q} \right| < c_T \right\}, \\
\mathcal{A}_{11} &= \left\{ \max_{m \in [p]} \max_{n \in [l_m] \cup \{0\}} \max_{q \in [d]} \left| \frac{1}{T} \sum_{t=1}^T z_{m,n,t} \epsilon_{t,q} \right| < c_T \right\}, \\
\mathcal{A}_{12} &= \left\{ \max_{m \in [p]} \max_{n \in [l_m] \cup \{0\}} \max_{i \in [d]} \max_{j \in [v]} \left| \frac{1}{T} \sum_{t=1}^T [z_{m,n,t} B_{t,ij} - \mathbb{E}(z_{m,n,t} B_{t,ij})] \right| < c_T \right\}, \\
\mathcal{A}_{13} &= \left\{ \max_{m \in [p]} \max_{n \in [l_m] \cup \{0\}} \left| \frac{1}{T} \sum_{t=1}^T z_{m,n,t} \right| < c_T \vee z_{\max} \right\},
\end{aligned} \tag{B.1}$$

where $\bar{B}_{\cdot,ij} := T^{-1} \sum_{t=1}^T B_{t,ij}$, $\bar{X}_{\cdot,ij} := T^{-1} \sum_{t=1}^T X_{t,ij}$, $\bar{\epsilon}_{\cdot,q} := T^{-1} \sum_{t=1}^T \epsilon_{t,q}$, $\mu_{b,\max} := \max_{i,j} |\mathbb{E}[B_{t,ij}]|$ being a constant implied by Assumption (M1), and $z_{\max}$ is the upper bound for $\{z_{j,k,t}\}$ implied in (M2) with $z_{\max} = 1$ for $\{z_{j,0,t}\}$ by default. Our main theoretical results depict the properties of estimators on the set $\mathcal{M}$ which holds with probability 1 as $T, d \rightarrow \infty$ by Assumption (R10), as shown in Lemma 2 which is similar to Theorem S.1 of Lam and Souza (2020).

To prove Lemma 2, we first quote a Nagaev-type inequality for functional dependent data from Theorem 2(ii), 2(iii) and Section 4 of Liu et al. (2013), presented as the following lemma.

Lemma 1 *For a zero mean time series process $\mathbf{x}_t = \mathbf{f}(\mathcal{F}_t)$ defined in (3.1) with dependence measure $\theta_{t,q,i}^x$ defined in (3.2), assume $\Theta_{m,2w}^x \leq Cm^{-\alpha}$ as in Assumption (M1). Then there exists constants C_1, C_2 and C_3 independent of n, T and the index i such that*

$$\mathbb{P}\left(\left|\frac{1}{T} \sum_{t=1}^T x_{t,i}\right| > n\right) \leq \frac{C_1 T^{w(1/2-\tilde{\alpha})}}{(Tn)^w} + C_2 \exp(-C_3 T^{\tilde{\beta}} n^2),$$

where $\tilde{\alpha} = \alpha \wedge (1/2 - 1/w)$ and $\tilde{\beta} = (3 + 2\tilde{\alpha}w)/(1 + w)$.

Furthermore, assume another zero mean time series process $\{\mathbf{e}_t\}$ (can be the same process $\{\mathbf{x}_t\}$) with $\Theta_{m,2w}^e$ as in Assumption (M1). Then provided $\max_i \|x_{t,i}\|_{2w}, \max_j \|e_{t,j}\|_{2w} \leq c_0 < \infty$ where c_0 is a constant, the above Nagaev-type inequality holds for the product process $\{x_{t,i}e_{t,j} - \mathbb{E}(x_{t,i}e_{t,j})\}$.

The above results also hold for any zero mean non-stationary process $\mathbf{x}_t = \mathbf{f}_t(\mathcal{F}_t)$ provided that $\max_i \|x_{t,i}\|_w < \infty$ and $\Theta_{m,2w}^{x,*} \leq Cm^{-\alpha}$, where $\Theta_{m,q}^{x,*}$ is uniform tail sum defined as

$$\Theta_{m,q}^{x,*} := \sum_{t=m}^{\infty} \max_i \theta_{t,q,i}^{x,*} := \sum_{t=m}^{\infty} \max_i \sup_t \|x_{t,i} - x'_{t,i}\|_q.$$

We present Lemma 2 below. Note that we assume $\alpha > 1/2 - 1/w$ which can be relaxed at the cost of more complicated rates and longer proofs presented here, and it simplifies the form of Lemma 1 as $w(1/2 - \tilde{\alpha}) = \tilde{\beta} = 1$.

Lemma 2 *Let Assumptions (M1), (M2) (or (M2')), and (R2) hold and $\alpha > 1/2 - 1/w$ in Assumption (M1). Suppose for the application of the Nagaev-type inequality in Lemma 1 for the processes in $\mathcal{M} = \bigcap_{i=1}^{13} \mathcal{A}_i$ where $\mathcal{A}_i$ is defined in (B.1), the constants C_1, C_2 and C_3 are the same. Then with $g \geq \sqrt{3/C_3}$ where g is the constant defined in $c_T = gT^{-1/2} \log^{1/2}(T \vee d)$, we have*

$$\mathbb{P}(\mathcal{M}) \geq 1 - 14C_1 r^2 v L \left(\frac{C_3}{3}\right)^{w/2} \frac{d^2}{T^{w/2-1} \log^{w/2}(T \vee d)} - \frac{14C_2 r^2 v L d^2}{T^3 \vee d^3} - \frac{2r}{T \vee d}.$$

Proof of Lemma 2. As shown in Theorem S.1 of Lam and Souza (2020), the tail condition in Assumption (M1) implies $\|\cdot\|_{2w}$ is bounded for the processes $\{B_{t,ij}X_{t,ql} - \mathbb{E}(B_{t,ij}X_{t,ql})\}$, $\{B_{t,ij}\epsilon_{t,q}\}$, $\{X_{t,ij}\}$ and $\{\epsilon_{t,q}\}$,

and further with Assumption (R2) we have

$$\begin{aligned}
\mathbb{P}(\mathcal{A}_1^c) &\leq C_1 r^2 \left(\frac{C_3}{3}\right)^{w/2} \frac{d^2}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 r^2 d^2}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_2^c) &\leq C_1 r \left(\frac{C_3}{3}\right)^{w/2} \frac{d^2}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 r d^2}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_3^c) &\leq C_1 \left(\frac{C_3}{3}\right)^{w/2} \frac{r}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 r}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_4^c) &\leq C_1 r \left(\frac{C_3}{3}\right)^{w/2} \frac{d}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 r d}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_5^c) &\leq C_1 \left(\frac{C_3}{3}\right)^{w/2} \frac{d}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 d}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_6^c) &\leq C_1 r \left(\frac{C_3}{3}\right)^{w/2} \frac{d}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 r d}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_7^c) &\leq \frac{2r}{T \vee d} + \mathbb{P}(\mathcal{A}_4^c) + \mathbb{P}(\mathcal{A}_5^c).
\end{aligned}$$

Consider the remaining sets. First let Assumption (M2) hold and it is easy to see $\|\cdot\|_{2w}$ is bounded for the processes $\{z_{m,n,t} B_{t,ij} X_{t,ql} - \mathbb{E}(z_{m,n,t} B_{t,ij} X_{t,ql})\}$, $\{z_{m,n,t} B_{t,ij} \epsilon_{t,q}\}$, $\{z_{m,n,t} X_{t,ij}\}$ and $\{z_{m,n,t} \epsilon_{t,q}\}$, and their uniform tail sums satisfy the condition in Lemma 1. Thus, apply Lemma 1 first on $\mathcal{A}_8^c$ and we have by the union bound,

$$\begin{aligned}
\mathbb{P}(\mathcal{A}_8^c) &\leq \sum_{m \in [p]} \sum_{n \in [l_m]} \sum_{i,q \in [d]} \sum_{j \in [v]} \sum_{l \in [r]} \mathbb{P} \left(\left| \frac{1}{T} \sum_{t=1}^T [z_{m,n,t} B_{t,ij} X_{t,ql} - \mathbb{E}(z_{m,n,t} B_{t,ij} X_{t,ql})] \right| \geq c_T \right) \\
&\leq d^2 r v L \left(\frac{C_1 T}{(T c_T)^w} + C_2 \exp(-C_3 T c_T^2) \right) \leq C_1 r v L \left(\frac{C_3}{3} \right)^{w/2} \frac{d^2}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 r v L d^2}{T^3 \vee d^3}.
\end{aligned}$$

Similarly using Lemma 1, we have

$$\begin{aligned}
\mathbb{P}(\mathcal{A}_9^c) &\leq C_1 r L \left(\frac{C_3}{3} \right)^{w/2} \frac{d}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 r L d}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_{10}^c) &\leq C_1 v L \left(\frac{C_3}{3} \right)^{w/2} \frac{d^2}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 v L d^2}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_{11}^c) &\leq C_1 L \left(\frac{C_3}{3} \right)^{w/2} \frac{d}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 L d}{T^3 \vee d^3}, \\
\mathbb{P}(\mathcal{A}_{12}^c) &\leq C_1 v L \left(\frac{C_3}{3} \right)^{w/2} \frac{d}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 v L d}{T^3 \vee d^3},
\end{aligned}$$

while $\mathbb{P}(\mathcal{A}_{13}^c) = 0$ by Assumption (M2). On the other hand, if we have Assumption (M2'), the above results remain valid, except that

$$\mathbb{P}(\mathcal{A}_{13}^c) \leq C_1 L \left(\frac{C_3}{3} \right)^{w/2} \frac{1}{T^{w/2-1} \log^{w/2}(T \vee d)} + \frac{C_2 L}{T^3 \vee d^3},$$

by applying Lemma 1 given the bounded support and tail sum assumption in (M2'). For any $\{z_{j,0,t}\}$, we may treat it as a non-stochastic basis as in (M2) and the result follows. Lastly, by $\mathbb{P}(\mathcal{M}) \geq 1 - \sum_{i=1}^{13} \mathbb{P}(\mathcal{A}_i^c)$ we complete the proof of Lemma 2. $\square$

We present the following lemma with a short proof as well, and we will frequently utilize the defined notation $\mathbf{V}_{\mathbf{H},K}$ (which is the same definition in Theorem 2) in the proof of main theorems.

Lemma 3 *For any $n \times d$ matrix $\mathbf{H} = (\mathbf{h}_1, \dots, \mathbf{h}_n)^\top$ and any $d \times K$ matrix $\mathbf{M}$, define*

$$\mathbf{V}_{\mathbf{H},K} = \begin{pmatrix} \mathbf{I}_K \otimes \mathbf{h}_1 \\ \vdots \\ \mathbf{I}_K \otimes \mathbf{h}_n \end{pmatrix}.$$

We then have $\mathbf{H}\mathbf{M} = (\mathbf{I}_n \otimes \text{vec}(\mathbf{M})^\top) \mathbf{V}_{\mathbf{H},K}$.

Proof of Lemma 3. Notice that

$$(\mathbf{I}_n \otimes \text{vec}(\mathbf{M})^\top) \mathbf{V}_{\mathbf{H},K} = \begin{pmatrix} \text{vec}(\mathbf{M})^\top (\mathbf{I}_K \otimes \mathbf{h}_1) \\ \vdots \\ \text{vec}(\mathbf{M})^\top (\mathbf{I}_K \otimes \mathbf{h}_n) \end{pmatrix},$$

whose j -th row (as a column vector) is hence $(\mathbf{I}_K \otimes \mathbf{h}_j^\top) \text{vec}(\mathbf{M}) = \text{vec}(\mathbf{h}_j^\top \mathbf{M}) = \text{vec}(\mathbf{M}^\top \mathbf{h}_j) = \mathbf{M}^\top \mathbf{h}_j$ which is the j -th row of $\mathbf{H}\mathbf{M}$ indeed. $\square$

Remark 2 *With the notation of $\mathbf{V}_{\mathbf{H},K}$, we may write any $d \times K$ matrix $\mathbf{M}$ as*

$$\mathbf{M} = \mathbf{I}_d \mathbf{M} = (\mathbf{I}_d \otimes \text{vec}(\mathbf{M})^\top) \mathbf{V}_{\mathbf{I}_d,K},$$

which will be useful if we are interested in the interaction only between $\mathbf{A}, \mathbf{M}$ in $\mathbf{ABM}$, with $\mathbf{A} \in \mathbb{R}^{l \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d}$, since we have

$$\mathbf{ABM} = (\mathbf{B}^\top \mathbf{A}^\top)^\top \mathbf{M} = \mathbf{V}_{\mathbf{B}^\top, l}^\top (\mathbf{I}_d \otimes \text{vec}(\mathbf{A}^\top)) \mathbf{M} = \mathbf{V}_{\mathbf{B}^\top, l}^\top (\mathbf{I}_d \otimes \text{vec}(\mathbf{A}^\top) \text{vec}(\mathbf{M})^\top) \mathbf{V}_{\mathbf{I}_d, K}.$$

Moreover, notice that in Lemma 3, if $K = 1$, i.e. $\mathbf{M}$ is a vector, then $\mathbf{V}_{\mathbf{H},1} = \text{vec}(\mathbf{H}^\top)$ and Lemma 3 simply coincides with the fact that

$$\mathbf{BM} = \text{vec}(\mathbf{M}^\top \mathbf{B}^\top) = \text{vec}(\mathbf{M}^\top \mathbf{B}^\top \mathbf{I}_r) = (\mathbf{I}_r \otimes \mathbf{M}^\top) \text{vec}(\mathbf{B}^\top) = (\mathbf{I}_r \otimes \text{vec}(\mathbf{M})^\top) \mathbf{V}_{\mathbf{B},1}.$$

Thus, we can interpret $\mathbf{V}_{\mathbf{H},K}$ to be a “ K -block vectorisation” of $\mathbf{H}$, as a generalisation of vectorisation.

Lemma 4 *Let the assumptions in Theorem 2 hold. Let $\mathbf{R}_\beta$ and Σ_β be defined in Theorem 2. For $I_2 = [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} T^{-2} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu$, we have I_2 asymptotically normal with rate $T^{-1/2} d^{-(1-b)/2}$ such that*

$$T^{1/2} (\mathbf{R}_\beta \Sigma_\beta \mathbf{R}_\beta^\top)^{-1/2} I_2 \xrightarrow{D} \mathcal{N}(\mathbf{0}, \mathbf{I}_r).$$

Proof of Lemma 4. Given any non-zero $\boldsymbol{\alpha} \in \mathbb{R}^r$ with $\|\boldsymbol{\alpha}\|_1 \leq c < \infty$, we construct below the asymptotic normality of $\boldsymbol{\alpha}^\top I_2$ which is $T^{1/2} d^{(1-b)/2}$ -convergent. First, we can further decompose

$$\begin{aligned} \boldsymbol{\alpha}^\top I_2 &= [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} (T^{-1} \mathbf{X}^\top \mathbf{B}^\nu - \mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)) T^{-1} (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu \\ &\quad + [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} \mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) T^{-1} (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu, \end{aligned}$$

with the second term dominating the first by (B.6). Recall that $\mathbf{R}_\beta = [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} \mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)$, then

$$\boldsymbol{\alpha}^\top I_2 = \frac{1}{T} \sum_{t=1}^T \boldsymbol{\alpha}^\top \mathbf{R}_\beta (\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top \boldsymbol{\epsilon}_t (1 + o_P(1)).$$

To construct the asymptotic normality of $\boldsymbol{\alpha}^\top I_2$, we want to show the following as in (B.33) that

$$\sum_{t \geq 0} \left\| P_0(\boldsymbol{\alpha}^\top \mathbf{R}_\beta (\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top \boldsymbol{\epsilon}_t) \right\|_2 < \infty, \quad (\text{B.2})$$

so that Theorem 3 (ii) of Wu (2011) can be applied. With the definition $s_2 := \boldsymbol{\alpha}^\top \mathbf{R}_\beta \Sigma_\beta \mathbf{R}_\beta^\top \boldsymbol{\alpha}$, we have $T^{1/2} s_2^{-1/2} \boldsymbol{\alpha}^\top I_2 \xrightarrow{D} \mathcal{N}(0, 1)$ and hence

$$T^{1/2} (\mathbf{R}_\beta \Sigma_\beta \mathbf{R}_\beta^\top)^{-1/2} I_2 \xrightarrow{D} \mathcal{N}(\mathbf{0}, \mathbf{I}_r). \quad (\text{B.3})$$

Similar to the proof of (B.33), we have (B.2) hold by Assumption (R7) and the following,

$$\begin{aligned} &\left\| P_0(\boldsymbol{\alpha}^\top \mathbf{R}_\beta (\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top \boldsymbol{\epsilon}_t) \right\|_2 \\ &= \left\| \boldsymbol{\alpha}^\top \mathbf{R}_\beta \{P_0((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top) \mathbb{E}_0(\boldsymbol{\epsilon}_t)\} + \boldsymbol{\alpha}^\top \mathbf{R}_\beta \{\mathbb{E}_{-1}((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top) P_0(\boldsymbol{\epsilon}_t)\} \right\|_2 \\ &\leq \left\{ 2\boldsymbol{\alpha}^\top \mathbf{R}_\beta \mathbb{E} \left\{ P_0((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top) \mathbb{E}_0(\boldsymbol{\epsilon}_t) \mathbb{E}_0(\boldsymbol{\epsilon}_t^\top) P_0(\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)) \right\} \mathbf{R}_\beta^\top \boldsymbol{\alpha} \right\}^{1/2} \\ &\quad + \left\{ 2\boldsymbol{\alpha}^\top \mathbf{R}_\beta \mathbb{E} \left\{ \mathbb{E}_{-1}((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))^\top) P_0(\boldsymbol{\epsilon}_t) P_0(\boldsymbol{\epsilon}_t^\top) \mathbb{E}_{-1}(\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)) \right\} \mathbf{R}_\beta^\top \boldsymbol{\alpha} \right\}^{1/2} \\ &= O\left(\|\boldsymbol{\alpha}\|_1 \|\mathbf{R}_\beta\|_\infty\right) \cdot \left(d \cdot \max_{j \in [d]} \mathbb{E}^{1/2}(\mathbb{E}_0^2(\epsilon_{t,j})) \cdot \max_{j \in [d]} \max_{k \in [v]} \|P_0(B_{t,jk})\|_2 + d \cdot \sigma_{\max} \max_{j \in [d]} \|P_0(\epsilon_{t,j})\|_2\right) \\ &= O\left(\max_{j \in [d]} \|P_0^\epsilon(\epsilon_{t,j})\|_2 + \max_{j \in [d]} \max_{k \in [v]} \|P_0^b(B_{t,jk})\|_2\right), \end{aligned}$$

where the second last equality used Assumption (R2), and the last used $\|\mathbf{R}_\beta\|_\infty = O(d^{-2} \cdot d) = O(d^{-1})$

by (B.4) and Assumption (R3).

It remains to show $\boldsymbol{\alpha}^\top I_2$ is indeed of order $T^{-1/2}d^{-(1-b)/2}$. To this end, we only need to show s_2 is of order $d^{-(1-b)}$. First, we have $\mathbf{R}_\beta \mathbf{R}_\beta^\top = [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1}$ which has all eigenvalues of order d^{-2} from (B.4) and Assumption (R3). Consider any j -th diagonal element of $\boldsymbol{\Sigma}_\beta$, we have

$$\begin{aligned} (\boldsymbol{\Sigma}_\beta)_{jj} &= \sum_{\tau} \mathbb{E} \{ (\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t))_{\cdot j}^\top \boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_{t+\tau}^\top (\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_t))_{\cdot j} \} \\ &= \sum_{\tau} \text{tr} \{ \mathbb{E}((\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_t))_{\cdot j} (\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_t))_{\cdot j}^\top) \mathbb{E}(\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_{t+\tau}^\top) \}, \end{aligned}$$

which is finite and has order exactly d^{1+b} by Assumption (R8). Notice $\boldsymbol{\Sigma}_\beta$ is $r \times r$, the order of eigenvalues of $\boldsymbol{\Sigma}_\beta$ is hence exactly d^{1+b} . The order of s_2 is d^{b-1} by

$$\|\boldsymbol{\alpha}\|_1^2 \lambda_{\min}(\mathbf{R}_\beta \mathbf{R}_\beta^\top) \lambda_{\min}(\boldsymbol{\Sigma}_\beta) \leq s_2 \leq \|\boldsymbol{\alpha}\|_1^2 \lambda_{\max}(\mathbf{R}_\beta \mathbf{R}_\beta^\top) \lambda_{\max}(\boldsymbol{\Sigma}_\beta).$$

This completes the proof of Lemma 4. $\square$

B.2 Proof of corollaries

Proof of Corollary 1. It is direct from Theorem 2. $\square$

Proof of Corollary 3. It is direct from Theorem 2. $\square$

B.3 Proof of theorems

Proof of Theorem 1. From (2.12) and (2.13), we have

$$\begin{aligned} \beta(\phi^*) &= \left(\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \left(\mathbf{y}^\nu - \sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k}^* \mathbf{W}_j^\otimes \mathbf{y}_{j,k}^\nu \right) \\ &= \left(\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \left(\mathbf{1}_T \otimes \boldsymbol{\mu}^* + \mathbf{X} \boldsymbol{\beta}^* + \boldsymbol{\epsilon}^\nu \right) \\ &= \boldsymbol{\beta}^* + \left(\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu. \end{aligned}$$

We now define a diagonal matrix $\mathbf{D}_{z_{j,k}} := \text{diag}(z_{j,k,1} \mathbf{I}_d, \dots, z_{j,k,T} \mathbf{I}_d) \in \mathbb{R}^{dT \times dT}$, with diagonal blocks $z_{j,k,1} \mathbf{I}_d, \dots, z_{j,k,T} \mathbf{I}_d$, and $\boldsymbol{\Pi}^{*\otimes} := (\mathbf{I}_{Td} - \sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k}^* \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}})^{-1}$. We then have $\mathbf{y}^\nu = \boldsymbol{\Pi}^{*\otimes} (\mathbf{1}_T \otimes \boldsymbol{\mu}^* +$

$\mathbf{X}\beta^* + \epsilon^\nu$). Thus,

$$\begin{aligned}
\beta(\tilde{\phi}) &= \left(\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \left(\mathbf{y}^\nu - \sum_{j=1}^p \sum_{k=0}^{l_j} \tilde{\phi}_{j,k} \mathbf{W}_j^\otimes \mathbf{y}_{j,k}^\nu \right) \\
&= \beta(\phi^*) + \left(\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \left(\sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \mathbf{W}_j^\otimes \mathbf{y}_{j,k}^\nu \right) \\
&= \beta(\phi^*) + \left(\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} \right)^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \left(\sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes} (\mathbf{1}_T \otimes \boldsymbol{\mu}^* + \mathbf{X}\beta^* + \epsilon^\nu) \right).
\end{aligned}$$

We can hence decompose $\beta(\tilde{\phi}) - \beta^* = \sum_{j=1}^5 I_j$ where

$$\begin{aligned}
I_1 &:= [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t) - T^{-2} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}] (\beta(\tilde{\phi}) - \beta^*), \\
I_2 &:= [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} T^{-2} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \epsilon^\nu, \\
I_3 &:= [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} T^{-2} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes} \mathbf{X} \beta^*, \\
I_4 &:= [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} T^{-2} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes} \epsilon^\nu, \\
I_5 &:= [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} T^{-2} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes} (\mathbf{1}_T \otimes \boldsymbol{\mu}^*).
\end{aligned}$$

Notice we can take any $t \in [T]$ for $\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)$ and $\mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)$ due to Assumption (M1). To bound I_1 to I_4 , we first have

$$\| [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} \|_1 \leq \frac{r^{1/2}}{\lambda_r [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]} \leq \frac{r^{1/2}}{d^2 u^2}, \quad (\text{B.4})$$

where $\lambda_r [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)] = \sigma_r^2(\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t)) \geq d^2 u^2$ with $u > 0$ being a constant by Assumption (R3), and $\lambda_j(\cdot)$ and $\sigma_j(\cdot)$ denoting the j -th largest eigenvalue and singular value of a matrix respectively.

Next, define

$$\mathbf{U} = \mathbf{I}_d \otimes T^{-1} \sum_{t=1}^T \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \mathbf{x}_t^\top, \quad \mathbf{U}_0 = \mathbf{I}_d \otimes \mathbb{E}(\mathbf{b}_t \mathbf{x}_t^\top).$$

By Lemma 3, we then have

$$\begin{aligned}
T^{-1} \mathbf{X}^\top \mathbf{B}^\nu &= T^{-1} \sum_{t=1}^T \mathbf{X}_t^\top (\mathbf{B}_t - \bar{\mathbf{B}}) = T^{-1} \sum_{t=1}^T \left\{ (\mathbf{I}_d \otimes \mathbf{x}_t^\top) \mathbf{V}_{\mathbf{I}_d, r} \right\}^\top \left\{ (\mathbf{I}_d \otimes \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}))^\top \mathbf{V}_{\mathbf{I}_d, v} \right\} \\
&= T^{-1} \sum_{t=1}^T \mathbf{V}_{\mathbf{I}_d, r}^\top (\mathbf{I}_d \otimes \mathbf{x}_t) (\mathbf{I}_d \otimes \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}))^\top \mathbf{V}_{\mathbf{I}_d, v} = \mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}^\top \mathbf{V}_{\mathbf{I}_d, v}.
\end{aligned}$$

Similarly we have $\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) = \mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}_0^\top \mathbf{V}_{\mathbf{I}_d, v}$. Thus on the set $\mathcal{M}$ in Lemma 2, with (B.4) it holds that

$$\begin{aligned}
\|I_1\|_1 &\leq \frac{r^{1/2}}{d^2 u^2} \cdot \|\mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}_0^\top \mathbf{V}_{\mathbf{I}_d, v} \mathbf{V}_{\mathbf{I}_d, v}^\top \mathbf{U}_0 - \mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}^\top \mathbf{V}_{\mathbf{I}_d, v} \mathbf{V}_{\mathbf{I}_d, v}^\top \mathbf{U}\|_1 \cdot \|\mathbf{V}_{\mathbf{I}_d, r}(\boldsymbol{\beta}(\tilde{\phi}) - \boldsymbol{\beta}^*)\|_1 \\
&= O\left(\frac{1}{d}\right) \left\{ \|\mathbf{V}_{\mathbf{I}_d, r}^\top (\mathbf{U}_0 - \mathbf{U})^\top \mathbf{V}_{\mathbf{I}_d, v} \mathbf{V}_{\mathbf{I}_d, v}^\top \mathbf{U}_0\|_1 + \|\mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}^\top \mathbf{V}_{\mathbf{I}_d, v} \mathbf{V}_{\mathbf{I}_d, v}^\top (\mathbf{U}_0 - \mathbf{U})\|_1 \right\} \cdot \|\boldsymbol{\beta}(\tilde{\phi}) - \boldsymbol{\beta}^*\|_1 \\
&= O\left(\frac{1}{d}\right) \left\{ d \cdot \|\mathbf{U}_0 - \mathbf{U}\|_{\max} \cdot \|\mathbf{U}_0\|_{\max} \right. \\
&\quad \left. + d \cdot \|\mathbf{U}_0 - \mathbf{U}\|_{\max} \cdot \left(\|\mathbf{U}_0 - \mathbf{U}\|_{\max} + \|\mathbf{U}_0\|_{\max} \right) \right\} \cdot \|\boldsymbol{\beta}(\tilde{\phi}) - \boldsymbol{\beta}^*\|_1 = O\left(c_T \|\boldsymbol{\beta}(\tilde{\phi}) - \boldsymbol{\beta}^*\|_1\right),
\end{aligned} \tag{B.5}$$

where the last equality used Assumption (R3) which implies $\|\mathbf{U}_0\|_{\max}$ is bounded by some constant, and Lemma 2 (using $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$) that

$$\begin{aligned}
\|\mathbf{U} - \mathbf{U}_0\|_{\max} &= \left\| \frac{1}{T} \sum_{t=1}^T [\text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \mathbf{x}_t^\top] - \mathbb{E}(\mathbf{b}_t \mathbf{x}_t^\top) \right\|_{\max} \\
&= \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{b}_t \mathbf{x}_t^\top - \mathbb{E}(\mathbf{b}_t \mathbf{x}_t^\top) - \text{vec}(\bar{\mathbf{B}}) \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t^\top \right\|_{\max} \leq c_T + \left\| \text{vec}(\bar{\mathbf{B}}) \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t^\top \right\|_{\max} \\
&\leq c_T + \left(\|\text{vec}(\bar{\mathbf{B}}) - \mathbb{E}[\mathbf{B}_t]\|_{\max} + \|\mathbb{E}[\mathbf{B}_t]\|_{\max} \right) \left\| \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t^\top \right\|_{\max} \leq c_T + c_T(c_T + \mu_{b, \max}).
\end{aligned} \tag{B.6}$$

Similarly for I_2 , we have on the set $\mathcal{M}$ that

$$\begin{aligned}
\|I_2\|_1 &\leq \frac{r^{1/2}}{d^2 u^2} \cdot \|\mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}^\top \mathbf{V}_{\mathbf{I}_d, v}\|_1 \cdot \|T^{-1}(\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu\|_1 \\
&= O\left(\frac{1}{d}\right) \|T^{-1}(\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu\|_1 = O\left(\frac{1}{d}\right) \left\| \frac{1}{T} \sum_{t=1}^T (\mathbf{B}_t^\top - \bar{\mathbf{B}}^\top) \boldsymbol{\epsilon}_t \right\|_1 \\
&= O_p\left(\frac{1}{d}\right) \left(c_T d^{\frac{1}{2} + \frac{1}{2w}} + c_T d^{1/2} \log^{1/2}(T \vee d) S_\epsilon(\mu_{b, \max} + c_T) \right) = O_p\left(c_T d^{-\frac{1}{2} + \frac{1}{2w}}\right),
\end{aligned} \tag{B.7}$$

where the first equality used $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$ in $\mathcal{M}$, the third used $\mathcal{A}_3, \mathcal{A}_7$ in $\mathcal{M}$, and the last used Assumption (R10).

For I_3 , first recall that

$$\boldsymbol{\Pi}_t^* = (\mathbf{I}_d - \mathbf{W}_t^*)^{-1} = \left(\mathbf{I}_d - \sum_{j=1}^p \sum_{k=0}^{l_j} \phi_{j,k}^* z_{j,k,t} \mathbf{W}_j \right)^{-1}, \tag{B.8}$$

and hence from Assumption (M2) (resp. (M2')) we have $\|\boldsymbol{\Pi}_t^*\|_\infty \leq 1/(1 - \eta) = O(1)$ (resp. $\|\boldsymbol{\Pi}_t^*\|_\infty =$

$O_P(1)$) using that $(\mathbf{I}_d - \mathbf{W}_t^*)$ is strictly diagonally dominant. Then by (B.4), we have on the set $\mathcal{M}$ that

$$\begin{aligned}
\|I_3\|_1 &\leq \frac{r^{1/2}}{d^2 u^2} \cdot \|\mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}^\top \mathbf{V}_{\mathbf{I}_d, v}\|_1 \cdot \left\| T^{-1}(\mathbf{B}^\nu)^\top \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes} \mathbf{X} \right\|_1 \cdot \|\boldsymbol{\beta}^*\|_1 \\
&= O\left(\frac{1}{d}\right) \left\| \frac{1}{T} \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) (\mathbf{B}^\nu)^\top \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes} \mathbf{X} \right\|_1 \\
&= O\left(\frac{1}{d}\right) \left\| \frac{1}{T} \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t^\top - \bar{\mathbf{B}}^\top) \mathbf{W}_j \mathbf{\Pi}_t^* \mathbf{X}_t \right\|_1 \\
&= O\left(\frac{1}{d}\right) \left\{ \max_{q \in [r]} \max_{s \in [v]} \left| \frac{1}{T} \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \sum_{t=1}^T \sum_{i=1}^d z_{j,k,t} \mathbf{W}_{j,\cdot i}^\top (\mathbf{B}_{t,\cdot s} - \bar{\mathbf{B}}_{\cdot s}) \mathbf{X}_{t,\cdot q}^\top \mathbf{\Pi}_{t,i}^* \right| \right\} \\
&= O\left(\frac{1}{d}\right) \left(\|\phi^* - \tilde{\phi}\|_1 \max_{q \in [r], s \in [v], j \in [p]} \max_{k \in [l_j] \cup \{0\}} \max_{m, n \in [d]} \left| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (B_{t,ms} - \bar{B}_{ms}) X_{t,nq} \right| \sum_{i=1}^d \|\mathbf{W}_{j,\cdot i}\|_1 \|\mathbf{\Pi}_{t,i}^*\|_1 \right) \\
&= O_p\left(\|\phi^* - \tilde{\phi}\|_1 [c_T + 1 + c_T(c_T + \mu_{b,\max})]\right) = O_p\left(\|\phi^* - \tilde{\phi}\|_1\right),
\end{aligned} \tag{B.9}$$

where the first equality used $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$ in $\mathcal{M}$, the second last used $\mathcal{A}_4, \mathcal{A}_8, \mathcal{A}_9$ in $\mathcal{M}$ and Assumptions (R1) and (R3). Similarly, for I_4 on the set $\mathcal{M}$,

$$\begin{aligned}
\|I_4\|_1 &\leq \frac{r^{1/2}}{d^2 u^2} \cdot \|\mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}^\top \mathbf{V}_{\mathbf{I}_d, v}\|_1 \cdot \left\| T^{-1}(\mathbf{B}^\nu)^\top \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes} \boldsymbol{\epsilon}^\nu \right\|_1 \\
&= O\left(\frac{1}{d}\right) \left\{ \max_{s \in [v]} \left| \frac{1}{T} \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \sum_{t=1}^T \sum_{i=1}^d z_{j,k,t} \mathbf{W}_{j,\cdot i}^\top (\mathbf{B}_{t,\cdot s} - \bar{\mathbf{B}}_{\cdot s}) \boldsymbol{\epsilon}_t^\top \mathbf{\Pi}_{t,i}^* \right| \right\} \\
&= O\left(\frac{1}{d}\right) \left\{ \|\phi^* - \tilde{\phi}\|_1 \max_{s \in [v]} \max_{j \in [p]} \max_{k \in [l_j] \cup \{0\}} \max_{m, q \in [d]} \left| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (B_{t,ms} - \bar{B}_{ms}) \epsilon_{t,q} \right| \sum_{i=1}^d \|\mathbf{W}_{j,\cdot i}\|_1 \|\mathbf{\Pi}_{t,i}^*\|_1 \right\} \\
&= O_p\left(\|\phi^* - \tilde{\phi}\|_1 [c_T + c_T(c_T + \mu_{b,\max})]\right) = O_p\left(c_T \|\phi^* - \tilde{\phi}\|_1\right),
\end{aligned} \tag{B.10}$$

where the first equality used $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$ in $\mathcal{M}$, the second last used $\mathcal{A}_4, \mathcal{A}_{10}, \mathcal{A}_{11}$ in $\mathcal{M}$ and Assumptions

(R1) and (R3). For I_5 we also have on the set $\mathcal{M}$,

$$\begin{aligned}
\|I_5\|_1 &\leq \frac{r^{1/2}}{d^2 u^2} \cdot \|\mathbf{V}_{\mathbf{I}_d, r}^\top \mathbf{U}^\top \mathbf{V}_{\mathbf{I}_d, v}\|_1 \cdot \|T^{-1}(\mathbf{B}^\nu)^\top \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \mathbf{W}_j^\otimes \mathbf{D}_{z_{j,k}} \mathbf{\Pi}^{*\otimes} (\mathbf{1}_T \otimes \boldsymbol{\mu}^*)\|_1 \\
&= O\left(\frac{1}{d}\right) \left\{ \max_{s \in [v]} \left| \frac{1}{T} \sum_{j=1}^p \sum_{k=0}^{l_j} (\phi_{j,k}^* - \tilde{\phi}_{j,k}) \sum_{t=1}^T \sum_{i=1}^d z_{j,k,t} \mathbf{W}_{j,i}^\top (\mathbf{B}_{t,s} - \bar{\mathbf{B}}_s) \boldsymbol{\mu}^{*\top} \mathbf{\Pi}_{t,i}^* \right| \right\} \\
&= O\left(\frac{1}{d}\right) \left\{ \|\phi^* - \tilde{\phi}\|_1 \max_{s \in [v]} \max_{j \in [p]} \max_{k \in [l_j] \cup \{0\}} \max_{m \in [d]} \left| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (B_{t,ms} - \bar{B}_{ms}) \right| \|\boldsymbol{\mu}^*\|_{\max} \sum_{i=1}^d \|\mathbf{W}_{j,i}\|_1 \|\mathbf{\Pi}_{t,i}^*\|_1 \right\} \\
&= O_p\left(\|\phi^* - \tilde{\phi}\|_1 [c_T + (z_{\max} c_T \vee c_T)]\right) = O_p\left(c_T \|\phi^* - \tilde{\phi}\|_1\right),
\end{aligned} \tag{B.11}$$

where the first equality used $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$ in $\mathcal{M}$, the second last used $\mathcal{A}_4, \mathcal{A}_{12}, \mathcal{A}_{13}$ in $\mathcal{M}$ and Assumptions (R1), (R3), and $\mathbb{E}(z_{j,k,t} \mathbf{B}_t) = \mathbf{0}$ from (M2') if (M2) is not satisfied.

From (B.5), (B.7), (B.9) and (B.10), combining with Lemma 2, we have

$$\|\beta(\tilde{\phi}) - \beta^*\|_1 \leq \sum_{j=1}^5 \|I_j\|_1 = O_p\left(\|\tilde{\phi} - \phi^*\|_1 + c_T d^{-\frac{1}{2} + \frac{1}{2w}}\right). \tag{B.12}$$

For the remaining of the proof of 1, we work on the rate for $\|\tilde{\phi} - \phi^*\|_1$. From (2.16), we have

$$\begin{aligned}
\tilde{\phi} &= [(\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)]^{-1} (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{y} - \Xi \mathbf{y}^\nu) \\
&= [(\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)]^{-1} (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{X}_{\beta^*} \text{vec}(\mathbf{I}_d) + \mathbf{B}^\top \boldsymbol{\epsilon} - \Xi \mathbf{y}^\nu) \\
&\quad + [(\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)]^{-1} (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\Xi \mathbf{Y}_W) \phi^* + \phi^* \\
&= \phi^* + [(\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)]^{-1} (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top \mathbf{B}^\top \boldsymbol{\epsilon} \\
&\quad - T^{-1/2} d^{-a/2} \cdot [(\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)]^{-1} (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top \\
&\quad \cdot \text{vec} \left\{ \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma}(\boldsymbol{\epsilon}^\nu)^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X} (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}_t^\top \right\},
\end{aligned}$$

where the second equality used (2.11), and the third used the fact that $\mathbf{B}^\top \mathbf{X}_{\beta(\phi)} \text{vec}(\mathbf{I}_d) = \Xi \mathbf{y}^\nu - \Xi \mathbf{Y}_W \phi$ from (A.1) and $\beta(\phi^*) = \beta^* + (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu$. Thus, we may decompose

$$\begin{aligned}
\tilde{\phi} - \phi^* &= D_1 - D_2, \text{ where} \\
D_1 &= [(\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)]^{-1} (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top \mathbf{B}^\top \boldsymbol{\epsilon}, \\
D_2 &= T^{-1/2} d^{-a/2} \cdot [(\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)]^{-1} (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top \\
&\quad \cdot \left\{ \sum_{t=1}^T \left(\mathbf{I}_d \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \right) \mathbf{X}_t \right\} (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu.
\end{aligned}$$

To bound the above, recall first the following definitions (from the statement of Theorem 2) for $j \in [p], k \in [l_j] \cup \{0\}$,

$$\begin{aligned}\mathbf{U}_{\mathbf{x},j,k} &:= \frac{1}{T} \sum_{t=1}^T z_{j,k,t} \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \mathbf{x}_t^\top, \quad \mathbf{U}_{\boldsymbol{\mu},j,k} := \frac{1}{T} \sum_{t=1}^T z_{j,k,t} \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\mu}^{*\top}, \\ \mathbf{U}_{\boldsymbol{\epsilon},j,k} &:= \frac{1}{T} \sum_{t=1}^T z_{j,k,t} \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\epsilon}_t^\top.\end{aligned}$$

Consider $\Xi \mathbf{Y}_W$, from its definition we have,

$$\begin{aligned}\Xi \mathbf{Y}_W &= T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \right) [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \\ &\quad \cdot (\mathbf{W}_1^\otimes \mathbf{y}_{1,0}^\nu, \dots, \mathbf{W}_1^\otimes \mathbf{y}_{1,l_1}^\nu, \dots, \mathbf{W}_p^\otimes \mathbf{y}_{p,0}^\nu, \dots, \mathbf{W}_p^\otimes \mathbf{y}_{p,l_p}^\nu) \\ &= T^{-1/2} d^{-a/2} \left(\sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \right) [\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X}]^{-1} \mathbf{X}^\top \mathbf{B}^\nu \\ &\quad \cdot \left\{ \sum_{t=1}^T z_{1,0,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_1 \mathbf{y}_t, \dots, \sum_{t=1}^T z_{p,l_p,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_p \mathbf{y}_t \right\}\end{aligned}$$

From (B.8), we have $\mathbf{y}_t = \boldsymbol{\Pi}_t^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}_t^* \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\Pi}_t^* \boldsymbol{\epsilon}_t$. It hence holds for any $j \in [p], k \in [l_j] \cup \{0\}$ by Lemma 3 that

$$\begin{aligned}\frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_j \mathbf{y}_t &= \frac{1}{T} \sum_{t=1}^T z_{j,k,t} \mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}})) (\boldsymbol{\Pi}_t^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}_t^* \mathbf{X}_t \boldsymbol{\beta}^* + \boldsymbol{\Pi}_t^* \boldsymbol{\epsilon}_t) \\ &= \mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \mathbf{U}_{\mathbf{x},j,k}) \mathbf{V}_{\boldsymbol{\Pi}_t^*, r} \boldsymbol{\beta}^* + \mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\mu},j,k}) \text{vec}(\boldsymbol{\Pi}_t^{*\top}) + \mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\epsilon},j,k}) \text{vec}(\boldsymbol{\Pi}_t^{*\top}).\end{aligned}\tag{B.13}$$

Consider also $\mathbf{B}^\top \mathbf{V}$, we have

$$\begin{aligned}\mathbf{B}^\top \mathbf{V} &= T^{-1/2} d^{-a/2} \left(\mathbf{I}_d \otimes \{(\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})(\mathbf{I}_T \otimes \boldsymbol{\gamma})\} \right) \\ &\quad \cdot \left\{ \left[\mathbf{I}_d \otimes (z_{1,0,1} \mathbf{y}_1, \dots, z_{1,0,T} \mathbf{y}_T)^\top \right] \text{vec}(\mathbf{W}_1^\top), \dots, \left[\mathbf{I}_d \otimes (z_{1,l_1,1} \mathbf{y}_1, \dots, z_{1,l_1,T} \mathbf{y}_T)^\top \right] \text{vec}(\mathbf{W}_1^\top), \right. \\ &\quad \left. \dots, \left[\mathbf{I}_d \otimes (z_{p,0,1} \mathbf{y}_1, \dots, z_{p,0,T} \mathbf{y}_T)^\top \right] \text{vec}(\mathbf{W}_p^\top), \dots, \left[\mathbf{I}_d \otimes (z_{p,l_p,1} \mathbf{y}_1, \dots, z_{p,l_p,T} \mathbf{y}_T)^\top \right] \text{vec}(\mathbf{W}_p^\top) \right\} \\ &= T^{-1/2} d^{-a/2} \left\{ \left[\mathbf{I}_d \otimes (\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})(\mathbf{I}_T \otimes \boldsymbol{\gamma})(z_{1,0,1} \mathbf{y}_1, \dots, z_{1,0,T} \mathbf{y}_T)^\top \right] \text{vec}(\mathbf{W}_1^\top), \right. \\ &\quad \left. \dots, \left[\mathbf{I}_d \otimes (\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})(\mathbf{I}_T \otimes \boldsymbol{\gamma})(z_{p,l_p,1} \mathbf{y}_1, \dots, z_{p,l_p,T} \mathbf{y}_T)^\top \right] \text{vec}(\mathbf{W}_p^\top) \right\} \\ &= T^{-1/2} d^{-a/2} \left\{ \left[\mathbf{I}_d \otimes \sum_{t=1}^T z_{1,0,t} (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \mathbf{y}_t^\top \right] \text{vec}(\mathbf{W}_1^\top), \dots, \left[\mathbf{I}_d \otimes \sum_{t=1}^T z_{p,l_p,t} (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \mathbf{y}_t^\top \right] \text{vec}(\mathbf{W}_p^\top) \right\}.\end{aligned}$$

Similar to $\Xi \mathbf{Y}_W$, for $j \in [p], k \in [l_p] \cup \{0\}$ we have

$$\begin{aligned}
& \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \mathbf{y}_t^\top = \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma (\boldsymbol{\mu}^{*\top} \boldsymbol{\Pi}_t^{*\top} + \boldsymbol{\beta}^{*\top} \mathbf{X}_t^\top \boldsymbol{\Pi}_t^{*\top} + \boldsymbol{\epsilon}_t^\top \boldsymbol{\Pi}_t^{*\top}) \\
&= \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\gamma^\top \otimes \mathbf{I}_d) \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\mu}^{*\top} \boldsymbol{\Pi}_t^{*\top} + \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\gamma^\top \otimes \mathbf{I}_d) \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \mathbf{x}_t^\top (\boldsymbol{\beta}^* \otimes \mathbf{I}_d) \boldsymbol{\Pi}_t^{*\top} \\
&\quad + \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\gamma^\top \otimes \mathbf{I}_d) \text{vec}(\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\epsilon}_t^\top \boldsymbol{\Pi}_t^{*\top} \\
&= (\gamma^\top \otimes \mathbf{I}_d) \mathbf{U}_{\boldsymbol{\mu},j,k} \boldsymbol{\Pi}_t^{*\top} + (\gamma^\top \otimes \mathbf{I}_d) \mathbf{U}_{\mathbf{x},j,k} (\boldsymbol{\beta}^* \otimes \mathbf{I}_d) \boldsymbol{\Pi}_t^{*\top} + (\gamma^\top \otimes \mathbf{I}_d) \mathbf{U}_{\boldsymbol{\epsilon},j,k} \boldsymbol{\Pi}_t^{*\top}.
\end{aligned} \tag{B.14}$$

With (B.13) and (B.14), recall the following definitions in the statement of Theorem 2,

$$\begin{aligned}
\mathbf{H}_{10} &= \left\{ \left(\mathbf{I}_d \otimes (\gamma^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\mathbf{x},1,0} (\boldsymbol{\beta}^* \otimes \mathbf{I}_d) \boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_1^\top) + \left(\mathbf{I}_d \otimes (\gamma^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},1,0} \boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_1^\top), \right. \\
&\quad \left. \dots, \left(\mathbf{I}_d \otimes (\gamma^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\mathbf{x},p,l_p} (\boldsymbol{\beta}^* \otimes \mathbf{I}_d) \boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_p^\top) + \left(\mathbf{I}_d \otimes (\gamma^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},p,l_p} \boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_p^\top) \right\}, \\
\mathbf{H}_{20} &= \mathbb{E}(\mathbf{X}_t \otimes \mathbf{B}_t \gamma) [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} \mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \\
&\quad \cdot \left\{ \mathbf{V}_{\mathbf{W}_1^\top, v}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\mathbf{x},1,0}) \mathbf{V}_{\boldsymbol{\Pi}_t^{*,r}}] \boldsymbol{\beta}^* + \mathbf{V}_{\mathbf{W}_1^\top, v}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\mu},1,0}) \text{vec}(\boldsymbol{\Pi}_t^{*\top})], \right. \\
&\quad \left. \dots, \mathbf{V}_{\mathbf{W}_p^\top, v}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\mathbf{x},p,l_p}) \mathbf{V}_{\boldsymbol{\Pi}_t^{*,r}}] \boldsymbol{\beta}^* + \mathbf{V}_{\mathbf{W}_p^\top, v}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\mu},p,l_p}) \text{vec}(\boldsymbol{\Pi}_t^{*\top})] \right\},
\end{aligned}$$

which are essentially $T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V}$ and $T^{-1/2} d^{a/2} \Xi \mathbf{Y}_W$ at the population level, respectively.

For the rest of the proof of Theorem 1, we find the rate of D_2 , followed by constructing the asymptotic normality of the dominating term in the expansion of D_1 . For D_2 , we further decompose $D_2 = F_1 + F_2 - F_3$ where

$$\begin{aligned}
F_1 &= [(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]^{-1} \left\{ (\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10}) \right. \\
&\quad \left. - T^{-1} d^a (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W) \right\} D_2, \\
F_2 &= [(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]^{-1} \left[(T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V} - \mathbf{H}_{10}) - (T^{-1/2} d^{a/2} \Xi \mathbf{Y}_W - \mathbf{H}_{20}) \right]^\top \\
&\quad \cdot \left\{ \frac{1}{T} \sum_{t=1}^T \left(\mathbf{I}_d \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \right) \mathbf{X}_t \right\} (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu, \\
F_3 &= [(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]^{-1} (\mathbf{H}_{20} - \mathbf{H}_{10})^\top \\
&\quad \cdot \left\{ \frac{1}{T} \sum_{t=1}^T \left(\mathbf{I}_d \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \gamma \right) \mathbf{X}_t \right\} (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu.
\end{aligned}$$

To bound the L1 norm of F_1 to F_3 , first observe that by Assumptions (R1) and (R4) we have

$$\begin{aligned}
\sigma_L^2(\mathbf{H}_{10}) &\geq \sigma_L^2 \left(\left\{ \left(\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\mathbf{x},1,0}(\boldsymbol{\beta}^* \otimes \mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_1^\top), \right. \right. \\
&\quad \left. \left. \dots, \left(\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\mathbf{x},p,l_p}(\boldsymbol{\beta}^* \otimes \mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}) \right) \text{vec}(\mathbf{W}_p^\top) \right\} \right) \\
&\geq \sigma_L^2(\mathbf{D}_W) \cdot \sigma_{d^2}^2 \left(\left\{ \left(\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\mathbf{x},1,0}(\boldsymbol{\beta}^* \otimes \mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}) \right), \right. \right. \\
&\quad \left. \left. \dots, \left(\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) \mathbb{E}(\mathbf{U}_{\mathbf{x},p,l_p}(\boldsymbol{\beta}^* \otimes \mathbf{I}_d)\boldsymbol{\Pi}_t^{*\top}) \right) \right\} \right) \geq Cd \cdot d^a = Cd^{1+a},
\end{aligned}$$

where $C > 0$ is a generic constant. Similarly, by Assumptions (R1), (R3), (R5) and (R6),

$$\begin{aligned}
\sigma_L(\mathbf{H}_{20}) &\geq \sigma_r \left(\mathbb{E}(\mathbf{X}_t \otimes \mathbf{B}_t \boldsymbol{\gamma}) \right) \sigma_r \left(\left[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t) \right]^{-1} \right) \sigma_r \left(\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \right) \\
&\quad \cdot \sigma_{\min} \left(\left\{ \mathbf{V}_{\mathbf{W}_1^\top, r}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\mathbf{x},1,0}) \mathbf{V}_{\boldsymbol{\Pi}_t^*, r}] \boldsymbol{\beta}^*, \dots, \mathbf{V}_{\mathbf{W}_p^\top, r}^\top \mathbb{E}[(\mathbf{I}_d \otimes \mathbf{U}_{\mathbf{x},p,l_p}) \mathbf{V}_{\boldsymbol{\Pi}_t^*, r}] \boldsymbol{\beta}^* \right\} \right) \\
&\geq \frac{Cd^{1+a} \cdot d}{\lambda_{\max}[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]} \cdot \sigma_v[\mathbb{E}(\ddot{\mathbf{G}})] \cdot \sigma_L(\mathbf{D}_W) \geq \frac{Cd^{1+a} \cdot d \cdot d^{1/2}}{\lambda_{\max}[\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]} \geq Cd^{1/2+a},
\end{aligned}$$

with some arbitrary constant $C > 0$. Notice $\mathbf{H}_{20}$ has the smallest singular value of order larger than that for $\mathbf{H}_{10}$, so $\sigma_L^2(\mathbf{H}_{20} - \mathbf{H}_{10}) \geq Cd^{1+a}$ for some $C > 0$. Thus,

$$\left\| [(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]^{-1} \right\|_1 \leq \frac{L^{1/2}}{\lambda_{\min}[(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]} \leq \frac{L^{1/2}}{Cd^{1+a}}. \quad (\text{B.15})$$

Consider F_1 first and we hence have on $\mathcal{M}$,

$$\begin{aligned}
\|F_1\|_1 &\leq \frac{L^{1/2} \cdot L}{Cd^{1+a}} \left\{ \left\| (\mathbf{H}_{20} - T^{-1/2} d^{a/2} \boldsymbol{\Xi} \mathbf{Y}_W) + (T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V} - \mathbf{H}_{10}) \right\|_{\max} \cdot \|\mathbf{H}_{20} - \mathbf{H}_{10}\|_1 \right. \\
&\quad \left. + \left\| T^{-1/2} d^{a/2} (\mathbf{B}^\top \mathbf{V} - \boldsymbol{\Xi} \mathbf{Y}_W) \right\|_{\max} \left\| (\mathbf{H}_{20} - T^{-1/2} d^{a/2} \boldsymbol{\Xi} \mathbf{Y}_W) + (T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V} - \mathbf{H}_{10}) \right\|_1 \right\} \|D_2\|_1 \\
&= O \left\{ \frac{L^{3/2}}{d^{1+a}} [c_T \cdot d^2 + 1 \cdot (c_T d^2 + c_T d^2)] \right\} \|D_2\|_1 = O \left(c_T L^{3/2} d^{1-a} \|D_2\|_1 \right),
\end{aligned} \quad (\text{B.16})$$

where the last line used the following rates to be shown later,

$$\left\| \mathbf{H}_{20} - T^{-1/2} d^{a/2} \boldsymbol{\Xi} \mathbf{Y}_W \right\|_{\max} = O_P(c_T), \quad (\text{B.17})$$

$$\left\| \mathbf{H}_{10} - T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V} \right\|_{\max} = O_P(c_T). \quad (\text{B.18})$$

For neat presentation, we define the following terms whose norms will be bounded and involved in

(B.17) and later,

$$\begin{aligned}
\mathbf{A}_1 &:= \frac{1}{T} \sum_{t=1}^T \mathbf{X}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma}, & \mathbf{A}_1^0 &:= \mathbb{E}(\mathbf{X}_t \otimes \mathbf{B}_t \boldsymbol{\gamma}), \\
\mathbf{A}_2 &:= \left(\frac{1}{T} \mathbf{X}^\top \mathbf{B}^\nu \frac{1}{T} (\mathbf{B}^\nu)^\top \mathbf{X} \right)^{-1}, & \mathbf{A}_2^0 &:= [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1}, \\
\mathbf{A}_3 &:= \frac{1}{T} \mathbf{X}^\top \mathbf{B}^\nu, & \mathbf{A}_3^0 &:= \mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t), \\
\mathbf{A}_{4,j,k} &:= \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_j (\boldsymbol{\Pi}_t^* \boldsymbol{\mu}^* + \boldsymbol{\Pi}_t^* \mathbf{X}_t \boldsymbol{\beta}^*), \\
\mathbf{A}_{4,j,k}^0 &:= \mathbb{E}(\mathbf{A}_{4,j,k}), & \mathbf{A}_{5,j,k} &:= \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_j \boldsymbol{\Pi}_t^* \boldsymbol{\epsilon}_t.
\end{aligned}$$

On $\mathcal{M}$, we immediately have from Lemma 2 (using $\mathcal{A}_1, \mathcal{A}_4, \mathcal{A}_6$),

$$\|\mathbf{A}_1 - \mathbf{A}_1^0\|_{\max} = O\left(c_T + c_T(c_T + \mu_{b,\max})\right) = O(c_T), \quad (\text{B.19})$$

which also gives $\|\mathbf{A}_1\|_{\max} \leq \|\mathbf{A}_1^0\|_{\max} + \|\mathbf{A}_1 - \mathbf{A}_1^0\|_{\max} = O(1 + c_T) = O(1)$. Hence with Assumptions (R5) and (R10), we also have on $\mathcal{M}$ that

$$\|\mathbf{A}_1\|_1 \leq \|\mathbf{A}_1^0\|_1 + \|\mathbf{A}_1 - \mathbf{A}_1^0\|_1 = O(d^{1+a} + c_T d^2) = O(d^{1+a}). \quad (\text{B.20})$$

Similarly, with Lemma 2 (using $\mathcal{A}_1$) and Assumption (R3) we have on $\mathcal{M}$,

$$\|\mathbf{A}_3 - \mathbf{A}_3^0\|_1 = O(c_T d), \quad \|\mathbf{A}_3^0\|_1 = O(d), \quad \|\mathbf{A}_3\|_1 \leq \|\mathbf{A}_3 - \mathbf{A}_3^0\|_1 + \|\mathbf{A}_3^0\|_1 = O(d). \quad (\text{B.21})$$

Rewrite $\mathbf{A}_2^0 = (\mathbf{A}_3^0 \mathbf{A}_3^{0\top})^{-1}$ and $\mathbf{A}_2 = (\mathbf{A}_3 \mathbf{A}_3^\top)^{-1}$, by Assumption (R3),

$$\|\mathbf{A}_2^0\|_1 \leq \frac{r^{1/2}}{\lambda_{\min}(\mathbf{A}_3^0 \mathbf{A}_3^{0\top})} = O(d^{-2}). \quad (\text{B.22})$$

Moreover, from (B.21) we have on $\mathcal{M}$

$$\begin{aligned}
\|(\mathbf{A}_2^0)^{-1} - \mathbf{A}_2^{-1}\|_1 &\leq \|\mathbf{A}_3^0 \mathbf{A}_3^{0\top} - \mathbf{A}_3 \mathbf{A}_3^\top\|_1 \leq \|\mathbf{A}_3^0 - \mathbf{A}_3\|_1 \|\mathbf{A}_3^{0\top}\|_1 + \|\mathbf{A}_3\|_1 \|\mathbf{A}_3^{0\top} - \mathbf{A}_3^\top\|_1 \\
&= O(c_T d \cdot d + d \cdot c_T d) = O(c_T d^2).
\end{aligned}$$

Thus, rewriting $\mathbf{A}_2 - \mathbf{A}_2^0 = (\mathbf{A}_2 - \mathbf{A}_2^0)[(\mathbf{A}_2^0)^{-1} - \mathbf{A}_2^{-1}]\mathbf{A}_2^0 + \mathbf{A}_2^0[(\mathbf{A}_2^0)^{-1} - \mathbf{A}_2^{-1}]\mathbf{A}_2^0$ and we have on $\mathcal{M}$,

$$\|\mathbf{A}_2 - \mathbf{A}_2^0\|_1 = o\left(\|\mathbf{A}_2 - \mathbf{A}_2^0\|_1\right) + O(c_T d^2 \cdot d^{-4}) = O(c_T d^{-2}). \quad (\text{B.23})$$

Consider now $\mathbf{A}_{4,j,k}^0$ for any $j \in [p], k \in [l_j] \cup \{0\}$. First, we have on $\mathcal{M}$,

$$\begin{aligned} & \left\| \mathbb{E} \left(\frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_j \boldsymbol{\Pi}_t^* \boldsymbol{\mu}^* \right) \right\|_1 \leq \left\| \mathbb{E} \left(\frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \right) \right\|_{\max} \|\mathbf{W}_j\|_1 \|\boldsymbol{\Pi}_t^* \boldsymbol{\mu}^*\|_1 \\ & = O(d \cdot \|\boldsymbol{\Pi}_t^*\|_\infty \|\boldsymbol{\mu}^*\|_{\max}) = O(d), \end{aligned}$$

where the last equality used Assumption (R1). Thus on $\mathcal{M}$,

$$\begin{aligned} \|\mathbf{A}_{4,j,k}^0\|_1 & \leq \|\mathbf{V}_{\mathbf{W}_j^\top, v}^\top [\mathbf{I}_d \otimes \mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})] \mathbf{V}_{\boldsymbol{\Pi}_t^*, r} \boldsymbol{\beta}^*\|_1 + \left\| \mathbb{E} \left(\frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \mathbf{W}_j \boldsymbol{\Pi}_t^* \boldsymbol{\mu}^* \right) \right\|_1 \\ & \leq \|\mathbf{V}_{\mathbf{W}_j^\top, v}^\top [\mathbf{I}_d \otimes \mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})]\|_{\max} \|\mathbf{V}_{\boldsymbol{\Pi}_t^*, r}\|_1 \|\boldsymbol{\beta}^*\|_1 + O(d) \\ & \leq \max_{i,n \in [d]} \max_{m \in [v]} \max_{q \in [r]} \mathbf{W}_{j,i}^\top \mathbb{E} \left\{ \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})_{\cdot m} X_{t,nq} \right\} \|\mathbf{V}_{\boldsymbol{\Pi}_t^*, r}\|_1 \|\boldsymbol{\beta}^*\|_1 + O(d) = O(d), \end{aligned} \tag{B.24}$$

where the last equality used Assumption (R1). Furthermore, we have on $\mathcal{M}$ that

$$\begin{aligned} & \|\mathbf{A}_{4,j,k} - \mathbf{A}_{4,j,k}^0\|_1 \\ & \leq \left\| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top - \mathbb{E} \left\{ \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \right\} \right\|_{\max} \|\mathbf{W}_j\|_1 \|\boldsymbol{\Pi}_t^* \boldsymbol{\mu}^*\|_1 \\ & \quad + \|\mathbf{V}_{\mathbf{W}_j^\top, v}^\top \{\mathbf{I}_d \otimes [\mathbf{U}_{\mathbf{x},j,k} - \mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})]\} \mathbf{V}_{\boldsymbol{\Pi}_t^*, r} \boldsymbol{\beta}^*\|_1 \\ & = O\left([c_T + 1 \cdot c_T + 1 \cdot c_T + 1 \cdot (c_T \vee 0)] d\right) + O\left(\|\mathbf{V}_{\boldsymbol{\Pi}_t^*, r}\|_1 \|\boldsymbol{\beta}^*\|_1\right) \\ & \quad \cdot \|\mathbf{W}_j^\top\|_1 \max_{n,m \in [d]} \max_{s \in [v]} \max_{q \in [r]} \left| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})_{ms} X_{t,nq} - \mathbb{E} \left(\frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})_{ms} X_{t,nq} \right) \right| = O(c_T d), \end{aligned} \tag{B.25}$$

where the second last equality used Assumption (R1) and $\mathcal{A}_4, \mathcal{A}_{12}, \mathcal{A}_{13}$, while the last used $\mathcal{A}_4, \mathcal{A}_8, \mathcal{A}_9$.

In particular, we used the following as an immediate result of $\mathcal{A}_{13}$,

$$\max_{m \in [p]} \max_{n \in [l_m] \cup \{0\}} \left| \frac{1}{T} \sum_{t=1}^T z_{m,n,t} - \mathbb{E} \left\{ \frac{1}{T} \sum_{t=1}^T z_{m,n,t} \right\} \right| \leq c_T \vee 0.$$

Lastly for any $j \in [p], k \in [l_j] \cup \{0\}$, similar to (B.10), we have on $\mathcal{M}$ that

$$\begin{aligned} \|\mathbf{A}_{5,j,k}\|_1 & = \|\mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\epsilon},j,k}) \mathbf{V}_{\boldsymbol{\Pi}_t^*, r}\|_1 \leq \|\mathbf{V}_{\mathbf{W}_j^\top, v}^\top (\mathbf{I}_d \otimes \mathbf{U}_{\boldsymbol{\epsilon},j,k})\|_{\max} \|\mathbf{V}_{\boldsymbol{\Pi}_t^*, r}\|_1 \\ & \leq \|\mathbf{W}_j^\top\|_1 \max_{m,n \in [d]} \max_{s \in [v]} \left| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})_{ms} \epsilon_{t,n} \right| \cdot \|\mathbf{V}_{\boldsymbol{\Pi}_t^*, r}\|_1 = O(c_T d). \end{aligned} \tag{B.26}$$

Consider now (B.17), we have

$$\begin{aligned}
& \left\| \mathbf{H}_{20} - T^{-1/2} d^{a/2} \mathbf{\Xi} \mathbf{Y}_W \right\|_{\max} = \max_{j,k} \left\| \mathbf{A}_1 \mathbf{A}_2 \mathbf{A}_3 (\mathbf{A}_{4,j,k} + \mathbf{A}_{5,j,k}) - \mathbf{A}_1^0 \mathbf{A}_2^0 \mathbf{A}_3^0 \mathbf{A}_{4,j,k}^0 \right\|_{\max} \\
& \leq \max_{j,k} \left\| \mathbf{A}_1 \right\|_{\max} \left\| \mathbf{A}_2 \right\|_1 \left\| \mathbf{A}_3 \right\|_1 \left\| \mathbf{A}_{5,j,k} \right\|_1 \\
& \quad + \max_{j,k} \left\{ \left\| \mathbf{A}_1 \right\|_{\max} \left\| \mathbf{A}_2 \mathbf{A}_3 \mathbf{A}_{4,j,k} - \mathbf{A}_2^0 \mathbf{A}_3^0 \mathbf{A}_{4,j,k}^0 \right\|_1 + \left\| \mathbf{A}_1 - \mathbf{A}_1^0 \right\|_{\max} \left\| \mathbf{A}_2^0 \mathbf{A}_3^0 \mathbf{A}_{4,j,k}^0 \right\|_1 \right\}, \text{ with} \\
& \max_{j,k} \left\| \mathbf{A}_2 \mathbf{A}_3 \mathbf{A}_{4,j,k} - \mathbf{A}_2^0 \mathbf{A}_3^0 \mathbf{A}_{4,j,k}^0 \right\|_1 \leq \max_{j,k} \left\{ \left\| \mathbf{A}_2 \right\|_1 \left\| \mathbf{A}_3 - \mathbf{A}_3^0 \right\|_1 \left\| \mathbf{A}_{4,j,k} \right\|_1 \right. \\
& \quad \left. + \left\| \mathbf{A}_2 \right\|_1 \left\| \mathbf{A}_3^0 \right\|_1 \left\| \mathbf{A}_{4,j,k} - \mathbf{A}_{4,j,k}^0 \right\|_1 + \left\| \mathbf{A}_2 - \mathbf{A}_2^0 \right\|_1 \left\| \mathbf{A}_3^0 \right\|_1 \left\| \mathbf{A}_{4,j,k}^0 \right\|_1 \right\}.
\end{aligned}$$

Together with all the rates from (B.19) to (B.26), we have (B.17) true on $\mathcal{M}$.

For (B.18), consider for any $j \in [p], k \in [l_j] \cup \{0\}$, we have on $\mathcal{M}$ that

$$\begin{aligned}
& \left\| \left[\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) \mathbf{U}_{\epsilon,j,k} \boldsymbol{\Pi}_t^{*\top} \right] \text{vec}(\mathbf{W}_j^\top) \right\|_{\max} = \max_{i \in [d]} \left\| (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) \mathbf{U}_{\epsilon,j,k} \boldsymbol{\Pi}_t^{*\top} \mathbf{W}_{j,i} \right\|_{\max} \\
& \leq \max_{i \in [d]} \left\| (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) \mathbf{U}_{\epsilon,j,k} \right\|_{\max} \left\| \boldsymbol{\Pi}_t^* \right\|_{\infty} \left\| \mathbf{W}_{j,i} \right\|_1 \\
& = O(1) \cdot \|\boldsymbol{\gamma}\|_1 \cdot \max_{m,n \in [d]} \max_{s \in [v]} \left| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})_{ms} \epsilon_{t,n} \right| = O(c_T),
\end{aligned} \tag{B.27}$$

where the second last equality used Assumption (R1) and the result below (B.8), and the last is similar to (B.10). In a similar way on $\mathcal{M}$,

$$\begin{aligned}
& \left\| \left[\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) (\mathbf{U}_{\boldsymbol{\mu},j,k} - \mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},j,k})) \boldsymbol{\Pi}_t^{*\top} \right] \text{vec}(\mathbf{W}_j^\top) \right\|_{\max} \\
& \leq \max_{i \in [d]} \left\| (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) (\mathbf{U}_{\boldsymbol{\mu},j,k} - \mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},j,k})) \right\|_{\max} \left\| \boldsymbol{\Pi}_t^* \right\|_{\infty} \left\| \mathbf{W}_{j,i} \right\|_1 \\
& = O\left(\|\boldsymbol{\gamma}\|_1 \|\boldsymbol{\mu}^*\|_{\max}\right) \cdot \left\| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top - \mathbb{E} \left\{ \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})^\top \right\} \right\|_{\max} = O(c_T),
\end{aligned} \tag{B.28}$$

with the last line similar to (B.25) which is also involved in the last line of the following,

$$\begin{aligned}
& \left\| \left[\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) (\mathbf{U}_{\mathbf{x},j,k} - \mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})) (\boldsymbol{\beta}^* \otimes \mathbf{I}_d) \boldsymbol{\Pi}_t^{*\top} \right] \text{vec}(\mathbf{W}_j^\top) \right\|_{\max} \\
& \leq \max_{i \in [d]} \left\| (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) (\mathbf{U}_{\mathbf{x},j,k} - \mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})) \right\|_{\max} \left\| \boldsymbol{\beta}^* \otimes \mathbf{I}_d \right\|_1 \left\| \boldsymbol{\Pi}_t^* \right\|_{\infty} \left\| \mathbf{W}_{j,i} \right\|_1 \\
& = O\left(\|\boldsymbol{\gamma}\|_1\right) \cdot \max_{m,n \in [d]} \max_{s \in [v]} \max_{q \in [r]} \left| \frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})_{ms} X_{t,nq} - \mathbb{E} \left(\frac{1}{T} \sum_{t=1}^T z_{j,k,t} (\mathbf{B}_t - \bar{\mathbf{B}})_{ms} X_{t,nq} \right) \right| = O(c_T).
\end{aligned} \tag{B.29}$$

Combining (B.27), (B.28) and (B.29), we have (B.18) true by the following decomposition,

$$\begin{aligned}
& \left\| \mathbf{H}_{10} - T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V} \right\|_{\max} \\
& \leq \max_{j,k} \left\| \left[\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) \mathbf{U}_{\boldsymbol{\epsilon},j,k} \boldsymbol{\Pi}_t^{*\top} \right] \text{vec}(\mathbf{W}_j^\top) \right\|_{\max} \\
& \quad + \max_{j,k} \left\| \left[\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) (\mathbf{U}_{\boldsymbol{\mu},j,k} - \mathbb{E}(\mathbf{U}_{\boldsymbol{\mu},j,k})) \boldsymbol{\Pi}_t^{*\top} \right] \text{vec}(\mathbf{W}_j^\top) \right\|_{\max} \\
& \quad + \max_{j,k} \left\| \left[\mathbf{I}_d \otimes (\boldsymbol{\gamma}^\top \otimes \mathbf{I}_d) (\mathbf{U}_{\mathbf{x},j,k} - \mathbb{E}(\mathbf{U}_{\mathbf{x},j,k})) (\boldsymbol{\beta}^* \otimes \mathbf{I}_d) \boldsymbol{\Pi}_t^{*\top} \right] \text{vec}(\mathbf{W}_j^\top) \right\|_{\max}.
\end{aligned}$$

Next for F_2 and F_3 , we consider first on $\mathcal{M}$,

$$\begin{aligned}
& \left\| \left\{ \frac{1}{T} \sum_{t=1}^T \left(\mathbf{I}_d \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \right) \mathbf{X}_t \right\} (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu \right\|_1 \\
& = \left\| \mathbf{A}_1 \mathbf{A}_2 \mathbf{A}_3 \frac{1}{T} \sum_{t=1}^T (\mathbf{B}_t - \bar{\mathbf{B}})^\top \boldsymbol{\epsilon}_t \right\|_1 \leq \|\mathbf{A}_1\|_1 \|\mathbf{A}_2\|_1 \|\mathbf{A}_3\|_1 \cdot v \max_{j \in [r]} \left| \frac{1}{T} \sum_{t=1}^T \sum_{q=1}^d B_{t,qj} \epsilon_{t,q} \right| \quad (\text{B.30}) \\
& = O(d^{1+a} \cdot d^{-2} \cdot d \cdot c_T d^{\frac{1}{2} + \frac{1}{2w} + a}) = O(c_T d^{\frac{1}{2} + \frac{1}{2w} + a}),
\end{aligned}$$

where the first equality used the fact that $\mathbf{X}_t = \mathbf{X}_t \otimes \mathbf{1}$, the second last used (B.20), (B.21), (B.22) and $\mathcal{A}_3$ in Lemma 2. Then for F_2 , we have on $\mathcal{M}$ that

$$\begin{aligned}
\|F_2\|_1 & \leq \frac{L^{1/2} \cdot L}{C d^{1+a}} \left(\left\| (\mathbf{H}_{20} - T^{-1/2} d^{a/2} \boldsymbol{\Xi} \mathbf{Y}_W) \right\|_{\max} + \left\| (T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V} - \mathbf{H}_{10}) \right\|_{\max} \right) \\
& \quad \cdot \left\| \left\{ \frac{1}{T} \sum_{t=1}^T \left(\mathbf{I}_d \otimes (\mathbf{B}_t - \bar{\mathbf{B}}) \boldsymbol{\gamma} \right) \mathbf{X}_t \right\} (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu \right\|_1 \quad (\text{B.31}) \\
& = O(L^{3/2} \cdot d^{-1-a} \cdot c_T \cdot c_T d^{\frac{1}{2} + \frac{1}{2w} + a}) = O(c_T^2 L^{3/2} d^{-\frac{1}{2} + \frac{1}{2w}}),
\end{aligned}$$

where the last line used (B.15), (B.17), (B.18) and (B.30). Similarly, we have $\|F_3\|_1 = O(c_T L^{3/2} d^{-\frac{1}{2} + \frac{1}{2w}})$ on $\mathcal{M}$, and hence together with $L = O(1)$, (B.16) and (B.31), it holds on $\mathcal{M}$ that

$$\|D_2\|_1 \leq \|F_1\|_1 + \|F_2\|_1 + \|F_3\|_1 = O(c_T d^{-\frac{1}{2} + \frac{1}{2w}}). \quad (\text{B.32})$$

Similar to the way we decompose D_2 , we can rewrite $D_1 = F_4 + F_5 - F_6$ where

$$\begin{aligned}
F_4 &= [(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]^{-1} \left\{ (\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10}) \right. \\
&\quad \left. - T^{-1} d^a (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W)^\top (\mathbf{B}^\top \mathbf{V} - \Xi \mathbf{Y}_W) \right\} D_1, \\
F_5 &= [(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]^{-1} \left[(T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V} - \mathbf{H}_{10}) - (T^{-1/2} d^{a/2} \Xi \mathbf{Y}_W - \mathbf{H}_{20}) \right]^\top \\
&\quad \cdot T^{-1/2} d^{a/2} \mathbf{B}^\top \boldsymbol{\epsilon}, \\
F_6 &= [(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]^{-1} (\mathbf{H}_{20} - \mathbf{H}_{10})^\top \cdot T^{-1/2} d^{a/2} \mathbf{B}^\top \boldsymbol{\epsilon}.
\end{aligned}$$

From (B.16), it is direct that $F_4 = O(c_T L^{3/2} d^{1-a} \|D_1\|_1)$ on $\mathcal{M}$. Moreover, (B.17) and (B.18) imply that F_5 has a smaller rate than that of F_6 . Given $L = O(1)$, we next construct the asymptotic normality of $\boldsymbol{\alpha}^\top F_6$ for any given non-zero $\boldsymbol{\alpha} \in \mathbb{R}^L$ with $\|\boldsymbol{\alpha}\|_1 \leq c < \infty$.

Denote by $\mathbf{R}_1 := [(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})]^{-1} (\mathbf{H}_{20} - \mathbf{H}_{10})^\top$, we have

$$\begin{aligned}
\boldsymbol{\alpha}^\top F_6 &= \frac{1}{T} \boldsymbol{\alpha}^\top \mathbf{R}_1 \left(\mathbf{I}_d \otimes \{(\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})(\mathbf{I}_T \otimes \boldsymbol{\gamma})\} \right) \text{vec}((\boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_T)^\top) \\
&= \frac{1}{T} \boldsymbol{\alpha}^\top \mathbf{R}_1 \left(\mathbf{I}_d \otimes \{(\mathbf{B}_1 - \bar{\mathbf{B}})\boldsymbol{\gamma}, \dots, (\mathbf{B}_T - \bar{\mathbf{B}})\boldsymbol{\gamma}\} \right) \sum_{t=1}^T \left\{ \boldsymbol{\epsilon}_t \otimes (\mathbb{1}_{\{j=t\}})_{j \in [T]} \right\} \\
&= \frac{1}{T} \sum_{t=1}^T \boldsymbol{\alpha}^\top \mathbf{R}_1 \left(\boldsymbol{\epsilon}_t \otimes (\mathbf{B}_t - \bar{\mathbf{B}})\boldsymbol{\gamma} \right) = \frac{1}{T} \sum_{t=1}^T \boldsymbol{\alpha}^\top \mathbf{R}_1 \left(\boldsymbol{\epsilon}_t \otimes [\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma} (1 + o_P(1)) \right),
\end{aligned}$$

where the vector $(\mathbb{1}_{\{j=t\}})_{j \in [T]} \in \mathbb{R}^T$ has value $\mathbb{1}_{\{j=t\}}$ at each j -th entry, and the last equality used $\mathcal{A}_4$ in Lemma 2. Hence to apply Theorem 3 (ii) of Wu (2011), we need to show

$$\sum_{t \geq 0} \left\| P_0 \left\{ \boldsymbol{\alpha}^\top \mathbf{R}_1 (\boldsymbol{\epsilon}_t \otimes [\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}) \right\} \right\|_2 < \infty, \tag{B.33}$$

where $P_0(\cdot) := \mathbb{E}_0(\cdot) - \mathbb{E}_{-1}(\cdot)$ and $\mathbb{E}_i(\cdot) := \mathbb{E}(\cdot \mid \sigma(\mathcal{G}_i, \mathcal{H}_i))$. Notice that

$$\begin{aligned}
&\left\| P_0 \left\{ \boldsymbol{\alpha}^\top \mathbf{R}_1 (\boldsymbol{\epsilon}_t \otimes [\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}) \right\} \right\|_2 \\
&= \left\| \boldsymbol{\alpha}^\top \mathbf{R}_1 \left\{ P_0(\boldsymbol{\epsilon}_t) \otimes \mathbb{E}_0([\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}) \right\} + \boldsymbol{\alpha}^\top \mathbf{R}_1 \left\{ \mathbb{E}_{-1}(\boldsymbol{\epsilon}_t) \otimes P_0([\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}) \right\} \right\|_2 \\
&\leq \left\{ 2\boldsymbol{\alpha}^\top \mathbf{R}_1 \left\{ \mathbb{E}(P_0(\boldsymbol{\epsilon}_t) P_0(\boldsymbol{\epsilon}_t)^\top) \otimes \mathbb{E}(\mathbb{E}_0([\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}) \mathbb{E}_0(\boldsymbol{\gamma}^\top [\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]^\top)) \right\} \mathbf{R}_1^\top \boldsymbol{\alpha} \right\}^{1/2} \\
&\quad + \left\{ 2\boldsymbol{\alpha}^\top \mathbf{R}_1 \left\{ \mathbb{E}(\mathbb{E}_{-1}(\boldsymbol{\epsilon}_t) \mathbb{E}_{-1}(\boldsymbol{\epsilon}_t)^\top) \otimes \mathbb{E}(P_0([\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]\boldsymbol{\gamma}) P_0(\boldsymbol{\gamma}^\top [\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)]^\top)) \right\} \mathbf{R}_1^\top \boldsymbol{\alpha} \right\}^{1/2} \\
&= O\left(\|\boldsymbol{\alpha}\|_1 \|\mathbf{R}_1\|_\infty\right) \cdot \left(\max_{j \in [d]} \|P_0(\boldsymbol{\epsilon}_{t,j})\|_2 \cdot \max_{j \in [d]} \text{Var}^{1/2}(\mathbf{B}_{t,j}^\top \boldsymbol{\gamma}) + \sigma_{\max} \max_{j \in [d]} \|P_0(\mathbf{B}_{t,j}^\top \boldsymbol{\gamma})\|_2 \right) \\
&= O\left(\max_{j \in [d]} \|P_0^\epsilon(\boldsymbol{\epsilon}_{t,j})\|_2 + \max_{j \in [d]} \max_{k \in [v]} \|P_0^b(B_{t,jk})\|_2 \right),
\end{aligned}$$

where the second last equality used $\text{Var}(\cdot) = \text{Var}(\mathbb{E}_i(\cdot)) + \mathbb{E}(\text{Var}_i(\cdot)) \geq \text{Var}(\mathbb{E}_i(\cdot))$, and the last used Assumption (R2) and $\|\mathbf{R}_1\|_\infty = O(1)$ which is implied from (B.15), $\|\mathbf{H}_{10}\|_1 = O(d)$ and $\|\mathbf{H}_{20}\|_1 = O(d^{1+a})$. With Assumption (R7), (B.33) is true. Therefore, with definition

$$s_1 := \boldsymbol{\alpha}^\top \mathbf{R}_1 \boldsymbol{\Sigma} \mathbf{R}_1^\top \boldsymbol{\alpha}, \quad \boldsymbol{\Sigma} := \sum_{\tau} \mathbb{E}(\boldsymbol{\epsilon}_t \boldsymbol{\epsilon}_{t+\tau}^\top) \otimes \mathbb{E}[(\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)) \boldsymbol{\gamma} \boldsymbol{\gamma}^\top (\mathbf{B}_{t+\tau} - \mathbb{E}(\mathbf{B}_{t+\tau}))^\top], \quad (\text{B.34})$$

we have by Theorem 3 (ii) of Wu (2011) that

$$T^{1/2} s_1^{-1/2} \boldsymbol{\alpha}^\top F_6 \xrightarrow{D} \mathcal{N}(0, 1).$$

Then equivalently we have

$$T^{1/2} (\mathbf{R}_1 \boldsymbol{\Sigma} \mathbf{R}_1^\top)^{-1/2} F_6 \xrightarrow{D} \mathcal{N}(\mathbf{0}, \mathbf{I}_L), \quad (\text{B.35})$$

so that F_6 is at least $T^{1/2} d^{(1+a-b)/2}$ -convergent which used $\lambda_{\max}(\mathbf{R}_1 \mathbf{R}_1^\top) = O(d^{-1-a})$ from (B.15), and all eigenvalues of $d^{-b} \boldsymbol{\Sigma}$ uniformly bounded from 0 and infinity by Assumption (R8). Hence, $\|D_1\|_1 = O(\|F_6\|_1) = O(T^{-1/2} d^{-(1+a-b)/2})$ on $\mathcal{M}$, and by (B.32) we have

$$\|\tilde{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_1 = O_P(\|D_1\|_1 + \|D_2\|_1) = O_P(T^{-1/2} d^{-(1+a-b)/2} + c_T d^{-\frac{1}{2} + \frac{1}{2w}}) = O_P(c_T d^{-\frac{1}{2} + \frac{1}{2w}}),$$

where the last equality used Assumption (R10). With the above plugged into (B.12), the proof of Theorem 1 is complete. $\square$

Proof of Theorem 2. By the KKT condition, $\hat{\boldsymbol{\phi}}$ is a solution to the adaptive LASSO problem in (2.17) if and only if there exists a subgradient

$$\mathbf{h} = \partial(\mathbf{u}^\top |\hat{\boldsymbol{\phi}}|) = \left\{ \mathbf{h} \in \mathbb{R}^L : \begin{cases} h_i = u_i \text{sign}(\hat{\phi}_i), & \hat{\phi}_i \neq 0; \\ |h_i| \leq u_i, & \text{otherwise.} \end{cases} \right\},$$

such that differentiating the expression in (2.17) with respect to $\boldsymbol{\phi}$, we have

$$T^{-1}(\boldsymbol{\Xi} \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\boldsymbol{\Xi} \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V}) \boldsymbol{\phi} + T^{-1}(\boldsymbol{\Xi} \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\mathbf{B}^\top \mathbf{y} - \boldsymbol{\Xi} \mathbf{y}^\nu) = -\lambda \mathbf{h}.$$

Substituting (2.11) in the above, we arrive at

$$\begin{aligned}
-\lambda \mathbf{h} &= T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V}) \phi \\
&\quad + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\mathbf{B}^\top \mathbf{V} \phi^* + \mathbf{B}^\top \mathbf{X}_{\beta^*} \text{vec}(\mathbf{I}_d) + \mathbf{B}^\top \boldsymbol{\epsilon} - \Xi \mathbf{y}^\nu) \\
&= T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \mathbf{V} (\phi^* - \phi) + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \boldsymbol{\epsilon} \\
&\quad + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \mathbf{X}_{\beta^*} \text{vec}(\mathbf{I}_d) + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\Xi \mathbf{Y}_W \phi^* - \Xi \mathbf{y}^\nu + \Xi \mathbf{Y}_W (\phi - \phi^*)) \\
&= T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top (\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V}) (\phi - \phi^*) + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \boldsymbol{\epsilon} \\
&\quad + T^{-1}(\Xi \mathbf{Y}_W - \mathbf{B}^\top \mathbf{V})^\top \mathbf{B}^\top \mathbf{X}_{\beta^* - \beta(\phi^*)} \text{vec}(\mathbf{I}_d),
\end{aligned}$$

where the last equality used the fact that $\mathbf{B}^\top \mathbf{X}_{\beta(\phi^*)} \text{vec}(\mathbf{I}_d) = \Xi \mathbf{y}^\nu - \Xi \mathbf{Y}_W \phi^*$ from (A.1). Then we may conclude that there exists a sign-consistent solution $\hat{\phi}$ if and only if

$$\left\{ \begin{aligned} -\lambda \mathbf{h}_H &= T^{-1}(\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)^\top (\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H) (\hat{\phi} - \phi^*) \\ &\quad + T^{-1}(\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)^\top \mathbf{B}^\top \mathbf{X}_{\beta^* - \beta(\phi^*)} \text{vec}(\mathbf{I}_d) + T^{-1}(\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)^\top \mathbf{B}^\top \boldsymbol{\epsilon}, \\ \lambda \mathbf{u}_{H^c} &\geq \left| T^{-1}(\Xi \mathbf{Y}_{W,H^c} - \mathbf{B}^\top \mathbf{V}_{H^c})^\top \mathbf{B}^\top \mathbf{X}_{\beta^* - \beta(\phi^*)} \text{vec}(\mathbf{I}_d) + T^{-1}(\Xi \mathbf{Y}_{W,H^c} - \mathbf{B}^\top \mathbf{V}_{H^c})^\top \mathbf{B}^\top \boldsymbol{\epsilon} \right|, \end{aligned} \right. \quad (\text{B.36})$$

where $\mathbf{A}_H$ and $\mathbf{a}_H$ denote the corresponding submatrix $\mathbf{A}$ with columns restricted on the set H and subvector $\mathbf{a}$ with entries restricted on the set H , respectively. Similarly $(\cdot)_{H^c}$ is defined. Consider the first equation in (B.36), similar to how D_2 is decomposed in the proof of Theorem 1, we write $\hat{\phi} - \phi^* = \sum_{j=1}^4 I_{\phi,j}$ where

$$\begin{aligned}
I_{\phi,1} &= [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} \left\{ (\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H \right. \\
&\quad \left. - T^{-1} d^a (\mathbf{B}^\top \mathbf{V}_H - \Xi \mathbf{Y}_{W,H})^\top (\mathbf{B}^\top \mathbf{V}_H - \Xi \mathbf{Y}_{W,H}) \right\} (\hat{\phi} - \phi^*), \\
I_{\phi,2} &= -[(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} d^a \lambda \mathbf{h}_H, \\
I_{\phi,3} &= [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} T^{-1} d^a (\Xi \mathbf{Y}_{W,H} - \mathbf{B}^\top \mathbf{V}_H)^\top \mathbf{B}^\top \mathbf{X}_{\beta(\phi^*) - \beta^*} \text{vec}(\mathbf{I}_d), \\
I_{\phi,4} &= [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} T^{-1} d^a (\mathbf{B}^\top \mathbf{V}_H - \Xi \mathbf{Y}_{W,H})^\top \mathbf{B}^\top \boldsymbol{\epsilon}.
\end{aligned}$$

Similar to F_1 in the proof of Theorem 1, we may derive that

$$\|I_{\phi,1}\|_{\max} = O_P\left(c_T d^{1-a} \|\hat{\phi} - \phi^*\|_{\max}\right) = o_P\left(\|\hat{\phi} - \phi^*\|_{\max}\right), \quad (\text{B.37})$$

where the first equality used the fact that (R1) implies for a positive constant u that $\sigma_{|H|}^2\{(\mathbf{D}_W)_H\} \geq du > 0$ uniformly as $d \rightarrow \infty$, and the conditions in the statement of Theorem 2, and the second used (R10).

Similarly, with Assumption (R9) we have

$$\|I_{\phi,2}\|_{\max} = O_P(d^{-1-a} \cdot d^a \cdot \lambda) = O_P(c_T d^{-1}). \quad (\text{B.38})$$

For $I_{\phi,4}$, we may decompose it as the following with the second term dominating the first term similarly to F_5 and F_6 in the proof of Theorem 1,

$$\begin{aligned} I_{\phi,4} = & \left([(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} \left[(T^{-1/2} d^{a/2} \mathbf{B}^\top \mathbf{V}_H - \mathbf{H}_{10,H}) - (T^{-1/2} d^{a/2} \mathbf{\Xi} \mathbf{Y}_{W,H} - \mathbf{H}_{20,H}) \right]^\top \right. \\ & \left. \cdot T^{-1/2} d^{a/2} \mathbf{B}^\top \boldsymbol{\epsilon} \right) - \left([(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} (\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top \cdot T^{-1/2} d^{a/2} \mathbf{B}^\top \boldsymbol{\epsilon} \right). \end{aligned}$$

The second term in the above has rate $T^{-1/2} d^{-(1+a-b)/2}$ by exactly the same way to construct asymptotic normality of F_6 in (B.35), except for the restriction to the set H here (proof omitted). Thus,

$$\|I_{\phi,4}\|_{\max} = O_P(T^{-1/2} d^{-(1+a-b)/2}). \quad (\text{B.39})$$

We next construct the asymptotic normality of $I_{\phi,3}$ and show its rate is of order $T^{-1/2} d^{-(1-b)/2}$ which is dominating over those of $I_{\phi,1}$, $I_{\phi,2}$ and $I_{\phi,4}$ by Assumption (R10). Recall $\mathbf{R}_H = [(\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top (\mathbf{H}_{20} - \mathbf{H}_{10})_H]^{-1} (\mathbf{H}_{20} - \mathbf{H}_{10})_H^\top$, and let non-zero $\boldsymbol{\alpha} \in \mathbb{R}^{|H|}$ such that $\|\boldsymbol{\alpha}\|_1 \leq c < \infty$. Then we have

$$\begin{aligned} \boldsymbol{\alpha}^\top I_{\phi,3} &= \boldsymbol{\alpha}^\top \mathbf{R}_H T^{-1} (\mathbf{B}_\gamma - \bar{\mathbf{B}}_\gamma)^\top \mathbf{X}_{\beta(\phi^*) - \beta^*} \text{vec}(\mathbf{I}_d) (1 + o_P(1)) \\ &= \boldsymbol{\alpha}^\top \mathbf{R}_H T^{-1} \left(\mathbf{I}_d \otimes \{(\mathbf{B}_1 - \bar{\mathbf{B}}, \dots, \mathbf{B}_T - \bar{\mathbf{B}})(\mathbf{I}_T \otimes \gamma(\beta(\phi^*) - \beta^*)^\top)(\mathbf{X}_1, \dots, \mathbf{X}_T)^\top\} \right) \text{vec}(\mathbf{I}_d) (1 + o_P(1)) \\ &= \boldsymbol{\alpha}^\top \mathbf{R}_H \frac{1}{T} \sum_{t=1}^T \text{vec}((\mathbf{B}_t - \mathbb{E}(\mathbf{B}_t)) \gamma(\beta(\phi^*) - \beta^*)^\top \mathbf{X}_t^\top) (1 + o_P(1)) \\ &= \boldsymbol{\alpha}^\top \mathbf{R}_H \text{vec} \left(\frac{1}{T} \sum_{t=1}^T \left[\gamma^\top(\mathbf{B}_{t,i\cdot} - \mathbb{E}(\mathbf{B}_{t,i\cdot})) \mathbf{X}_{t,j\cdot}^\top (\beta(\phi^*) - \beta^*) \right]_{i,j \in [d]} \right) (1 + o_P(1)) \\ &= \boldsymbol{\alpha}^\top \mathbf{R}_H \mathbf{S}_\gamma (\beta(\phi^*) - \beta^*) (1 + o_P(1)), \end{aligned}$$

where the third last equality used $\mathcal{A}_4$ in Lemma 2 and the last used $\mathcal{A}_1$. From (B.6) and Lemma 4, we have

$$\begin{aligned} T^{1/2} (\mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top)^{-1/2} (\beta(\phi^*) - \beta^*) &= T^{1/2} (\mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top)^{-1/2} \left\{ (\mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu \right\} \\ &= T^{1/2} (\mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top)^{-1/2} \left\{ [\mathbb{E}(\mathbf{X}_t^\top \mathbf{B}_t) \mathbb{E}(\mathbf{B}_t^\top \mathbf{X}_t)]^{-1} T^{-2} \mathbf{X}^\top \mathbf{B}^\nu (\mathbf{B}^\nu)^\top \boldsymbol{\epsilon}^\nu (1 + o(1)) \right\} \xrightarrow{D} \mathcal{N}(\mathbf{0}, \mathbf{I}_r). \end{aligned}$$

Define $s_3 := \boldsymbol{\alpha}^\top \mathbf{R}_H \mathbf{S}_\gamma \mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top \mathbf{S}_\gamma^\top \mathbf{R}_H^\top \boldsymbol{\alpha}$, we hence have $T^{1/2} s_3^{-1/2} \boldsymbol{\alpha}^\top I_{\phi,3} \xrightarrow{D} \mathcal{N}(0, 1)$ and equivalently

$$T^{1/2} (\mathbf{R}_H \mathbf{S}_\gamma \mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top \mathbf{S}_\gamma^\top \mathbf{R}_H^\top)^{-1/2} I_{\phi,3} \xrightarrow{D} \mathcal{N}(\mathbf{0}, \mathbf{I}_{|H|}).$$

As shown in the proof of Lemma 4, the eigenvalues of $\mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top$ are of order d^{b-1} . Similar to $\mathbf{R}_1$ in the proof of Theorem 1, $\lambda_{\max}(\mathbf{R}_H \mathbf{R}_H^\top) = O(d^{-1-a})$. We also have $\lambda_{\max}(\mathbf{S}_\gamma^\top \mathbf{S}_\gamma) = O(d^{1+a})$ by Assumption (R5). Combining them, we have

$$\|\boldsymbol{\alpha}\|_1^2 \lambda_{\min}(\mathbf{R}_H \mathbf{S}_\gamma \mathbf{S}_\gamma^\top \mathbf{R}_H^\top) \lambda_{\min}(\mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top) \leq s_3 \leq \|\boldsymbol{\alpha}\|_1^2 \lambda_{\max}(\mathbf{R}_H \mathbf{R}_H^\top) \lambda_{\max}(\mathbf{S}_\gamma^\top \mathbf{S}_\gamma) \lambda_{\max}(\mathbf{R}_\beta \boldsymbol{\Sigma}_\beta \mathbf{R}_\beta^\top),$$

with the right hand side of order d^{b-1} . The left hand side is of the same order by the assumption in the statement of Theorem 2 that $\mathbf{R}_H \mathbf{S}_\gamma \mathbf{S}_\gamma^\top \mathbf{R}_H^\top$ has the smallest eigenvalue of constant order. Thus, s_3 is of order exactly d^{b-1} and hence $\boldsymbol{\alpha}^\top I_{\phi,3}$ has order exactly $T^{-1/2} d^{-(1-b)/2}$. It implies $I_{\phi,3}$ is the leading term in $\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}^*$ whose asymptotic normality therefore holds.

As $I_{\phi,1}$ to $I_{\phi,4}$ are all $o_P(1)$, we conclude $\text{sign}(\hat{\boldsymbol{\phi}}_H) = \text{sign}(\boldsymbol{\phi}_H^*)$. It remains to show the second part in (B.36) for the zero consistency of $\hat{\boldsymbol{\phi}}_{H^c}$.

To this end, notice similar to $I_{\phi,3}$ but with restriction on the set H^c , we have

$$\begin{aligned} & \|T^{-1}(\boldsymbol{\Xi} \mathbf{Y}_{W,H^c} - \mathbf{B}^\top \mathbf{V}_{H^c})^\top \mathbf{B}^\top \mathbf{X}_{\beta^* - \beta(\phi^*)} \text{vec}(\mathbf{I}_d)\|_{\max} \\ &= O_P(T^{-1/2} d^{-(1-b)/2} \cdot d^{-a} \cdot d^{1+2a}) = O_P(T^{-1/2} d^{\frac{1}{2} + \frac{b}{2} + a}), \end{aligned}$$

which used $\|(\mathbf{H}_{20} - \mathbf{H}_{10})^\top (\mathbf{H}_{20} - \mathbf{H}_{10})\|_{\max} \leq \sigma_1^2(\mathbf{H}_{20} - \mathbf{H}_{10}) = O(d^{1+2a})$ similarly to the steps above (B.15). In the same manner, we also have from $I_{\phi,4}$ that

$$\|T^{-1}(\boldsymbol{\Xi} \mathbf{Y}_{W,H^c} - \mathbf{B}^\top \mathbf{V}_{H^c})^\top \mathbf{B}^\top \boldsymbol{\epsilon}\|_{\max} = O_P(T^{-1/2} d^{\frac{1}{2} + \frac{b}{2} + \frac{a}{2}}).$$

The left hand side of the second inequality in (B.36) has minimum value of

$$\frac{\lambda}{\|\tilde{\boldsymbol{\phi}}_{H^c}\|_{\max}} \geq \frac{\lambda}{\|\tilde{\boldsymbol{\phi}}_{H^c} - \boldsymbol{\phi}_{H^c}^*\|_{\max}} \geq \frac{\lambda}{\|\tilde{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_{\max}},$$

so it suffices to show

$$(T^{-1/2} d^{\frac{1}{2} + \frac{b}{2} + a}) \cdot \|\tilde{\boldsymbol{\phi}} - \boldsymbol{\phi}^*\|_{\max} = o_P(c_T),$$

which is true by Assumption (R10) and Theorem 1 in which each entry of F_6 can be shown to be asymptotically normal. This completes the proof of Theorem 2. $\square$

Proof of Theorem 3. We have

$$\begin{aligned}\|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\|_\infty &= \left\| \sum_{j=1}^p \left[(\widehat{\phi}_{j,0} - \phi_{j,0}^*) + \sum_{k=1}^{l_j} (\widehat{\phi}_{j,k} - \phi_{j,k}^*) z_{j,k,t} \right] \mathbf{W}_j \right\|_\infty \\ &= O_P\left(\|\widehat{\phi} - \phi^*\|_1 \cdot \max_j \|\mathbf{W}_j\|_\infty\right) = O_P(T^{-1/2} d^{-(1-b)/2}),\end{aligned}\tag{B.40}$$

where the last equality used Theorem 2, Assumptions (M2) (or (M2')) and (R1). Observe that we have similarly $\|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\|_1 = O_P(T^{-1/2} d^{-(1-b)/2})$ by Assumption (R1), and hence

$$\|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\| \leq \left(\|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\|_1 \|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\|_\infty \right)^{1/2} = O_P(T^{-1/2} d^{-(1-b)/2}).$$

With $\mathbf{\Pi}_t^*$ defined in (B.8), we can decompose

$$\begin{aligned}\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^* &= \frac{1}{T} \sum_{t=1}^T \left\{ (\mathbf{I}_d - \mathbf{\Lambda}_t \widehat{\boldsymbol{\Phi}}) \mathbf{y}_t - \mathbf{X}_t \widehat{\boldsymbol{\beta}} \right\} - \frac{1}{T} \sum_{t=1}^T \left\{ (\mathbf{I}_d - \mathbf{\Lambda}_t \boldsymbol{\Phi}^*) \mathbf{y}_t - \mathbf{X}_t \boldsymbol{\beta}^* - \boldsymbol{\epsilon}_t \right\} \\ &= \frac{1}{T} \sum_{t=1}^T \left\{ (\mathbf{W}_t^* - \widehat{\mathbf{W}}_t) (\mathbf{\Pi}_t^* \boldsymbol{\mu}^* + \mathbf{\Pi}_t^* \mathbf{X}_t \boldsymbol{\beta}^* + \mathbf{\Pi}_t^* \boldsymbol{\epsilon}_t) \right\} + \bar{\mathbf{X}}(\boldsymbol{\beta}^* - \widehat{\boldsymbol{\beta}}) + \bar{\boldsymbol{\epsilon}},\end{aligned}$$

so that combining (B.40), Lemma 2 and Theorem 1, we have

$$\|\widehat{\boldsymbol{\mu}} - \boldsymbol{\mu}^*\|_{\max} = O_P \left\{ \max_t \|\widehat{\mathbf{W}}_t - \mathbf{W}_t^*\|_\infty \left(\|\boldsymbol{\mu}^*\|_{\max} + c_T \|\boldsymbol{\beta}^*\|_{\max} + c_T \right) + c_T \|\boldsymbol{\beta}^* - \widehat{\boldsymbol{\beta}}\|_1 + c_T \right\} = O_P(c_T).$$

This completes the proof of Theorem 3. $\square$

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